Elimination of the GGHV Normal Forms at Degree Pair $(72, 108)$, and Closure of the Two-Dicritical Line:

A Certified Computational Audit in Three Parts

Branch-(a,b) Elimination · Branch-(c) Elimination · Dicriticals at Infinity of Keller Maps

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Computation, Lean and LaTeX Source:
https://github.com/brandonmccraryresearch-cloud/2D-Jacobian-Conjecture-Workspace

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Abstract

Part I. We present a computational elimination of the branch (a,b) normal form arising from Proposition 4.3 of Guccione–Guccione–Horruitiner–Valqui [1] for a hypothetical $(72, 108)$ counterexample to the planar Jacobian conjecture. Every base object is source-gated directly against GGHV v1 before any proof is drafted.

Starting from the verified Newton polygons

$$N(P) = \mathrm{conv}\{(0,0),(1,0),(8,14),(8,16)\},$$
$$N(Q) = \mathrm{conv}\{(0,0),(2,1),(12,21),(12,24)\},$$

with $[P, Q] = x^2$ post-φ, we reduce the 72-unknown bilinear system to five one-variable layers $E_1, \dots, E_5$ totaling 92 equations, verified as an exact symbolic polynomial identity.

The top layer E_5 is the Davenport–Stothers extremal equation $\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta) = 1$ [2, 3]. The Belyi classification gives the geometric upper bounds 3, 10, 35 chart points for $m = 3, 5, 7$; for $m = 7$ the bound is sharp, with the degree-35 eliminating polynomial $W(a_6)$ irreducible over $\mathbb{Q}$, a quintic in $S = a_6^7$ with nonzero discriminant and exactly one real root. The 35 solutions are distinct and form a single Galois orbit: one real solution and 17 conjugate pairs, grouped in five S-orbits. The classification is proved twice: the Belyi bound combined with the 35 exhibited points gives a complete classification (relying on Riemann's existence theorem and the computed msolve parametrization), while §6.4 gives an independent algebraic proof, valid over every field of characteristic 0, which is the route formalized in Lean. In reversed coordinates the chart equations are the coefficient recursion of $(1 + y_1 v + \cdots + y_7 v^7)^{3/2}$. Six weighted-homogeneous relations, of weights $4, 5, 5, 6, 6, 7$ and with explicit certificates, cut out the five torus orbits. We formalize this proof in Lean.

For $m = 3$ and $m = 5$ the bounds 3 and 10 are also sharp. These two classifications are kernel-checked in Lean as well, over every field of characteristic 0. In the normalized chart, the system holds if and only if two things hold: the eliminant vanishes ($3a_2^3 - 32$ for $m = 3$, $9a_4^{10} + 37200a_4^5 + 95051008$ for $m = 5$), and the remaining unknowns take explicit values. In each case the eliminant has no repeated root. So over every algebraically closed field of characteristic 0 there are exactly 3 solutions for $m = 3$ and 10 for $m = 5$, and this count

is kernel-checked too. Both eliminants are irreducible over $\mathbb{Q}$, and this is kernel-checked as well. Theorem 1.1 does not need these two cases.

The descent proceeds through E_4 (an 18×19 matrix of rank 17, kernel dimension 2), E_3 (a 19×20 matrix of rank 18, kernel dimension 2, unconditionally solvable for every direction t), and E_2 (injective 19×12, rank 12), leaving seven explicit compatibility conditions. We expand these as $b_i(t) + s_1 M_i(t) + s_2 L_i(t) = 0$ with b_i cubic and M_i, L_i linear, eliminate (s_1, s_2) to obtain 35 quintic 3×3 minors. The entire descent is computed exactly over the quintic field K_5 in characteristic 0: the 35 minors span all binary quintics, forcing $t = 0$. The case $t = 0$ forces B_0 constant, contradicting the required vertex $(12, 24)$. The Lean formalization quantifies over all roots of R and all torus parameters, covering all 35 top-layer solutions.

Hence no polynomials P, Q satisfying NewtonNF2 have $[P, Q] = \lambda x^2$ with $\lambda \neq 0$ (Theorem 1.1). In Lean this NewtonNF2 form is a single kernel-checked theorem, `main_theorem`, on the standard axioms only, independent of GGHV Proposition 4.3. Its top-layer input is `chartClassification_holds`, a formalized algebraic proof of the $m = 7$ top-layer classification of Proposition 6.1, in the normalized chart (§6.4). The further rigidity $a_{8,16} = 0$ (Corollary 1.2) is kernel-checked in Lean as well, together with the vanishing of every other coefficient of P and Q off the top edges except the constant terms (`lower_edge_rigidity`); it uses the layer E_1. A second proof of $a_{8,16} = 0$, outside Lean and independent of the descent, is an explicit certificate over K_5: the identity $a_{8,16}^2 = \sum_{k=1}^{75} H_k e_k$, where the e_k are the 75 equations of the layers $E_4, \dots, E_1$. Two independent programs check it exactly. The application to the Jacobian conjecture is conditional on the correctness and exhaustiveness of [1, Proposition 4.3] (Corollary 1.4).

Part II. We present a computational elimination of the branch-(c) normal form arising from case (1) of Proposition 4.3 of Guccione–Guccione–Horruitiner–Valqui [1] for a hypothetical $(72, 108)$ counterexample to the planar Jacobian conjecture. Every base object is source-gated directly against GGHV v1 before any proof is drafted. This paper is the branch-(c) companion to the branch-(a,b) elimination [9], whose format it follows.

Starting from the verified Newton polygons

$$N(P) = \mathrm{conv}\{(0,0), (1,0), (8,14), (8,16), (0,8)\},$$
$$N(Q) = \mathrm{conv}\{(0,0), (2,1), (12,21), (12,24), (0,12)\},$$

with $[P, Q] = \lambda x^2$ post-φ and $\lambda \neq 0$, we reduce the bracket condition to layer identities E_n in the grading $w(x) = 2$, $w(y) = -1$, extract 132 exact scalar equations, and eliminate the layers $E_4, \dots, E_{-2}$ to twelve chart conditions $\Omega, \Psi, \Phi_1, \Phi_2, \Theta_1, \Theta_2, \Theta_3$ plus five pure rows, over the quintic field $K_5 = \mathbb{Q}[w]/(w^5 - w^4 + 3w^3 + 3w^2 + 26)$. The top layer is the branch-(a,b) problem, whose exact K_5 classification is inherited. The descent is proved twice: symbolically, as 600 kernel-checked reflective polynomial identities in Lean 4 (no `sorry`, standard axioms only), and numerically, by exact finite-field linear algebra.

The chart is empty. On the stratum $b_{11,20} = 0$ this is proved in the Lean kernel from an exact K_5 certificate (degree 2, 75 terms). On the whole chart it follows from a rank lemma: full row rank of the fixed-degree Macaulay matrix lifts from $\mathbb{F}_p$ to K_5 through a nonzero minor (retained as independent corroboration; the Lean certificate of Theorem 1.5 is now the load-bearing route). At weight $W = 24$ the 3199×6054 Macaulay matrix has full row rank 3199 at two primes ($p = 1000003$, $p = 32003$), verified by two independent implementations (FLINT and NumPy), with negative controls. An earlier reduction lemma used in a preliminary version of this work is false as stated and is retracted here; the rank lemma is its correct replacement, and the distinction—rank lifts, mere consistency does not—is proved by counterexample.

In Lean, with all 790 modules built and `#print axioms` reporting only `propext`, `Classical.choice`, `Quot.sound`,

$$\texttt{ChartEmptyC_T1ne0}\ L \Rightarrow \neg\exists\, P\, Q\, \lambda,\ \lambda \neq 0 \wedge \texttt{NewtonNFc}\ P\, Q \wedge [P, Q] = \lambda x^2,$$

and, since November 4, 2026, `ChartEmptyC_T1ne0` is proved inside Lean with no hypotheses (`challenge_target`; Theorem 1.5 of Part II), so branch-(c) elimination is unconditional in Lean. The rank computation remains as independent corroboration. The application to the Jacobian conjecture is conditional on the correctness and exhaustiveness of [1, Proposition 4.3] (Corollary 1.4).

Part III. Let $F = (P, Q) \colon \mathbb{A}^2 \to \mathbb{A}^2$ be a Keller map, that is, $[P, Q] := P_x Q_y - P_y Q_x \in K^\times$ for K algebraically closed of characteristic 0, and let $\lambda P + \mu Q$ be its pencil. The October 1 draft of this paper claimed that no Keller map admits two distinct dicritical divisors, unconditionally. That claim is false: every Keller map of degree at least 2 has at least two exceptional dicriticals in total (Corollary 1.5), and the draft's E_1 degree obstruction theorem is false as stated, with an explicit Keller counterexample in §3.

The corrected target is **Theorem R** (§1), which is **open**: for every Keller map, the pencil $\lambda P + \mu Q$ has at most one dicritical divisor **at infinity**. Theorem R is equivalent to the planar Jacobian conjecture (Proposition 1.9). This paper proves, or records as sketch or fixed-degree computation, only the following steps toward it, each labelled with its grade:

- the count of dicriticals (Proposition 1.3, grade B) and the position a counterexample's dicriticals would occupy (§1.7);
- the line at infinity is not dicritical for a non-affine Keller map (Lemma 2.1, grade B), and no first blow-up at infinity is dicritical (Lemma 2.3, algebra grade A, chart step grade B), so every dicritical at infinity of a non-affine Keller map has depth at least 2;
- the $m = 1$ bridge equations collapse under two stated premises (P1) and (P2) (Theorem 4.1, equations grade A, geometric reading grade B), where (P2) is **not** derived and remains open;
- the constant-leading-form argument for $m \geq 2$ on E_k, $k \geq 2$, as a **proof sketch for free centers only** (Proposition 2.5, grade C);
- the adjacent $m = 1$ case as a computation at $d = 3$ only (grade C).

What remains open is collected in §5 (obligations O1–O6) and the tests any proof must pass (T1–T4). §6 maps each archived script to the exact statement it certifies, under the corrected statements.

Validation note — November 4, 2026 (Part II update)

This note records the deepest-research validation (fact-check mode: verification of a provided text against its record, not a systematic review) run on the November 4, 2026 draft before any repo incorporation. It is not part of the mathematical argument.

- **What changed.** Part II no longer rests on a grade-B premise. On November 4, 2026, `challenge_target` was proved in Lean: `ChartEmptyC_T1ne0` holds for every field of characteristic 0 with no hypotheses, with `#print axioms` exactly `[propext, Classical.choice, Quot.sound]`; `branch_c_eliminated` follows from it by `main_theorem_c_of_chartEmpty_T1ne0` with the same axiom triple (Theorem 1.5). The rank-lemma route of §10 is retained as independent corroboration at grade B, not as the load-bearing step.
- **Scope guard.** This eliminates one case — case (1) of GGHV Proposition 4.3, branch (c) — unconditionally in Lean. It does *not* by itself exclude $(72, 108)$ unconditionally and does *not* prove the planar Jacobian conjecture: those still require the correctness and exhaustiveness of GGHV Proposition 4.3 and the reduction chain, which were ledgered, not re-derived, in this validation.
- **Case numbering.** Verified throughout: Proposition 4.3 case (1) is the polygon set *with* vertices $(0, 8)$ resp. $(0, 12)$ = `NewtonNFc` = branch (c); case (2) is the set *without* them = `NewtonNF2` = branches (a),(b). No occurrence of the older shorthand "(a),(b) = case (1)" appears in this paper. Two Part II line citations were corrected in this finalization (GGHV v1 lines 495/599–600 → 494/617–618) after re-extraction of the v1 text, which shows case (1)

at line 494 and the post-φ case c list at lines 617–618.

- **Provenance caveat.** `challenge_target` and `branch_c_eliminated` were proved November 4, 2026 as scratch files (`bridge_attempt_2026_11_04.lean`, `composition_2026-11-04.lean`, run on lean-build-2 with logs `bridge_challenge_target_2026-11-04.log` and `branch_c_eliminated_2026-11-04.log`); they are not yet tracked repo modules. Incorporation as tracked modules is a separate, later step and is listed as open item (5) in §12.
- **Title reading.** "Closure of the Two-Dicritical Line" in the title means closure (withdrawal and correction) of the October 1 draft's false two-dicritical claim, not a proof of Theorem R: Theorem R remains **open** (Part III, §1).

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Part I

Elimination of the GGHV Branch-(a,b) Normal Form at Degree Pair (72,108)

1 Introduction

1.1 Context and prior work

The planar Jacobian conjecture asserts that a polynomial map $(P, Q) : \mathbb{C}^2 \to \mathbb{C}^2$ with $[P, Q] := P_x Q_y - P_y Q_x \in \mathbb{C}^\times$ is invertible. In [1], Guccione, Guccione, Horruitiner, and Valqui raise the lower bound on $\max\{\deg P, \deg Q\}$ for a hypothetical counterexample from 100 (Moh [4]) to 108, leaving open only the degree pair $(72, 108)$ and its symmetric $(108, 72)$.

Santibáñez-Leal has published machine evidence related to the $(72, 108)$ candidate [5, 6], explicitly not claiming elimination. The candidate itself is documented in [7].

For the degree pair $(72, 108)$, [1] (an unrefereed preprint, arXiv:2204.14178v1) identifies two cases. The corner $(9, 27)$ is discarded in [1, §5]; the open case is the corner $(8, 28)$ with $(m, n) = (3, 2)$. For this case, [1, Proposition 4.3] shows that any counterexample gives rise, after an explicit (non-invertible) morphism φ, to one of two Newton polygon configurations. Case (1), with additional vertices $(0, 8)$ resp. $(0, 12)$, is branch (c). Case (2), the subject of this paper, is branches (a) and (b):

$$\begin{aligned} N(P) &= \operatorname{conv}\{(0,0), (1,0), (8,14), (8,16)\}, \\ N(Q) &= \operatorname{conv}\{(0,0), (2,1), (12,21), (12,24)\}, \end{aligned} \tag{1}$$

with $[P, Q] = x^2$ in post-φ coordinates. See Figure 1 for the polygon shapes.

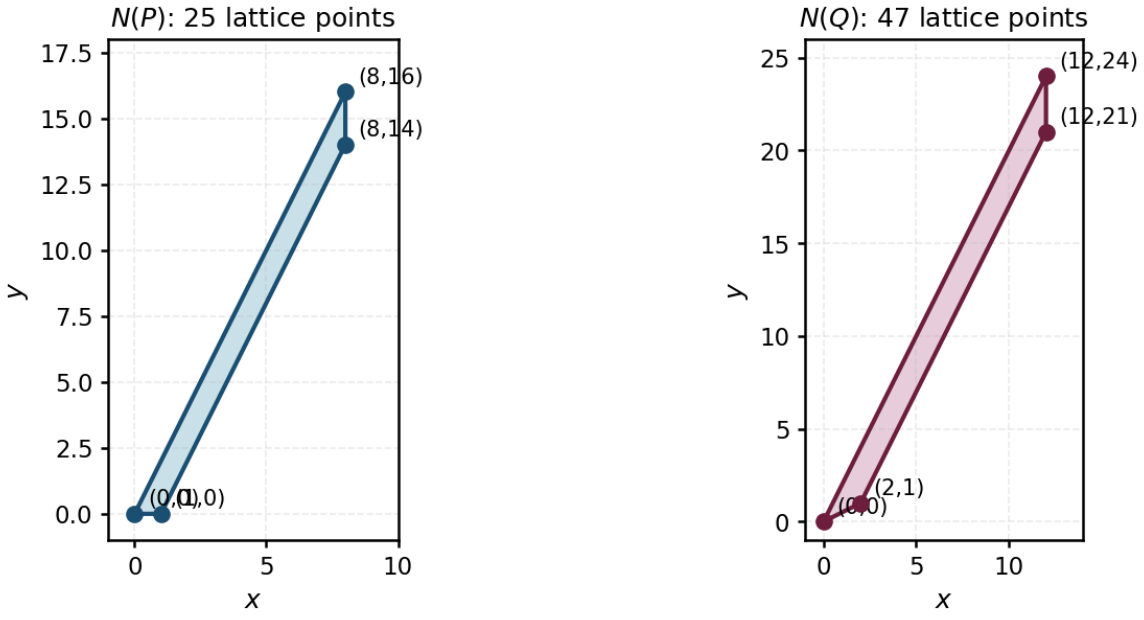


Figure 1: The branch (a,b) Newton polygons from [1, Proposition 4.3(2)], source-gated at v1 lines 495 and 599–600. $N(P)$ has 25 lattice points; $N(Q)$ has 47.

1.2 Main result

Theorem 1.1 (Elimination of the GGHV branch-(a,b) normal form)**.** *Let K be a field of characteristic 0. Let $P, Q \in K[x, y]$ have $\operatorname{supp} P \subset N_P$ and $\operatorname{supp} Q \subset N_Q$, with $a_{1,0} \cdot a_{8,14} \cdot b_{2,1} \cdot b_{12,21} \neq 0$ and $[P, Q] = \lambda x^2$ for some $\lambda \neq 0$. Then $b_{12,24} = 0$. In particular, no P, Q satisfy* NewtonNF2(P, Q) *(listing 11; the polygons are in §2) with $[P, Q] = \lambda x^2$.*

Corollary 1.2 (Lower-edge rigidity)**.** *Under the hypotheses of Theorem 1.1, $a_{8,16} = 0$ as well. More precisely, every coefficient $a_{i,j}$ with $j \geq 2i - 1$ and every coefficient $b_{i,j}$ with $j \geq 2i - 2$ vanishes, except $a_{0,0}$ and $b_{0,0}$: $P = P_2 + a_{0,0}$ and $Q = Q_3 + b_{0,0}$. In Lean this is* ***`lower_edge_rigidity`*** *(listing 21), kernel-checked on the standard axioms; $\lambda \neq 0$ is not needed. A second proof of $a_{8,16} = 0$, outside Lean, is an explicit certificate over K_5 (`scripts/a816_certificate/`), checked exactly by* SINGULAR *and by python-flint; see §5. The corollary is not needed for the elimination: $b_{12,24} = 0$ already contradicts the required Newton vertex $(12, 24)$.*

Remark 1.3 (Lean-to-mathematics correspondence)*.* The Lean predicate `NewtonNF2` (listing 11) is

$$\texttt{NewtonNF2}(P, Q) \iff \begin{cases} \operatorname{supp} P \subseteq N_P,\ \operatorname{supp} Q \subseteq N_Q \text{ (via } \texttt{inNP}, \texttt{inNQ}), \\ a_{0,0}, a_{1,0}, a_{8,14}, a_{8,16} \neq 0, \\ b_{0,0}, b_{2,1}, b_{12,21}, b_{12,24} \neq 0, \end{cases}$$

i.e. exactly the support conditions of Theorem 1.1 together with the nonvanishing of the eight vertex coefficients. The half-plane predicates are tied to the vertex sets by `decide` checks run on duplicate predicates `latticeNP`/`latticeNQ` (which restate the same inequalities): the lattice-point counts (25 and 47) and the vertex-membership check confirm the eight vertices lie on the polygons. This bridge is what lets the kernel-checked `main_theorem` speak about the same P, Q as Theorem 1.1.

Corollary 1.4 (Conditional on GGHV)**.** *Assuming [1, Prop. 4.3], no counterexample to the plane Jacobian conjecture arises in case $(8, 28)$, sub-cases a) and b).*

The proof is computational but exact. We reduce the 72-unknown system to triangular layers (Figure 2), determine the top layer via the Belyi-map/Riemann-existence argument (Figure 3), and independently by an algebraic argument formalized in Lean (§6.4). We then obstruct the descent via the Lean-formalized layer certificates (§7). Key steps were verified in SINGULAR 4.4.1 and Python/SYMPY; scripts are archived with this draft.

1.3 Logical status

Remark 1.5. Theorem 1.1 does not depend on GGHV. Its proof uses Proposition 6.1 for $m = 7$, which is proved twice. On paper it follows from Riemann's existence theorem (§6). In §6.4 the classification it provides is proved algebraically, in the normalized chart, and that proof is formalized in Lean as `chartClassification_holds` (listing 16). In Lean, `main_theorem_of_chart` (listing 7) derives the NewtonNF2 form of Theorem 1.1 from the explicit statement `ChartClassification` (listing 12). `main_theorem` (listing 17) combines it with `chartClassification_holds` into a single kernel-checked theorem independent of GGHV Proposition 4.3, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The $b_{12,24} = 0$ part follows from `no_completion_K5_PQ` together with `topLayerClassification_of_chart` and `chartClassification_holds`; the $a_{8,16} = 0$ part (Corollary 1.2) is `a816_eq_zero`, and the whole corollary is `lower_edge_rigidity`

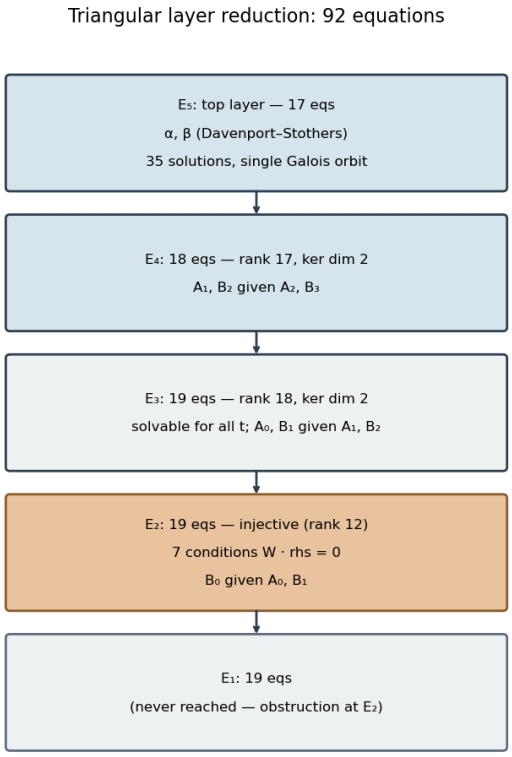


Figure 2: The triangular layer structure: 92 equations in five layers. Each layer is affine-linear in its unknowns given the solution above it.

(listing 21); both are kernel-checked on the same three axioms and reuse the descent up to $t = 0$. Outside Lean, an explicit certificate $a_{8,16}^2 = \sum_k H_k e_k$ over K_5, stored as the 76 cofactors of $1 = \sum_k z^2 H_k e_k - (1 + a_{8,16}z)(a_{8,16}z - 1)$ in `scripts/a816_certificate/a816_lift.txt` and checked exactly by SINGULAR and by python-flint, proves $a_{8,16} = 0$ without the descent (see §5). Only Corollary 1.4 also assumes GGHV Proposition 4.3.

1.4 Outline

Section 2 source-gates all base objects against GGHV v1. Section 3 sets up the 72-unknown system. Section 4 gives the exact layer reduction. Section 5 fixes normalizations. Section 6 classifies the top layer. Section 7 carries out the descent. Section 8 proves the final obstruction. Section 9 lifts to characteristic zero. Section 10 concludes; branch (c) is eliminated in Part II (Theorem 1.5). Section 11 covers prior work, attribution, and independence.

The proof is organized around four certificates. *Certificate I* (Newton reduction): $[P, Q] = \lambda x^2 \Rightarrow E_5, \ldots, E_1$ (§4). *Certificate II* (top-layer completeness): $E_5 \Rightarrow$ five torus orbits over K_5 (§6, §6.4). *Certificate III* (descent obstruction): $E_4, E_3, E_2 \Rightarrow$ the compatibility ideal equals $(t_1, t_2)^5$, hence $t = 0$ (§7, §8). *Certificate IV* (vertex destruction): $t = 0 \Rightarrow B_0' = 0 \Rightarrow b_{12,24} = 0$ (Corollary 8.6). The rigidity $a_{8,16} = 0$ (Corollary 1.2) is a supplementary result; Theorem 1.1 does not use it. The dependency graph is

$$\text{Counterexample in case } (8,28) \xrightarrow{\text{GGHV 4.3}} \text{NewtonNF2, } [P,Q] = \lambda x^2 \xrightarrow{\text{Thm. 1.1}} \text{contradiction}$$

with Theorem 1.1 itself independent of GGHV:

Result	Depends on GGHV?
Layer reduction	No
Top-layer classification	No
K_5 descent	No
$b_{12,24} = 0$	No
$a_{8,16} = 0$	No
Theorem 1.1	No
Corollary 1.4 (Jacobian application)	Yes

2 Source data

We verify every base object directly against version 1 of [1] (arXiv:2204.14178v1), downloaded from `https://arxiv.org/pdf/2204.14178v1` and extracted with `pdftotext -layout`. No proof is drafted until the polygons, bracket, and φ are confirmed character-by-character against GGHV's text.

Proposition 2.1 ([1, Proposition 4.3], source-gated)**.** *If there is a counterexample to the Jacobian conjecture in case* $(8, 28)$*, then there exist* $P, Q \in L^{(1)}$ *with* $[P, Q] = x^2$ *and one of:*

(1) $N(P) = \{(0,0), (1,0), (8,14), (8,16), (0,8)\}$,
$N(Q) = \{(0,0), (2,1), (12,21), (12,24), (0,12)\}$;
(2) $N(P) = \{(0,0), (1,0), (8,14), (8,16)\}$,
$N(Q) = \{(0,0), (2,1), (12,21), (12,24)\}$.

Source locations. The `pdftotext -layout` extraction of v1 yields:

- Line 33: $[P, Q] := P_x Q_y - P_y Q_x \in K^\times$ (bracket convention).
- Line 492: "Proposition 4.3 (Case $(8, 28)$)."
- Line 493: "with $[P, Q] = x^2$" (rendered as x2 in the extraction).
- Line 495: Case (2) polygons
$N(P) = \{(0,0), (1,0), (8,14), (8,16)\}$, $N(Q) = \{(0,0), (2,1), (12,21), (12,24)\}$.
- Lines 596–597: The morphism $\varphi(x) = x^{-1}$, $\varphi(y) = x^4 y$; "this is not an automorphism and the chain rule gives $[\varphi(P), \varphi(Q)] = -[P, Q]x^2$."
- Lines 599–600: "in the cases a) and b) the Newton Polygons of P and Q become $N(P) = \{(0,0), (1,0), (8,14), (8,16)\}$, $N(Q) = \{(0,0), (2,1), (12,21), (12,24)\}$."

A verification script (`source_gate_verify.py`) asserts each of these lines automatically; it exits non-zero if any check fails. □

We work exclusively with case (2) (branches a,b).

3 The polynomial system

3.1 Lattice points

The Newton polygons (1) contain:

$$|N(P) \cap \mathbb{Z}^2| = 25, \qquad |N(Q) \cap \mathbb{Z}^2| = 47, \tag{2}$$

verified by direct enumeration (cross-checked with Pick's theorem: area$(N(P))$ = 15, boundary 18, interior 7). This gives 25 + 47 = 72 unknown coefficients. The two constant terms do not appear in $[P, Q]$, leaving 70 effective unknowns, but we retain all 72 for the layer structure.

3.2 The bracket equations

Writing $P = \sum_{(i,j)\in N(P)} a_{ij} x^i y^j$ and $Q = \sum_{(i,j)\in N(Q)} b_{ij} x^i y^j$, the condition $[P, Q] = x^2$ expands to bilinear equations in the coefficients. Solving the full 72-unknown bilinear system directly is impractical; we use the exact layer reduction of §4. (Direct expansion remains feasible for *verification* of the reduction, as in Proposition 4.1.)

4 Exact layer reduction

Set $u = xy^2$ and decompose into y-layers:

$$P = \sum_k y^{-k} A_k(u), \qquad Q = \sum_\ell y^{-\ell} B_\ell(u), \tag{3}$$

where $A_k, B_\ell \in \mathbb{C}[u]$ are one-variable polynomials whose coefficients are linear forms in the a_{ij}, b_{ij}. The bracket $[P, Q]$ becomes a sum over y-layers, each a one-variable identity in u.

Proposition 4.1 (Exact layer reduction)**.** *The system $[P, Q] = x^2$ on the 72 coefficients implies five one-variable polynomial identities $E_1, \dots, E_5$ (the converse is not needed):*

Layer	***Equations***	***Unknowns (given previous layers)***
E_5	17	*Top:* α, β
E_4	18	A_1, B_2 *given* A_2, B_3
E_3	19	A_0, B_1 *given* A_1, B_2
E_2	19	B_0 *given* A_0, B_1
E_1	19	*Lowest coefficients*
Total	92	

The layer reduction was verified as a polynomial identity (script ***C1_layer_reduction.sing****) with all 72 coefficients treated as symbolic indeterminates; the residual is identically zero.*

Computational verification. Substituting (3) into $[P, Q]-x^2$ and collecting by powers of y yields the five identities $E_1, \dots, E_5$. Expanding each E_i in the 72 indeterminates and comparing coefficients with the direct bilinear expansion of $[P, Q] - x^2$ gives zero difference as a polynomial in 72 variables. Script: `C1_layer_reduction.sing` (SINGULAR). □

Listing 1: Mechanically extracted from ***Jacobian/BranchAbLayers.lean*** *(sha256* ***7c16141b4b126b3e****); statement verbatim, proof elided.*

```
theorem layers_of_jac (lam : K) (A₀ A₁ A₂ B₀ B₁ B₂ B₃ : Polynomial K)
    (p₀ p₁ p₂ q₀ q₁ q₂ q₃ : MvPolynomial (Fin 2) K)
    (hp₀ : X 1 ^ 0 * p₀ = evH A₀) (hp₁ : X 1 ^ 1 * p₁ = evH A₁) (hp₂ : X 1 ^ 2 * p₂ =
     ↪  evH A₂)
```

```
    (hq₀ : X 1 ^ 0 * q₀ = evH B₀) (hq₁ : X 1 ^ 1 * q₁ = evH B₁) (hq₂ : X 1 ^ 2 * q₂ =
     ↪  evH B₂)
    (hq₃ : X 1 ^ 3 * q₃ = evH B₃)
    (hJ : jac (p₀ + p₁ + p₂) (q₀ + q₁ + q₂ + q₃) = C lam * X 0 ^ 2) :
    layerTerm 2 3 A₂ B₃ = Polynomial.C lam * Polynomial.X ^ 2 /\
    layerTerm 2 2 A₂ B₂ + layerTerm 1 3 A₁ B₃ = 0 /\
    layerTerm 2 1 A₂ B₁ + layerTerm 1 2 A₁ B₂ + layerTerm 0 3 A₀ B₃ = 0 /\
    layerTerm 2 0 A₂ B₀ + layerTerm 1 1 A₁ B₁ + layerTerm 0 2 A₀ B₂ = 0 /\
    layerTerm 1 0 A₁ B₀ + layerTerm 0 1 A₀ B₁ = 0 := by
  -- proof body elided (see artifact)
```

The basic definitions used above are printed verbatim here, since only a human reader (not the Lean kernel) checks definitions:

*Listing 2: Mechanically extracted from **Jacobian/BranchAbLayers.lean** (sha256 **7c16141b4b126b3e**); definition printed verbatim in full.*

```
def jac (P Q : MvPolynomial (Fin 2) K) : MvPolynomial (Fin 2) K :=
  pderiv 0 P * pderiv 1 Q - pderiv 1 P * pderiv 0 Q
```

*Listing 3: Mechanically extracted from **Jacobian/BranchAbLayers.lean** (sha256 **7c16141b4b126b3e**); definition printed verbatim in full.*

```
def uu : MvPolynomial (Fin 2) K := X 0 * X 1 ^ 2
```

*Listing 4: Mechanically extracted from **Jacobian/BranchAbLayers.lean** (sha256 **7c16141b4b126b3e**); definition printed verbatim in full.*

```
def evH : Polynomial K →ₐ[K] MvPolynomial (Fin 2) K := Polynomial.aeval (uu (K := K))
```

*Listing 5: Mechanically extracted from **Jacobian/BranchAbLayers.lean** (sha256 **7c16141b4b126b3e**); definition printed verbatim in full.*

```
def layerTerm (a b : Nat) (A B : Polynomial K) : Polynomial K :=
  (a : Polynomial K) * A * Polynomial.derivative B - (b : Polynomial K) *
   ↪  Polynomial.derivative A * B
```

The system is triangular (Figure 2): E_5 involves only the top-layer unknowns, and each E_i ($i < 5$) is affine-linear in its unknowns given a solution of the layers above. This is the key structural fact that makes the problem tractable. Conceptually, the Newton polygons induce a weighted grading, and the Jacobian equation respects that grading: the five E_i are the homogeneous components of $[P, Q] = x^2$ with respect to it. The triangular structure is thus not a computational accident but a structural consequence of the grading, and the operator $\mathrm{LT}_{a,b}(A, B) = aAB' - bA'B$ is its homogeneous bracket.

5 Normalization

The top-layer unknowns admit rescaling symmetries from the torus action. We fix, without loss of generality,

$$a_{1,0} = a_{8,14} = 1, \tag{4}$$

leaving $a_{8,16}$ free. The remaining 1-dimensional torus acts by the Y-weight grading used in the layer decomposition (§4); the finite subgroup μ_7 gives 7 chart copies per orbit.

Remark 5.1 (Lean chart normalization). The Lean formalization uses chart coordinates $(a_1, \dots, a_7, b_0, \dots, b_9)$ with the torus normalization $a_7^3 b_0^2 = 1$ (see `ChartClassification` in `Jacobian/BranchAbChart.lean`). After fixing $\alpha_0 = 1$ and $\beta_{10} = 1$, the remaining scaling $u \mapsto \kappa u$ acts on a_7 with weight 7 and on b_0 with weight −10, hence on $a_7^3 b_0^2$ with weight $21 - 20 = 1$. The condition $a_7^3 b_0^2 = 1$ therefore fixes κ completely, with no 7th-root ambiguity: the Lean chart meets each torus orbit *exactly once*, giving 5 points, not 35. The §5 chart ($a_{1,0} = a_{8,14} = 1$) meets each orbit 7 times. The two charts are related by a torus element requiring a 7th root.

Vertex conditions used. The proof uses $(1, 0)$ and $(2, 1)$ (equivalently $\lambda \neq 0$), $(8, 14)$ (normalization and top-layer degree), and $(12, 24)$ (the $t = 0$ exclusion). The vertex $(8, 16)$ is left free in `main_theorem`. Once E_1 is included, $a_{8,16} = 0$ is forced (Corollary 1.2). This gives a second contradiction, from the lower edge: `main_theorem_lower_edge` (listing 22) refutes NewtonNF2 without using the vertex $(12, 24)$. It shares the descent up to $t = 0$ with `main_theorem`. The explicit certificate over K_5 (`scripts/a816_certificate/`, checked exactly by SINGULAR and by python-flint) reaches $a_{8,16} = 0$ without the descent, outside Lean.

Listing 6: Mechanically extracted from ***`Jacobian/BranchAbTorus.lean`*** *(sha256* ***`97918673d5c4dc09`****); statement verbatim, proof elided.*

```
theorem layers_transport (α β κ : K) (A₀ A₁ A₂ B₀ B₁ B₂ B₃ : K[X])
    (E4 : layerTerm 2 2 A₂ B₂ + layerTerm 1 3 A₁ B₃ = 0)
    (E3 : layerTerm 2 1 A₂ B₁ + layerTerm 1 2 A₁ B₂ + layerTerm 0 3 A₀ B₃ = 0)
    (E2 : layerTerm 2 0 A₂ B₀ + layerTerm 1 1 A₁ B₁ + layerTerm 0 2 A₀ B₂ = 0) :
    layerTerm 2 2 (C α * sc κ A₂) (C β * sc κ B₂) + layerTerm 1 3 (C α * sc κ A₁) (C
     ↪  β * sc κ B₃) = 0 /\
    layerTerm 2 1 (C α * sc κ A₂) (C β * sc κ B₁) + layerTerm 1 2 (C α * sc κ A₁) (C
     ↪  β * sc κ B₂) +
      layerTerm 0 3 (C α * sc κ A₀) (C β * sc κ B₃) = 0 /\
    layerTerm 2 0 (C α * sc κ A₂) (C β * sc κ B₀) + layerTerm 1 1 (C α * sc κ A₁) (C
     ↪  β * sc κ B₁) +
      layerTerm 0 2 (C α * sc κ A₀) (C β * sc κ B₂) = 0 := by
  -- proof body elided (see artifact)
```

6 The top layer

6.1 The Davenport–Stothers equation

The top identity E_5 takes the form

$$\alpha\beta + u\,(2\alpha\beta' - 3\alpha'\beta) = 1, \tag{5}$$

where $\alpha = A_2/u$ and $\beta = B_3/u^2$ are polynomials of degree 7 and 10 respectively, the degrees being forced by the vertices $(8, 14) \in N(P)$ and $(12, 21) \in N(Q)$.

Setting $u = s^2$ with $f(s) = \alpha(s^2)$ and $g(s) = s\beta(s^2)$, equation (5) becomes the Davenport–Stothers extremal equation [2, 3]

$$2fg' - 3f'g = c \neq 0. \tag{6}$$

Indeed $2fg' - 3f'g = 2(\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta))$. With $\deg f = 14 = 2k$ and $\deg g = 21 = 3k$ ($k = 7$), we have $\deg(g^2 - c \cdot f^3) = 8 = k + 1$, which is exactly Davenport's bound: the pair is extremal. with f even and g odd. Here $\deg_s f = 2m$ where $m = \deg_u \alpha$; the relation $1 + 2n - 3m = 0$ (with $n = \deg_u \beta$) forces m odd. The cases $m = 1, 3, 5, 7$ admit solutions; $m = 7$ is the relevant case for (1). The symmetric (parity-constrained) classification is computed here via elimination; we do not claim it follows from the literature.

6.2 Classification

Proposition 6.1 (Top-layer solutions)**.** *The system E_5 has:*

- $m = 3$*: 3 complex solutions; eliminant* $3a_2^3 - 32$*, irreducible over* $\mathbb{Q}$*;*
- $m = 5$*: 10 complex solutions; eliminant* $9a_4^{10} + 37200a_4^5 + 95051008$*, irreducible over* $\mathbb{Q}$*;*
- $m = 7$*: 35 complex solutions; eliminant* $W(a_6)$ *of degree* 35*.*

The $m = 7$ case is the one relevant to (1)*.*

Proof. We work over $\mathbb{C}$: the Belyi classification is topological, hence valid over $\mathbb{C}$, and the resulting count of solutions transfers to any field of characteristic 0 by base change (the equations are defined over $\mathbb{Q}$). We normalize the third branch point to $c = 1$ by the scaling $u \mapsto c\,u$, which preserves the torus action. We prove the counts in characteristic 0 via the Belyi-map classification. For the top layer, E_5 is $\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta) = 1$ with $\deg\alpha = m$, $\deg\beta = (3m - 1)/2$ ($m = 3, 5, 7$, so $\deg\beta = 4, 7, 10$). Put $\varphi = u\beta^2/\alpha^3$. Then $\varphi' = \beta E/\alpha^4 = \beta/\alpha^4$, so φ is a Belyi map: it ramifies only over $0, \infty$, and $c = \varphi(\infty) \neq 0$. The ramification is forced as follows. At a root r of α, $E(r) = -3r \cdot \alpha'(r)\beta(r) = 1$. So α has simple nonzero roots, none shared with β. At a root s of β, $E(s) = 2s \cdot \alpha(s)\beta'(s) = 1$, so β also has simple nonzero roots. Hence $\deg\varphi = 3m$, and $\varphi' = \beta/\alpha^4$ vanishes in $\mathbb{C}$ only at the roots of β, which forces the passport

$m = 3$	$[2^4 1]$ $(n = 9)$	$[3^3]$	$[7 \cdot 1^2]$
$m = 5$	$[2^7 1]$ $(n = 15)$	$[3^5]$	$[12 \cdot 1^3]$
$m = 7$	$[2^{10} 1]$ $(n = 21)$	$[3^7]$	$[17 \cdot 1^4]$

By Riemann's existence theorem, covers with this passport correspond to S_n-conjugacy classes of transitive triples in $C_0 \times C_1 \times C_\infty$ with product 1 (here $n = 3m$). The Frobenius character formula (computed exactly in `belyi_counts_m357.py` via the Murnaghan–Nakayama rule) counts N triples with $N = 9!, 2 \cdot 15!, 5 \cdot 21!$ for $m = 3, 5, 7$; since every such triple has trivial centralizer, each S_n-conjugacy class has $n!$ elements, giving $N/n! = 1, 2, 5$ covers. Every triple counted is transitive, for general m: a non-transitive triple would have an orbit on which g_∞ acts trivially, so on that orbit $g_0 = g_1^{-1}$; but g_1 has cycle type 3^m (hence fixes no point, and so does g_1^{-1}), while g_0 has cycle type $2^{(3m-1)/2}1$ and contains no 3-cycle—a contradiction. Hence all counted triples are genuine covers. Each cover has trivial automorphism group: an

automorphism $u \mapsto \kappa u$ acts freely on the m poles and on the $(m+1)/2$ unramified points over c, so its order divides $\gcd(m,(m+1)/2) = 1$.

Now suppose two solutions give isomorphic covers. The isomorphism fixes the unique unramified point over 0 ($u = 0$) and the unique maximally ramified point over c ($u = \infty$). So it is $u \mapsto \kappa u$, and comparing zeros and poles shows the two solutions differ by the torus action. Hence distinct orbits give non-isomorphic covers, and there are at most $m \cdot (\#\text{covers})$ chart points: 3, 10, 35.

For $m = 7$, the exactly verified parametrization (script `verify_msolve_param.py`, shipped with inputs `e5_m7.ms` and `e5_m7.out` in `scripts/`) exhibits 35 distinct points, so the classification is complete in characteristic 0. For $m = 3$ and $m = 5$ the Belyi bound is likewise sharp. In the normalization $\alpha_0 = \beta_0 = \alpha_m = 1$ the system is zero-dimensional, of vector-space dimension 3 for $m = 3$ and 10 for $m = 5$; hence the solution set is nonempty and has at most d points, where $d = 3$ resp. 10, by the dimension alone (independent of the Belyi chart). Eliminating all variables but a_{m-1} gives, up to a nonzero rational factor, the stated eliminant. Each eliminant is irreducible and separable over $\mathbb{Q}$, so the Galois group acts transitively on its d distinct roots; since the system is defined over $\mathbb{Q}$, the solution set is Galois-stable, and every root occurs as the a_{m-1}-coordinate of a solution. Thus there are at least d distinct solutions, and with the upper bound of d, exactly d. All of this is checked by script `e5_exact_counts_m35.py` in `scripts/`. Both classifications are also kernel-checked in Lean (`lean/Jacobian/B26.lean`, over any field of characteristic 0): in this chart the system holds if and only if $T(a_{m-1}) = 0$ and the other unknowns are explicit functions of a_{m-1} (`m3_chart_iff`: $8a_1 = 5a_2^2$; `m5_chart_iff`: $362a_4a_1 = 3a_4^5 + 13078$, $181a_4^2a_2 = 15a_4^5 + 32448$, $2172a_4^3a_3 = 861a_4^5 + 492128$; in both cases β is an explicit polynomial in $a_1, \dots, a_{m-1}$), and T has no repeated root (`m3_T_squarefree`, `m5_T_squarefree`). This gives exactly d solutions over $\overline{\mathbb{Q}}$ without the Galois argument. The count is kernel-checked as well, over every algebraically closed field of characteristic 0 (`m3_chart_card`, `m5_chart_card` in `lean/Jacobian/B26Count.lean`).

The $m = 3$ eliminant $3a_2^3 - 32$ and $m = 5$ eliminant $9a_4^{10} + 37200a_4^5 + 95051008$ are verified irreducible over $\mathbb{Q}$ by factorization (same script). Their irreducibility is also kernel-checked in Lean (`T3poly_irreducible`, `T5poly_irreducible` in `lean/Jacobian/B26Irred.lean`). A rational root of $3a_2^3 - 32$ would satisfy $a_2^3 = 32/3$, but the 3-adic valuation -1 of $32/3$ is not a multiple of 3; a cubic without a rational root is irreducible. For $m = 5$ write the eliminant as $F(a_4^5)$ with $F(u) = 9u^2 + 37200u + 95051008 = (3u + 6200)^2 + 56611008$. This quadratic has no real root, so it is irreducible over $\mathbb{Q}$. Let u be a root of F. The norm of u from $\mathbb{Q}(u)$ to $\mathbb{Q}$ is $95051008/9 = 2^8 \cdot 13^5/3^2$, with 3-adic valuation -2. The norm of a fifth power is a fifth power, and -2 is not a multiple of 5, so u is not a fifth power in $\mathbb{Q}(u)$. Hence $X^5 - u$ is irreducible over $\mathbb{Q}(u)$, by the criterion for $X^p - a$ with p prime. Let $\theta^5 = u$. Then $\mathbb{Q}(\theta)$ contains $u = \theta^5$, so it equals $\mathbb{Q}(u)(\theta)$ and has degree $2 \cdot 5 = 10$ over $\mathbb{Q}$. Since θ is a root of the eliminant, which has degree 10, the eliminant is irreducible. For $m = 7$, the eliminating polynomial in a_6 has degree 35 (Gröbner elimination via SINGULAR `eliminate`). □

Remark 6.2 (Status of `ChartClassification`). `ChartClassification` (in `Jacobian/BranchAbChart.lean`) encodes the 17 equations of the E_5 chart system: the 16 bilinear equations plus the torus normalization $a_7^3 b_0^2 = 1$. Up to v16 of the artifact it was an unproven hypothesis standing for the case $m = 7$ of Proposition 6.1. It is now a theorem, `chartClassification_holds` (§6.4), so the implication `ChartClassification` $L \to \neg\exists \dots$ of `main_theorem_of_chart` is discharged and cannot be vacuous. The statement itself is unchanged. Its 17 equations were verified against the `chart_system_Q.sing` Singular script and by the mechanical chart audit (see the computational archive). The generator of the proof

in §6.4 reads this same statement from the Lean source.

Listing 7: Mechanically extracted from ***Jacobian/BranchAbChart.lean*** *(sha256* ***15907ffeffb6d274****); statement verbatim, proof elided.*

```
theorem main_theorem_of_chart {L : Type*} [Field L] [CharZero L] (hc :
↪  ChartClassification L) :
    ¬ exists (P Q : MvPolynomial (Fin 2) L) (lam : L), lam ≠ 0 /\ NewtonNF2 P Q /\
     ↪  jac P Q = MvPolynomial.C lam * MvPolynomial.X 0 ^ 2 :=
  -- proof term elided (see artifact)
```

The two definitions on which `main_theorem_of_chart` rests are printed verbatim in full:

Listing 8: Mechanically extracted from ***Jacobian/BranchAbNewton.lean*** *(sha256* ***fda6fce4c16abbe8****); definition printed verbatim in full.*

```
abbrev mono (i j : Nat) : Fin 2 →₀ Nat := Finsupp.single 0 i + Finsupp.single 1 j

@[simp] lemma mono_apply_zero (i j : Nat) : mono i j 0 = i := by simp [mono]
@[simp] lemma mono_apply_one (i j : Nat) : mono i j 1 = j := by simp [mono]
```

Listing 9: Mechanically extracted from ***Jacobian/BranchAbNewton.lean*** *(sha256* ***fda6fce4c16abbe8****); definition printed verbatim in full.*

```
def inNP (m : Fin 2 →₀ Nat) : Prop := m 0 ≤ 8 /\ m 1 ≤ 2 * m 0 /\ 2 * m 0 ≤ m 1 + 2
```

Listing 10: Mechanically extracted from ***Jacobian/BranchAbNewton.lean*** *(sha256* ***fda6fce4c16abbe8****); definition printed verbatim in full.*

```
def inNQ (m : Fin 2 →₀ Nat) : Prop := m 0 ≤ 12 /\ m 1 ≤ 2 * m 0 /\ 2 * m 0 ≤ m 1 + 3
↪  /\ m 0 ≤ 2 * m 1
```

Listing 11: Mechanically extracted from ***Jacobian/BranchAbNewton.lean*** *(sha256* ***fda6fce4c16abbe8****); definition printed verbatim in full.*

```
def NewtonNF2 (P Q : MvPolynomial (Fin 2) K) : Prop :=
  (forall m ∈ P.support, inNP m) /\ (forall m ∈ Q.support, inNQ m) /\
  P.coeff (mono 0 0) ≠ 0 /\ P.coeff (mono 1 0) ≠ 0 /\ P.coeff (mono 8 14) ≠ 0 /\
   ↪  P.coeff (mono 8 16) ≠ 0 /\
  Q.coeff (mono 0 0) ≠ 0 /\ Q.coeff (mono 2 1) ≠ 0 /\ Q.coeff (mono 12 21) ≠ 0 /\
   ↪  Q.coeff (mono 12 24) ≠ 0
```

Listing 12: Mechanically extracted from ***Jacobian/BranchAbChart.lean*** *(sha256* ***15907ffeffb6d274****); definition printed verbatim in full.*

```
def ChartClassification (L : Type*) [Field L] : Prop :=
  forall a1 a2 a3 a4 a5 a6 a7 b0 b1 b2 b3 b4 b5 b6 b7 b8 b9 : L,
    (3 : L) * b1 + (-2 : L) * a1 * b0 = 0 →
    (5 : L) * b2 + (-5 : L) * a2 * b0 = 0 →
    (7 : L) * b3 + (2 : L) * a1 * b2 + (-3 : L) * a2 * b1 + (-8 : L) * a3 * b0 = 0 →
    (9 : L) * b4 + (4 : L) * a1 * b3 + (-1 : L) * a2 * b2 + (-6 : L) * a3 * b1 + (-11
     ↪  : L) * a4 * b0 = 0 →
    (11 : L) * b5 + (6 : L) * a1 * b4 + (1 : L) * a2 * b3 + (-4 : L) * a3 * b2 + (-9
     ↪  : L) * a4 * b1 + (-14 : L) * a5 * b0 = 0 →
    (13 : L) * b6 + (8 : L) * a1 * b5 + (3 : L) * a2 * b4 + (-2 : L) * a3 * b3 + (-7
     ↪  : L) * a4 * b2 + (-12 : L) * a5 * b1 + (-17 : L) * a6 * b0 = 0 →
    (15 : L) * b7 + (10 : L) * a1 * b6 + (5 : L) * a2 * b5 + (-5 : L) * a4 * b3 +
     ↪  (-10 : L) * a5 * b2 + (-15 : L) * a6 * b1 + (-20 : L) * a7 * b0 = 0 →
    (17 : L) * b8 + (12 : L) * a1 * b7 + (7 : L) * a2 * b6 + (2 : L) * a3 * b5 + (-3
     ↪  : L) * a4 * b4 + (-8 : L) * a5 * b3 + (-13 : L) * a6 * b2 + (-18 : L) * a7 *
     ↪  b1 = 0 →
    (19 : L) * b9 + (14 : L) * a1 * b8 + (9 : L) * a2 * b7 + (4 : L) * a3 * b6 + (-1
     ↪  : L) * a4 * b5 + (-6 : L) * a5 * b4 + (-11 : L) * a6 * b3 + (-16 : L) * a7 *
     ↪  b2 = 0 →
    (21 : L) + (16 : L) * a1 * b9 + (11 : L) * a2 * b8 + (6 : L) * a3 * b7 + (1 : L) *
     ↪  a4 * b6 + (-4 : L) * a5 * b5 + (-9 : L) * a6 * b4 + (-14 : L) * a7 * b3 = 0 →
    (18 : L) * a1 + (13 : L) * a2 * b9 + (8 : L) * a3 * b8 + (3 : L) * a4 * b7 + (-2
     ↪  : L) * a5 * b6 + (-7 : L) * a6 * b5 + (-12 : L) * a7 * b4 = 0 →
    (15 : L) * a2 + (10 : L) * a3 * b9 + (5 : L) * a4 * b8 + (-5 : L) * a6 * b6 +
     ↪  (-10 : L) * a7 * b5 = 0 →
    (12 : L) * a3 + (7 : L) * a4 * b9 + (2 : L) * a5 * b8 + (-3 : L) * a6 * b7 + (-8
     ↪  : L) * a7 * b6 = 0 →
    (9 : L) * a4 + (4 : L) * a5 * b9 + (-1 : L) * a6 * b8 + (-6 : L) * a7 * b7 = 0 →
    (6 : L) * a5 + (1 : L) * a6 * b9 + (-4 : L) * a7 * b8 = 0 →
    (3 : L) * a6 + (-2 : L) * a7 * b9 = 0 →
    a7 ^ 3 * b0 ^ 2 = 1 →
    exists w : L, w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0 /\
      a1 = (((512502980) / 1428159681 : L) + ((8080) / 109858437 : L) * w +
       ↪  ((877688) / 1428159681 : L) * w ^ (2 : Nat) + ((127760) / 84009393 : L) *
       ↪  w ^ (3 : Nat) + ((-531068) / 1428159681 : L) * w ^ (4 : Nat)) /\
      a2 = (((7293893720809472) / 226626674937113529 : L) + ((285855137792) /
       ↪  5810940383002911 : L) * w + ((-47390789046272) / 226626674937113529 : L) *
       ↪  w ^ (2 : Nat) + ((-90487897751552) / 226626674937113529 : L) * w ^ (3 :
       ↪  Nat) + ((21860765739008) / 226626674937113529 : L) * w ^ (4 : Nat)) /\
      a3 = (((6464293233975242927636480) / 32365907978427875263742424 9 : L) +
       ↪  ((-311599571772213100544) / 24896852291098365587494173 : L) * w +
       ↪  ((-13107803800410363756544) / 32365907978427875263742424 9 : L) * w ^ (2 :
       ↪  Nat) + ((-39360612513016248860672) / 32365907978427875263742424 9 : L) * w
       ↪  ^ (3 : Nat) + ((9886191189342570790912) / 32365907978427875263742424 9 : L)
       ↪  * w ^ (4 : Nat)) /\
      a4 = (((401489080334849991328866 1860352) /
       ↪  154078949379156364060580574704501523 : L) +
       ↪  ((-19247583898558620695724556288) / 39507422917732401041174506334487 57 :
       ↪  L) * w + ((-14781345296916798503187 4871296) /
       ↪  154078949379156364060580574704501523 : L) * w ^ (2 : Nat) +
       ↪  ((-21329313178098532487186612224 00) /
       ↪  154078949379156364060580574704501523 : L) * w ^ (3 : Nat) +
       ↪  ((58039857816239632626286055178240) / 154078949379156364060580574704501523
       ↪  : L) * w ^ (4 : Nat)) /\
      a5 = (((-47195336258536808324667608668861104128) /
       ↪  7334978106471703364862620608159252145056 4721 : L) +
       ↪  ((-29932787320250950904651205753668894 72) /
       ↪  5642290851132079511432785083199424726966517 : L) * w +
       ↪  ((32500098714604785191372567805805199 36) /
       ↪  8149975673857448183180689564621391272284969 : L) * w ^ (2 : Nat) +
       ↪  ((-35972900451489262851635484544208994304) /
       ↪  7334978106471703364862620608159252145056 4721 : L) * w ^ (3 : Nat) +
       ↪  ((42124958173030895116478408190549032 96) /
       ↪  2444992702157234454954206869386417381685 4907 : L) * w ^ (4 : Nat)) /\
```

```
a6 = (((-353637771581420436331059704199622022880296 96) /
 ↪  241742769061860274917434465920909466323440390492032 31 : L) +
 ↪  ((-244358877861941369482776189370917360542679 04) /
 ↪  185955976201430980705718819939161127941107992686 1787 : L) * w +
 ↪  ((348725961520751776828947066063346243175710720) /
 ↪  241742769061860274917434465920909466323440390492032 31 : L) * w ^ (2 : Nat)
 ↪  + ((-891434267787294593043481518636230483797606 4) /
 ↪  142201628859917808774961450541711450778494347348 2543 : L) * w ^ (3 : Nat)
 ↪  + ((677989855609902867411530105006060279347281 92) /
 ↪  241742769061860274917434465920909466323440390492032 31 : L) * w ^ (4 :
 ↪  Nat)) /\
a7 = (((342097029127255765817099097920847945090722114306048) /
 ↪  885249425506264203827321815866695986293243258737477684336649 : L) +
 ↪  ((-568105419722006463099987956174489022887517749248 0) /
 ↪  226987032181093385596749183555563073408523912496789 14982991 : L) * w +
 ↪  ((206885581446139393490715159158183040429422553333760) /
 ↪  885249425506264203827321815866695986293243258737477684336649 : L) * w ^ (2
 ↪  : Nat) + ((-633482589598705098139540581219264850266194790 31808) /
 ↪  885249425506264203827321815866695986293243258737477684336649 : L) * w ^ (3
 ↪  : Nat) + ((253322561059155215789084089746168458666056079 11424) /
 ↪  885249425506264203827321815866695986293243258737477684336649 : L) * w ^ (4
 ↪  : Nat)) /\
b0 = (((241787651899027059649022111227674440235744649167926923464101195333242
    8638485407) /
 ↪  811441542599883814485270953564641654639931576344287204467623854 08 : L) +
 ↪  ((147114702866918150301106182425486691319761000221649016655576552446507 3
 ↪  2041478549) /
 ↪  811441542599883814485270953564641654639931576344287204467623854 08 : L) *
 ↪  w + ((617854596280970291332161410119563350111260243577025705635188990137 8
 ↪  76816776295) /
 ↪  811441542599883814485270953564641654639931576344287204467623854 08 : L) *
 ↪  w ^ (2 : Nat) + ((-366378570719933994254445864210576333868526055032758879
 ↪  4221990329866702931989049) /
 ↪  811441542599883814485270953564641654639931576344287204467623854 08 : L) *
 ↪  w ^ (3 : Nat) + ((283919616043133221256996475572763279716168016664606915 3
 ↪  9514166693155875345 3503) /
 ↪  405720771299941907242635476782320827319965788172143602233811927 04 : L) *
 ↪  w ^ (4 : Nat)) /\
b1 = (((213129345216263992177813311926195028654591343373099779726908014927974
    58781) / 206452661968217945879623181753674347303056069698831468671795 2 :
 ↪  L) + ((907860591902046261556432976412843546973938642324006831011933983788
 ↪  54556479) / 206452661968217945879623181753674347303056069698831468671795 2
 ↪  : L) * w + ((367683922090893100561414709129515035875503403273788620768039
 ↪  4068928684725) /
 ↪  206452661968217945879623181753674347303056069698831468671795 2 : L) * w ^
 ↪  (2 : Nat) + ((-210875061122372142827023757868450079885049551692627677710 7
 ↪  9750788915150635) /
 ↪  206452661968217945879623181753674347303056069698831468671795 2 : L) * w ^
 ↪  (3 : Nat) + ((180447192994234227283436464239178979183551309852872229435 04
 ↪  00675635340349) /
 ↪  103226330984108972939811590876837173651528034849415734335897 6 : L) * w ^
 ↪  (4 : Nat)) /\
```

```
b2 = (((-144544033917986925185258501166756369390602635518529573940467345287) /
 ↪  41036826318611406019350603181624792344421065146603929 6 : L) +
 ↪  ((575311520507585192568556318243232964279467466281194477346330819301) /
 ↪  10259206579652851504837650795406198086105266286650982 4 : L) * w +
 ↪  ((54330691268897368779422216815165957896195264760328594263494900495) /
 ↪  20518413159305703009675301590812396172210532573301964 8 : L) * w ^ (2 :
 ↪  Nat) +
 ↪  ((-17418788001837933630619826390105924004718047264922620305812814089 5) /
 ↪  10259206579652851504837650795406198086105266286650982 4 : L) * w ^ (3 :
 ↪  Nat) +
 ↪  ((80166557140337989909132856542161474521909286203155803758626110033) /
 ↪  41036826318611406019350603181624792344421065146603929 6 : L) * w ^ (4 :
 ↪  Nat)) /\
b3 = (((-74254584050343031010100327956018662424742104029996359431 31) /
 ↪  20392339559214813347634883685912610051826122752 : L) +
 ↪  ((67443155429140090353909812378418781701102502436228141280 87) /
 ↪  20392339559214813347634883685912610051826122752 : L) * w +
 ↪  ((47844074923574423613487288100643626182739349341609881785 3) /
 ↪  20392339559214813347634883685912610051826122752 : L) * w ^ (2 : Nat) +
 ↪  ((-37003441214903100177186361694540920084341786584254332271 71) /
 ↪  20392339559214813347634883685912610051826122752 : L) * w ^ (3 : Nat) +
 ↪  ((62233097938932572222702608499502859806885134430639402117) /
 ↪  10196169779607406673817441842956305025913061376 : L) * w ^ (4 : Nat)) /\
b4 = (((-19698212433453796683487798065412975573876758031724 7) /
 ↪  40534081214727694198655996538823597096 96 : L) +
 ↪  ((73561629390378434641754255497164844993622904322 53) /
 ↪  10133520303681923549663999134705899274 24 : L) * w +
 ↪  ((33342147832168882830933893468612523198171552858 63) /
 ↪  20267040607363847099327998269411798548 48 : L) * w ^ (2 : Nat) +
 ↪  ((-13592771536859666166850556543437368777444921613959) /
 ↪  10133520303681923549663999134705899274 24 : L) * w ^ (3 : Nat) +
 ↪  ((-17431675224264311010531744232209775637115617032 71) /
 ↪  40534081214727694198655996538823597096 96 : L) * w ^ (4 : Nat)) /\
b5 = (((-59453262799114593004279671862282369546808 73) /
 ↪  16114009235195405603857112403804 16 : L) +
 ↪  ((-12074147960445986377782466413655465170024 99) /
 ↪  16114009235195405603857112403804 16 : L) * w +
 ↪  ((14739790683222069070539748868284852592790 3) /
 ↪  16114009235195405603857112403804 16 : L) * w ^ (2 : Nat) +
 ↪  ((-10030464739554601758391959380915946133063 05) /
 ↪  16114009235195405603857112403804 16 : L) * w ^ (3 : Nat) +
 ↪  ((-57989457976321180426899430535315371961481) /
 ↪  80570046175977028019285562019020 8 : L) * w ^ (4 : Nat)) /\
b6 = (((-40717564079347560855720292779927573) / 32029996221749467301072076 8 :
 ↪  L) + ((-44319563411879624998266751612718 25) / 80074990554373668252680192 :
 ↪  L) * w + ((75574733796761682344936632071 57) /
 ↪  16014998110874733650536038 4 : L) * w ^ (2 : Nat) +
 ↪  ((-86947127489203867647410751459119 7) / 80074990554373668252680192 : L) *
 ↪  w ^ (3 : Nat) + ((-93883857777777701654732412834599 7) /
 ↪  32029996221749467301072076 8 : L) * w ^ (4 : Nat)) /\
b7 = (((-42258994952000509880862795) / 31833190703543877632 : L) +
 ↪  ((-36376188014455196253194745) / 31833190703543877632 : L) * w +
 ↪  ((55273196192234647649820 93) / 31833190703543877632 : L) * w ^ (2 : Nat) +
 ↪  ((36224632464902536997290 05) / 31833190703543877632 : L) * w ^ (3 : Nat) +
 ↪  ((-71703963487061132872164 3) / 15916595351771938816 : L) * w ^ (4 : Nat))
 ↪  /\
b8 = (((89529406222349457) / 6327518887936 : L) + ((-9772922162070171) /
 ↪  1581879721984 : L) * w + ((8308630462897287) / 3163759443968 : L) * w ^ (2
 ↪  : Nat) + ((8039403284956497) / 1581879721984 : L) * w ^ (3 : Nat) +
 ↪  ((-2811959945010567) / 6327518887936 : L) * w ^ (4 : Nat)) /\
b9 = (((1461767229) / 5030912 : L) + ((82735263) / 5030912 : L) * w +
 ↪  ((34254981) / 5030912 : L) * w ^ (2 : Nat) + ((182339829) / 5030912 : L) *
 ↪  w ^ (3 : Nat) + ((-10377315) / 2515456 : L) * w ^ (4 : Nat))
```

The top-layer classification statement and its witnesses:

*Listing 13: Mechanically extracted from **`Jacobian/BranchAbClassification.lean`** (sha256 **`29fdecd5a9fda336`**); definition printed verbatim in full.*

```
def TopLayerClassification (L : Type*) [Field L] : Prop :=
  forall (A₂ B₃ : L[X]) (lam : L),
    (forall i, A₂.coeff i ≠ 0 → 1 ≤ i /\ i ≤ 8) → (forall i, B₃.coeff i ≠ 0 → 2 ≤ i
     ↪  /\ i ≤ 12) →
    A₂.coeff 1 ≠ 0 → A₂.coeff 8 ≠ 0 → B₃.coeff 2 ≠ 0 → B₃.coeff 12 ≠ 0 →
    layerTerm 2 3 A₂ B₃ = C lam * X ^ 2 →
    exists w ρ σ ε : L, w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0 /\ ρ ≠ 0 /\ σ ≠
     ↪  0 /\ ε ≠ 0 /\
      (forall i, A₂.coeff i = ρ * ε ^ (i - 1) * topA w i) /\ (forall k, B₃.coeff k =
       ↪  σ * ε ^ (k - 2) * topB w k)
```

*Listing 14: Mechanically extracted from **`Jacobian/BranchAbSharp.lean`** (sha256 **`9eb4eb229d2047f6`**); definition printed verbatim in full.*

```
noncomputable def topA {L : Type*} [Field L] (w : L) : Nat → L
  | 1 => ((1 : L))
  | 2 => (((303) / 16 : L) + ((879) / 128 : L) * w + ((-389) / 128 : L) * w ^ (2 :
   ↪  Nat) + ((-51) / 128 : L) * w ^ (3 : Nat) + ((49) / 128 : L) * w ^ (4 : Nat))
  | 3 => (((315) / 16 : L) + ((693) / 8 : L) * w + ((-147) / 8 : L) * w ^ (2 : Nat) +
   ↪  ((-189) / 8 : L) * w ^ (3 : Nat) + ((147) / 16 : L) * w ^ (4 : Nat))
  | 4 => (((-79653) / 64 : L) + ((29463) / 64 : L) * w + ((7371) / 128 : L) * w ^ (2 :
   ↪  Nat) + ((-4767) / 16 : L) * w ^ (3 : Nat) + ((8001) / 128 : L) * w ^ (4 : Nat))
  | 5 => (((-33519) / 4 : L) + ((-6843) / 4 : L) * w + ((8181) / 8 : L) * w ^ (2 :
   ↪  Nat) + ((-1713) / 2 : L) * w ^ (3 : Nat) + ((-1305) / 8 : L) * w ^ (4 : Nat))
  | 6 => (((-8805) / 8 : L) + ((-76965) / 8 : L) * w + ((11523) / 16 : L) * w ^ (2 :
   ↪  Nat) + ((6927) / 4 : L) * w ^ (3 : Nat) + ((-18579) / 16 : L) * w ^ (4 : Nat))
  | 7 => (((389740) / 17 : L) + ((-51922) / 17 : L) * w + ((-48572) / 17 : L) * w ^
   ↪  (2 : Nat) + ((63239) / 17 : L) * w ^ (3 : Nat) + ((-6475) / 17 : L) * w ^ (4 :
   ↪  Nat))
  | 8 => (((476437) / 68 : L) + ((314405) / 68 : L) * w + ((-174571) / 136 : L) * w ^
   ↪  (2 : Nat) + ((1574) / 17 : L) * w ^ (3 : Nat) + ((82783) / 136 : L) * w ^ (4 :
   ↪  Nat))
  | _ => 0
```

*Listing 15: Mechanically extracted from **`Jacobian/BranchAbSharp.lean`** (sha256 **`9eb4eb229d2047f6`**); definition printed verbatim in full.*

```
noncomputable def topB {L : Type*} [Field L] (w : L) : Nat → L
  | 2 => ((1 : L))
  | 3 => (((101) / 8 : L) + ((293) / 64 : L) * w + ((-389) / 192 : L) * w ^ (2 : Nat)
   ↪  + ((-17) / 64 : L) * w ^ (3 : Nat) + ((49) / 192 : L) * w ^ (4 : Nat))
  | 4 => (((315) / 16 : L) + ((693) / 8 : L) * w + ((-147) / 8 : L) * w ^ (2 : Nat) +
   ↪  ((-189) / 8 : L) * w ^ (3 : Nat) + ((147) / 16 : L) * w ^ (4 : Nat))
  | 5 => (((-11379) / 8 : L) + ((4209) / 8 : L) * w + ((1053) / 16 : L) * w ^ (2 :
   ↪  Nat) + ((-681) / 2 : L) * w ^ (3 : Nat) + ((1143) / 16 : L) * w ^ (4 : Nat))
  | 6 => (((-695299) / 48 : L) + ((-48367) / 48 : L) * w + ((52679) / 32 : L) * w ^
   ↪  (2 : Nat) + ((-45661) / 24 : L) * w ^ (3 : Nat) + ((-2211) / 32 : L) * w ^ (4 :
   ↪  Nat))
  | 7 => (((-309895) / 8 : L) + ((-251275) / 8 : L) * w + ((101481) / 16 : L) * w ^ (2
   ↪  : Nat) + ((1811) / 4 : L) * w ^ (3 : Nat) + ((-60489) / 16 : L) * w ^ (4 : Nat))
```

```
| 8 => (((1733348) / 13 : L) + ((-2368725) / 26 : L) * w + ((-184976) / 13 : L) * w
↪  ^ (2 : Nat) + ((1819939) / 52 : L) * w ^ (3 : Nat) + ((-569071) / 52 : L) * w ^
↪  (4 : Nat))
| 9 => (((18736247) / 39 : L) + ((1087133) / 13 : L) * w + ((-5656777) / 78 : L) *
↪  w ^ (2 : Nat) + ((1993384) / 39 : L) * w ^ (3 : Nat) + ((971365) / 78 : L) * w
↪  ^ (4 : Nat))
| 10 => (((305098) / 17 : L) + ((5784066) / 17 : L) * w + ((-308335) / 17 : L) * w
↪  ^ (2 : Nat) + ((-950608) / 17 : L) * w ^ (3 : Nat) + ((754021) / 17 : L) * w ^
↪  (4 : Nat))
| 11 => (((-149721182) / 323 : L) + ((26391690) / 323 : L) * w + ((21560033) / 323
↪  : L) * w ^ (2 : Nat) + ((-24378412) / 323 : L) * w ^ (3 : Nat) + ((2786935) /
↪  323 : L) * w ^ (4 : Nat))
| 12 => (((-91912444) / 969 : L) + ((-20715500) / 323 : L) * w + ((17474146) / 969
↪  : L) * w ^ (2 : Nat) + ((-1623428) / 969 : L) * w ^ (3 : Nat) + ((-8783302) /
↪  969 : L) * w ^ (4 : Nat))
| _ => 0
```

6.3 Irreducibility and reality

Theorem 6.3 (Top-layer structure). *Let $W \in \mathbb{Z}[a_6]$ be the degree-35 eliminating polynomial for a_6 at $m = 7$. Then:*

*(i) W is irreducible over $\mathbb{Q}$ (SINGULAR **`factorize`** over $\mathbb{Q}$ yields a nonzero constant times a single degree-35 factor);*
(ii) $W(a_6) = V(a_6^7)$ for a quintic $V \in \mathbb{Z}[S]$;
(iii) V is separable (nonzero discriminant) and has exactly one real root, $S \approx 3208.612478738048$;
(iv) W is separable modulo 101 ($\gcd(W, W') = 1$ in $\mathbb{F}_{101}[a_6]$), so the 35 complex solutions are distinct.

Consequently the 35 solutions form a single Galois orbit over $\mathbb{Q}$, comprising one real solution and 17 complex-conjugate pairs. See Figure 3. The map $a_6 \mapsto a_6^7$ sends the 35 roots of W onto the 5 roots of V, 7 to 1, and both sets are single Galois orbits.

Proof. (i) Direct factorization in SINGULAR over $\mathbb{Q}$. (ii) By inspection: every exponent of W is divisible by 7. (iii) Discriminant computed exactly in SYMPY (nonzero integer); real-root count via Sturm sequence gives exactly one real S. (iv) Direct gcd computation in $\mathbb{F}_{101}[a_6]$. □

Remark 6.4 (Characteristic-zero/modular consistency). Reducing the integer quintic V modulo 101 gives 58 times the quintic obtained directly from the mod-101 Gröbner computation (equivalently, the mod-101 quintic is 54 times V), confirming consistency between the characteristic-zero and modular routes.

6.4 An algebraic proof of the top-layer classification for $m = 7$, and its formalization

The proof of Proposition 6.1 above rests on Riemann's existence theorem, which is not in Mathlib (at least not in the version pinned by the artifact). For $m = 7$ we now give a second proof of the classification it provides, in the normalized chart of §5: every solution is one of the points $(P_j(w), Q_k(w))$ with $R(w) = 0$, the five K_5-conjugates. This is the form in which the main theorem uses Proposition 6.1. The eliminant W and the cases $m = 3, 5$ are not reproved here. The proof is purely algebraic and valid over every field of characteristic 0, and it is formalized: Theorem 6.5 is the Lean theorem `chartClassification_holds` (listing 16). We

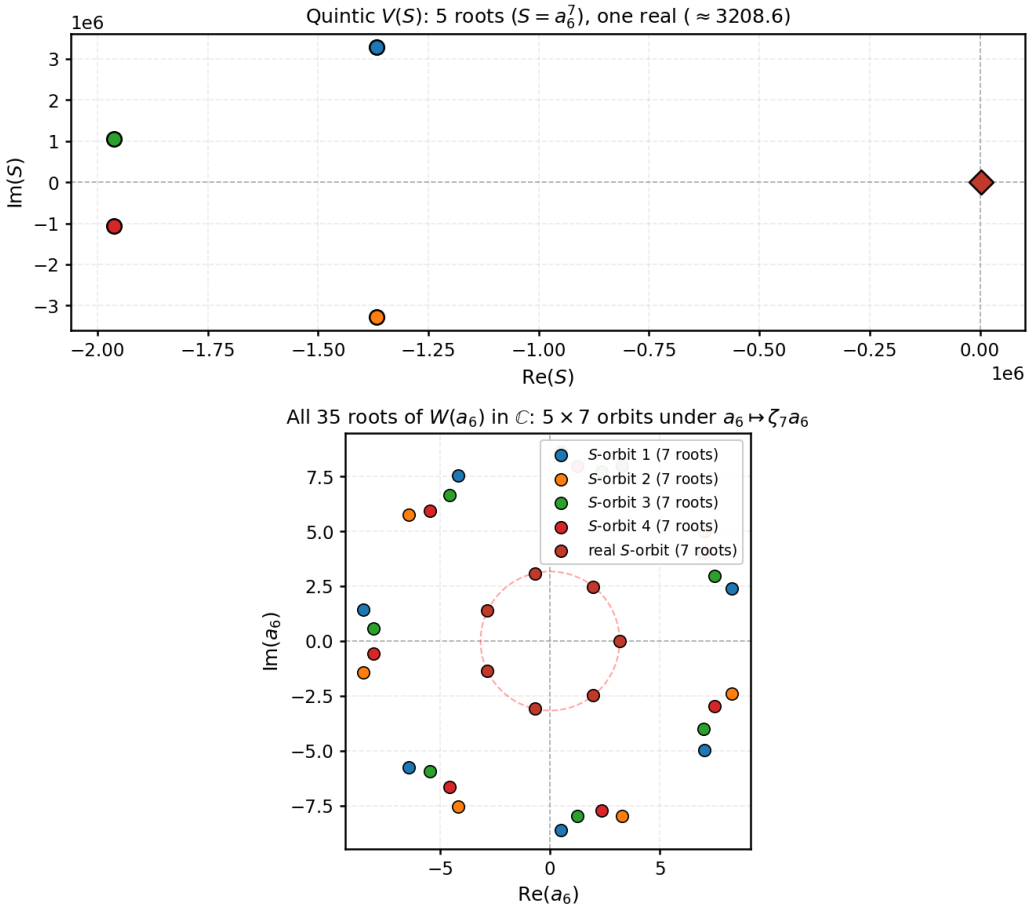


Figure 3: Top: the five roots of the quintic $V(S)$ in the complex S-plane, computed from the integer coefficients; the real root $S \approx 3208.61$ is marked with a diamond. Bottom: all 35 roots of $W(a_6)$ in $\mathbb{C}$, colored by S-orbit; each $S \neq 0$ lifts to 7 values of a_6 via 7th roots of unity. The dashed circle marks the real S-orbit. The 35 roots form a single Galois orbit over $\mathbb{Q}$.

work in the Lean chart of §5: $\alpha = 1 + a_1 u + \cdots + a_7 u^7$, $\beta = b_0 + b_1 u + \cdots + b_9 u^9 + u^{10}$, and $E_n = \sum_{i+k=n}(1 + 2k - 3i)\, a_i b_k$ $(1 \le n \le 16)$ is the coefficient of u^n in $\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta)$.

Theorem 6.5 (`chartClassification_holds`). *Let L be a field of characteristic 0. Suppose $(a_1, \dots, a_7, b_0, \dots, b_9) \in L^{17}$ satisfies $E_1 = \cdots = E_{16} = 0$ and $a_7^3 b_0^2 = 1$. Then $(a_j, b_k) = (P_j(w), Q_k(w))$ for a root $w \in L$ of R. Here $P_j, Q_k \in \mathbb{Q}[w]$ are the coordinates in listing 12.*

Proof. Step 1: reversed coordinates. Since $a_7^3 b_0^2 = 1$, we have $a_7 \neq 0$. Put $y_i = a_{7-i}/a_7$ for $1 \le i \le 6$ and $y_7 = 1/a_7$, and give y_i the weight i. Define $S_k \in \mathbb{Q}[y_1, \dots, y_7]$, weighted homogeneous of weight k, by

$$(1 + y_1 v + \cdots + y_7 v^7)^{3/2} = \sum_{k \ge 0} S_k\, v^k,$$

or, equivalently, by $S_0 = 1$ and $2k\, S_k = \sum_{i=1}^{7} (5i - 2k)\, y_i\, S_{k-i}$. Write $\tilde\beta_k = b_{10-k}$ for $0 \le k \le 10$ (so $\tilde\beta_0 = 1$) and $\tilde\beta_k = 0$ for $k > 10$; in both recursions, terms with a negative index are zero. Reindexing E_{17-k} by $i \mapsto 7 - i$, $k \mapsto 10 - k$ gives

$$E_{17-k}/a_7 = -2k\,\tilde\beta_k + \sum_{i=1}^{7} (5i - 2k)\, y_i\, \tilde\beta_{k-i} \qquad (1 \le k \le 16).$$

This is the recursion for S_k. Induction on k gives $b_{10-k} = S_k(y)$ for $1 \le k \le 10$, and

$$S_{11}(y) = \cdots = S_{16}(y) = 0, \qquad S_{10}(y)^2 = y_7^3. \tag{7}$$

The last equation holds because $b_0 = S_{10}(y)$ and $a_7^3 b_0^2 = 1$. In the language of §6, the reversed β is the truncation of the power-series square root of the reversed α^3/a_7^3. This is the Davenport–Stothers condition.

Step 2: the orbit relations. Consider the six weighted-homogeneous relations

$$\begin{aligned}
y_4 &= \tfrac{2431}{18240} y_1^4 - \tfrac{1027}{1710} y_1^2 y_2 + \tfrac{143}{380} y_2^2 + \tfrac{41}{60} y_1 y_3,\\
y_5 &= \tfrac{1154351}{25313472} y_1^5 - \tfrac{3833107}{18985104} y_1^3 y_2 + \tfrac{6512}{43947} y_1 y_2^2 + \tfrac{24449}{166536} y_1^2 y_3,\\
y_6 &= \tfrac{4206749569}{4242031637760} y_1^6 + \tfrac{8589189473}{1590761864160} y_1^4 y_2 - \tfrac{440013827}{9819517680} y_1^2 y_2^2\\
&\quad + \tfrac{7454351}{734423760} y_1^3 y_3 + \tfrac{253}{5054} y_2^3,\\
y_7 &= -\tfrac{4539711243127}{16353031963564800} y_1^7 + \tfrac{13373800868041}{6132386986336800} y_1^5 y_2 - \tfrac{214713769679}{37854240656400} y_1^3 y_2^2\\
&\quad + \tfrac{81661753}{707800898700} y_1^4 y_3 + \tfrac{92333}{19483170} y_1 y_2^3,\\
y_2 y_3 &= \tfrac{3738317}{46630080} y_1^5 - \tfrac{16882087}{34972560} y_1^3 y_2 + \tfrac{115021}{161910} y_1 y_2^2 + \tfrac{2277881}{5828760} y_1^2 y_3,\\
y_3^2 &= \tfrac{740761013029}{10605079094400} y_1^6 - \tfrac{754440973541}{1988452330200} y_1^4 y_2 + \tfrac{3827878261}{8182931400} y_1^2 y_2^2\\
&\quad + \tfrac{137816863}{918029700} y_1^3 y_3 + \tfrac{289}{12635} y_2^3.
\end{aligned}$$

Write $g = 0$ for any of them. Each satisfies

$$y_7^3 \cdot g \in (S_{11}, \dots, S_{16}) \subset \mathbb{Q}[y_1, \dots, y_7],$$

with explicit cofactors of 341 to 694 terms, whose numerators and denominators have at most 42 digits. Since $y_7 \neq 0$, every solution of (7) satisfies all six relations.

The relations were found by exact interpolation on the K_5 point, as all weighted-homogeneous forms of weight at most 7 that vanish on the five orbits. The cofactors come from exact weighted-homogeneous linear algebra over $\mathbb{Q}$. Let (g) be the ideal of the six relations. A Gröbner computation over $\mathbb{Q}$ (`saturation_check.py`) shows that $(S_{11},\dots,S_{16}) \subseteq (g)$ and $(g) : y_7 = (g)$. With the certificates this gives

$$(S_{11},\dots,S_{16}) : y_7^{\infty} = (S_{11},\dots,S_{16}) : y_7^{3} = (g).$$

The ideal (g) has Krull dimension 1 and needs all six generators, so it is a weighted complete intersection. It has exactly five solutions in the chart $y_1 = 1$ and none with $y_1 = 0$ other than $y = 0$, in agreement with weighted Bézout, $4 \cdot 5 \cdot 5 \cdot 6 \cdot 6 \cdot 7/7! = 5$. These facts explain the choice of relations but are not used in the proof.

Step 3: a resultant. The relation for y_7 shows $y_7 \in y_1\,\mathbb{Q}[y_1, y_2, y_3]$, so $y_1 \neq 0$. Put $t = y_2/y_1^2$ and $s = y_3/y_1^3$. The relation for $y_2 y_3$ becomes $(t - c)\,s = N(t)$ with $c = \frac{2277881}{5828760}$ and $\deg N = 2$, and the relation for y_3^2 becomes monic quadratic in s. Their Sylvester resultant in s is $\kappa\, m(t)$ with $\kappa \in \mathbb{Q}^{\times}$ and

$$\begin{aligned} m(t) = {} & 287548593020928\,t^5 - 688401965085696\,t^4 + 640652914818432\,t^3 \\ & - 292066554895024\,t^2 + 65563255857792\,t - 5817852446211, \end{aligned}$$

which is irreducible over $\mathbb{Q}$. Hence $t - c$ is invertible modulo m, and $s = q_3(t)$. The first four relations then give $y_i/y_1^i = q_i(t)$ for $4 \le i \le 7$, with $q_i \in \mathbb{Q}[t]$ reduced modulo m.

Step 4: the torus. Let $z = 1/y_1$. By weighted homogeneity,

$$z^{10} S_{10}(y) = S_{10}\big(1, t, q_3(t), \dots, q_7(t)\big) \equiv \sigma(t) \pmod{m} \qquad \text{and} \qquad z^7 y_7 = q_7(t).$$

Multiplying $S_{10}^2 = y_7^3$ by z^{21} gives $z\,\sigma(t)^2 = q_7(t)^3$, that is,

$$y_1 = \sigma(t)^2\, q_7(t)^{-3} \bmod m(t).$$

This fixes the torus without a seventh root, as in §5, and makes every y_i a polynomial in t modulo m. The substitution $w = \omega(t)$, with $\omega \in \mathbb{Q}[t]$ of degree 4, induces an isomorphism $\mathbb{Q}[t]/(m) \cong \mathbb{Q}[w]/(R) = K_5$ and gives $R(w) = 0$ and $y_i = Y_i(w)$.

Step 5: back to the chart. The formulas $a_7 = 1/y_7$ and $a_{7-i} = y_i/y_7$, together with the recursion of Step 1 for b_{10-k}, give $a_j = P_j(w)$ and $b_k = Q_k(w)$. □

Lemma 6.6 (Characteristic-zero transfer)**.** *Let F be a field of characteristic 0. Every F-valued solution of the normalized $m = 7$ chart system is obtained by specializing the K_5-point at a root $w \in F$ of R; its five specializations at the five roots of R (over a splitting field of R) are the five solutions.*

Proof. Steps 1–5 are proved algebraically over $\mathbb{Q}$ and hence hold over every characteristic-zero field by base change. The Lean formalization quantifies over all L with `CharZero`. The five solutions are the K_5-conjugates $(P_j(w), Q_k(w))$ with $R(w) = 0$. □

Remark 6.7 (Roles of the two classifications)*.* The $m = 7$ case of Proposition 6.1 is proved twice. The Belyi route combines the geometric upper bound (at most 35 chart points over $\mathbb{C}$) with the 35 exhibited points from the msolve parametrization, giving a complete classification in characteristic 0; it depends on Riemann's existence theorem and the computed parametrization. The algebraic certificate of this section gives an independent classification: every solution in characteristic 0 is one of the five K_5-orbits (Lemma 6.6). Lean's `main_theorem` uses only the algebraic route.

Remark 6.8 (Formalization). The proof is formalized in `lean/Jacobian/ChartProof/`: 12 modules, all generated by `lean/certgen/chartproof/` except the hand-written `Reflect.lean`.

- *Steps.* There are 111 steps. In 105 of them a polynomial g vanishes because $g = \sum_i c_i e_i$ for polynomials e_i already known to vanish. In the other six, one for each relation of Step 2, the identity $y_7^3 g = e$ holds with e already known to vanish, and y_7^3 is cancelled using $y_7 \neq 0$.
- *Kernel checking.* Every identity is checked by the Lean kernel. `lc_zero` (listing 18) and `eq_of_toPolyK` reduce it to an equality of normal forms. These are computed by the verified normalizer of `Lean.Grind.CommRing` from Lean core, with a kernel-friendly multiplication defined by recursors (`toPolyK`, soundness theorem `denote_toPolyK`), and `decide +kernel` evaluates them.
- *Axioms.* No `native_decide` is used. `#print axioms` for `chartClassification_holds` and for `main_theorem` reports only `propext`, `Classical.choice` and `Quot.sound`.
- *Batching.* The certificates of Step 2 have between 27,747 and 56,999 monomial products each. They are split into batches of at most 9,000 products, one declaration per batch.
- *Independent checks.* Every certificate was also verified by exact SymPy expansion before it was written, and the generator reproduces all eleven generated modules byte-identically. A second checker, written independently of the generator (`independent_check.py`), parses the Lean text and re-verifies all 111 identities over $\mathbb{Z}$. This checker verifies the arithmetic of the certificates only; it does not re-prove the mathematical statements. The Lean kernel remains solely responsible for verifying that these certificates prove the stated formal propositions. Six controls behave as expected: four one-integer perturbations are rejected by `decide`, a planted `sorry` is flagged, and an unmodified module is accepted.
- *Cost.* The build takes about 25 minutes sequentially and needs at most 5.4 GB of memory for any one module.

Remark 6.9 (Why these coordinates). Earlier formulations failed to yield certificates of practical size because of the degenerate Davenport–Stothers components of lower degree (the solutions for $m = 1, 3, 5$). With $a_0 = b_0 = 1$ and b eliminated, these components force saturation by a_7^7 or a_7^8, so the certificates have weight 61. In the chart $a_0 = b_{10} = 1$, the sub-case $a_1 = 0$ alone needed 503,254 cofactor terms modulo a prime. Replayed over $\mathbb{Q}$, its Buchberger trace reached coefficients of 2,722 digits after 929 of its 2,065 elements (`lean/CLASSIFICATION_STATUS.md`, §5). In the reversed coordinates the same components lie on $y_7 = 0$, and y_7^3 suffices, at weight at most 28. The five orbits then form a weighted complete intersection that can be written down explicitly. The six certificates of Step 2 have 341 to 694 cofactor terms each over $\mathbb{Q}$, roughly a thousand times fewer than the modular lift for $a_1 = 0$ alone.

Listing 16: Mechanically extracted from ***Jacobian/ChartProof/Final.lean*** *(sha256* ***638b644ffb58cb8b****); statement verbatim, proof elided.*

```
theorem chartClassification_holds (L : Type*) [Field L] [CharZero L] :
↪  ChartClassification L := by
  -- proof body elided (see artifact)
```

Listing 17: Mechanically extracted from ***Jacobian/ChartProof/Final.lean*** *(sha256* ***638b644ffb58cb8b****); statement verbatim, proof elided.*

```
theorem main_theorem {L : Type*} [Field L] [CharZero L] :
    ¬ exists (P Q : MvPolynomial (Fin 2) L) (lam : L), lam ≠ 0 /\ NewtonNF2 P Q /\
     ↪  jac P Q = MvPolynomial.C lam * MvPolynomial.X 0 ^ 2 :=
  -- proof term elided (see artifact)
```

Listing 18: Mechanically extracted from ***Jacobian/ChartProof/Reflect.lean*** *(sha256* ***9e3ea428ffe7236b****); statement verbatim, proof elided.*

```
theorem lc_zero (ctx : Context α) (g : Expr) (l : List (Expr × Expr))
    (hcert : Poly.beq' (toPolyK (.sub g (lcExpr l))) (.num 0) = true) (h : AllZero
     ↪  ctx l) :
    g.denote ctx = 0 := by
  -- proof body elided (see artifact)
```

7 Descent

Fix a top-layer solution (α, β). The descent solves E_4, E_3, E_2 in turn. We use the scaled variables $A_2 = u\alpha$ and $B_3 = u^2\beta$.

7.1 Layer E_4

Proposition 7.1 (E_4 structure)**.** *E_4 is a linear system in* 19 *unknowns (A_1 on $u^1, \dots, u^8$; B_2 on $u^2, \dots, u^{12}$) with* 18 *equations, rank* 17*, and kernel dimension* 2*. The kernel is parametrized by $t = (t_1, t_2)$, giving $(A_1(t), B_2(t))$ linear in t. Computed exactly in characteristic zero over the quintic field $K_5 = \mathbb{Q}[w]/(w^5 - w^4 + 3w^3 + 3w^2 + 26)$ (script* ***exact_ranks_K5.py****); the modular evaluation at $p = 101$, $a_6 = 55$ is an independent cross-check.*

Proof. Rank computation on the 18×19 coefficient matrix, with residual asserts verifying the kernel. Scripts: `exact_ranks_K5.py` (exact rank and kernel over K_5) and `C5_E2_obstruction_corrected.py` (which verifies E_4 with E_4-kernel residuals zero; mod-101 cross-check). □

7.2 Layer E_3

Proposition 7.2 (E_3 structure and unconditional solvability)**.** *E_3 is a* 19×20 *linear system of rank* 18 *with* 2*-dimensional kernel. Moreover, E_3 is solvable for* every *direction $t = (t_1, t_2)$: writing the system as $M(t)x = K(t)$ with M constant and K quadratic in t, the left nullvector v (with $v^{\mathsf{T}}M = 0$) satisfies*

$$v^{\mathsf{T}}K_{11} \equiv 0, \qquad v^{\mathsf{T}}K_{22} \equiv 0, \qquad v^{\mathsf{T}}K_{12} \equiv 0 \tag{8}$$

identically as polynomials in t, where $K = t_1^2K_{11} + t_1t_2K_{12} + t_2^2K_{22}$. Hence no direction is obstructed at E_3. All computations are exact over K_5; see Lemma 9.2(i).

Proof. The 19×20 matrix M has rank 18; its left nullspace is 1-dimensional, spanned by v. Decomposing the inhomogeneous part and computing $v^T K_{ij}$ symbolically gives zero for all i, j. Script: `C5_E2_obstruction_corrected.py` (which verifies E_3 with residual asserts). □

The general solution of E_3 is thus the 2-parameter affine family

$$x(t,s) = t_1^2 x_{11} + t_1 t_2 x_{12} + t_2^2 x_{22} + s_1 w_1 + s_2 w_2, \tag{9}$$

encoding (A_0, B_1) in terms of $(t,s) = (t_1, t_2, s_1, s_2) \in \mathbb{C}^4$.

7.3 Layer E_2

Proposition 7.3 (E_2 injectivity)**.** *The homogeneous operator of E_2 in the unknown B_0' is a 19×12 matrix of rank 12, hence injective. Its left nullspace has dimension $19 - 12 = 7$. Exact over K_5; see Lemma 9.2(i)–(ii).*

Therefore B_0' (and hence B_0 up to an additive constant) is uniquely fixed by (A_0, B_1, A_1, B_2). The seven left-nullvector pairings then give seven compatibility conditions:

$$W \cdot \mathrm{rhs}(t,s) = 0, \tag{10}$$

where W is 7×19 and $\mathrm{rhs}(t,s)$ is the E_2 inhomogeneous term. These seven conditions are the *only* remaining obstructions; E_1 is never needed.

*Listing 19: Mechanically extracted from **`Jacobian/Descent/Main.lean`** (sha256 **`e18a3a9e531df6d6`**); statement verbatim, proof elided.*

```
theorem descent_K5 {L : Type*} [Field L] [CharZero L]
    (w : L)
    (hw : w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0)
    (a_1_0 a_2_2 a_3_4 a_4_6 a_5_8 a_6_10 a_7_12 a_8_14 b_2_1 b_3_3 b_4_5 b_5_7
     ↪  b_6_9 b_7_11 b_8_13 b_9_15 b_10_17 b_11_19 b_12_21 : L)
    (h_a_1_0 : a_1_0 = ((1 : L)))
    -- ... (18 h4_*, 19 h3_*, 19 h2_*, 8 h_a_*, 11 h_b_*, 51 lower-layer unknowns in
     ↪  one binder)
    (hv : b_12_24 ≠ 0) :
    False := by
  -- proof body elided (see artifact)
```

8 The final obstruction

We use the homogeneous formulation: E_3 with A_0 on $u^1, \ldots, u^8$ ($a_{8,16}$ free, no constant term) and B_1 on $u^1, \ldots, u^{12}$. This gives the genuine 2-dimensional kernel with two independent directions $s = (s_1, s_2)$, both moving actual coefficients. (An earlier formulation froze $a_{8,16} = 1$ and included the constant term as an unknown; the constant is killed by A_0' and contributes only a trivial direction.)

Remark 8.1 (Free parameter). §5 leaves $a_{8,16}$ free from the start. The E_2/E_3 analysis therefore applies to every slice, including $a_{8,16} = 1$, without a superset argument.

8.1 Explicit form of the seven conditions

Expanding (10) symbolically in (t_1, t_2, s_1, s_2) gives:

Lemma 8.2. *Each of the seven conditions has the form*

$$b_i(t_1, t_2) + s_1 \cdot M_i(t_1, t_2) + s_2 \cdot L_i(t_1, t_2) = 0, \qquad i = 1, \dots, 7, \tag{11}$$

where b_i is homogeneous cubic and M_i, L_i are linear in (t_1, t_2). Both s_1 and s_2 appear genuinely.

Proof. Exact expansion over K_5 in python-flint (script `exact_obstruction_K5.py`) of $W \cdot \mathrm{rhs}(t, s)$ with the homogeneous E_3 parametrization $x(t, s) = t_1^2 x_{11} + t_1 t_2 x_{12} + t_2^2 x_{22} + s_1 w_1 + s_2 w_2$ substituted; the SymPy expansion over $\mathbb{F}_{101}$ (script `C5_E2_obstruction_corrected.py`) is a cross-check. □

8.2 Elimination of (s_1, s_2)

For fixed $t \neq 0$, the seven equations are affine-linear in (s_1, s_2). A solution can exist only if the 7×3 matrix $[b(t) \mid M(t) \mid L(t)]$ has rank at most 2, i.e. only if all 3×3 minors vanish:

$$\det \begin{pmatrix} b_i & M_i & L_i \\ b_j & M_j & L_j \\ b_k & M_k & L_k \end{pmatrix} = 0, \qquad 1 \le i < j < k \le 7. \tag{12}$$

There are $\binom{7}{3} = 35$ such minors, each a homogeneous binary quintic in (t_1, t_2) (cubic × linear × linear).

Theorem 8.3 (No nonzero direction)**.** *The 35 quintics (12) span the degree-five homogeneous component $K_5[t_1, t_2]_5$ and generate the ideal $(t_1, t_2)^5$, i.e. they have Gröbner basis*

$$(t_1, t_2)^5 = \left\{ t_1^5,\ t_1^4 t_2,\ t_1^3 t_2^2,\ t_1^2 t_2^3,\ t_1 t_2^4,\ t_2^5 \right\} \tag{13}$$

exactly over K_5 (the 35×6 coefficient matrix has rank 6, spanning all binary quintics). Hence their common projective zero locus is empty and their only common affine zero in $\overline{K_5}^2$ is $(0, 0)$: there is no *nonzero t admitting (s_1, s_2) satisfying the seven E_2 conditions. The mod-101 Gröbner computation (lex order) is an independent cross-check.*

Proof. Exact computation over K_5: the 35 minors are homogeneous quintics, and their 35×6 coefficient matrix has rank 6, so

$$\mathrm{span}_{K_5}\{M_{ijk}\} = K_5[t_1, t_2]_5,$$

the full six-dimensional space of binary quintics. Hence every degree-five monomial lies in the ideal I they generate, i.e. $(t_1, t_2)^5 \subseteq I$; conversely each generator lies in $(t_1, t_2)^5$, so $I = (t_1, t_2)^5$. The radical is (t_1, t_2), so the affine variety over the algebraic closure is $\{(0, 0)\}$. Script: `exact_obstruction_K5.py`, which also writes the 35×6 matrix and a 6×6 nonzero minor to `logs/k5_minor_certificate.json` as a machine-readable certificate. The SymPy Gröbner basis over $GF(101)$ (script `C5_E2_obstruction_corrected.py`, saved, rerunnable; log archived) is a cross-check. □

Remark 8.4 (Interpretation). The top layer admits five characteristic-zero orbits, but none extends to a full solution: the E_4 kernel gives a two-dimensional space of directions $t \in K^2$, the E_3 layer imposes no further restriction, and then the seven E_2 compatibility conditions force $t = 0$ via the 35 quintic minors. The remaining case $t = 0$ is ruled out separately (Corollary 8.6). In other words, the top-layer solutions exist in isolation, but none extends to a Newton-normal-form pair with $[P, Q] = \lambda x^2$.

8.3 The case $t = 0$

Lemma 8.5 (Collapse at $t = 0$). *At $t = (0, 0)$ the E_2 layer equation reduces to*

$$2A_2B_0' = 0.$$

Proof. Because the target bracket identity $[P, Q] = x^2 = u^2y^{-4}$ contains only a y^{-4} component, the layer equation E_4 at y^{-3} is homogeneous:

$$2(A_2B_2' - A_2'B_2) + (A_1B_3' - 3A_1'B_3) = 0.$$

Consequently $t = (0, 0)$ forces $(A_1, B_2) \equiv (0, 0)$. In layer E_3 the quadratic driving term $K(t)$ vanishes identically at $t = 0$, leaving only the homogeneous kernel. In layer E_2 the terms $(A_1B_1' - A_1'B_1) - 2A_0'B_2$ then vanish identically, leaving $2A_2B_0' = 0$. □

Corollary 8.6 ($t = 0$ impossible). *By Lemma 8.5, at $t = 0$ we have $2A_2B_0' = 0$. Since $A_2 = u\alpha \neq 0$, this requires $B_0' = 0$, forcing $B_0(u)$ to be constant. Hence $b_{12,24} = 0$; in particular, $(12, 24)$ cannot be a vertex of $N(Q)$.*

Remark 8.7. The Lean formalization `e2_t0` (listing 20) proves $b_{12,24} = 0$ via a *different route*: a left-inverse certificate on the reduced E_2 equations with $t = 0$ supplied as hypotheses, rather than the $2A_2B_0' = 0$ argument above. Both routes reach the same conclusion.

*Listing 20: Mechanically extracted from **`Jacobian/Descent/E2/T0.lean`** (sha256 **`3559bcbdbff42d3c`**); statement verbatim, proof elided.*

```
theorem e2_t0 {L : Type*} [Field L] [CharZero L]
    (w : L)
    (hw : w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0)
    (b_11_20 b_12_22 b_11_21 b_12_23 b_1_2 b_2_4 b_3_6 b_4_8 b_5_10 b_6_12 b_7_14
     ↪  b_8_16 b_9_18 b_10_20 b_11_22 b_12_24 : L)
    -- 19 reduced E2 equations (r2_1_1) ... (r2_19_37), hypotheses elided for brevity
    (ht1 : b_11_20 = 0)
    (ht2 : b_12_22 = 0) :
    b_12_24 = 0 := by
  -- proof body elided (see artifact)
```

8.4 Proof of the main theorem

Proof of Theorem 1.1. By Proposition 4.1, the layers satisfy $E_5, \dots, E_1$. Since $a_{1,0} \cdot a_{8,14} \cdot b_{2,1} \cdot b_{12,21} \neq 0$, Proposition 6.1 puts (A_2, B_3) in the torus orbit of the K_5 point for some root w of R. By torus transport (Listing 6) we may take it to be that point. Propositions 7.1–7.3 and Theorem 8.3 force $t = 0$. At $t = 0$, Lemma 8.5 and Corollary 8.6 give $B_0' = 0$, so $b_{12,24} = 0$

(in Lean, `e2_t0`, Listing 20). The last sentence of the theorem follows because NewtonNF2 requires $b_{12,24} \neq 0$. □

Proof of Corollary 1.2. By the proof of Theorem 1.1 we may take the top layer to be the K_5 point, where E_4, E_3, E_2 force $t = 0$; torus transport multiplies each coefficient by a nonzero scalar. At $t = 0$ we have $A_1 = B_2 = 0$ (Proposition 7.1). By Proposition 7.2, the coefficients of A_0 other than $a_{0,0}$ and the coefficients of B_1 are K_5-linear forms in $s = (b_{11,21}, b_{12,23})$, the two free unknowns of E_3. In particular $a_{8,16} = \kappa\, b_{12,23}$ with $\kappa \in K_5^\times$, and $b_{11,21}$ occurs in $a_{1,2}$ with a nonzero coefficient. Since $A_1 = 0$, the layer E_1, $A_1 B_0' - A_0' B_1 = 0$, reduces to $A_0' B_1 = 0$ in $K[u]$. If $B_1 = 0$, then $s = 0$. If $A_0' = 0$, then $a_{1,2} = a_{8,16} = 0$, so again $s = 0$. Hence $A_0 = a_{0,0}$ and $B_1 = 0$; in particular $a_{8,16} = 0$. Finally B_0 is constant by Corollary 8.6. In Lean (`lower_edge_rigidity`, generated by `lean/certgen/gen_a816.py`), the E_1 step uses the coefficients of $x^{17}y^{34}$, $x^{18}y^{36}$ and $x^{19}y^{38}$ in E_1 instead. At $t = 0$ they are quadratic forms in s that span all three quadratic monomials, and explicit K_5 multipliers give $b_{12,23}^2 = 0$ and $b_{11,21}^2 = 0$. The last one alone gives $a_{8,16} = 0$: it is $12\, a_{8,15} b_{12,24} - 8\, a_{8,16} b_{12,23} = -8\kappa\, b_{12,23}^2$.

A second proof of $a_{8,16} = 0$, outside Lean and independent of the descent, is an explicit certificate. At the K_5 point, let $e_1, \dots, e_{75}$ be the coefficients of $[P, Q] - x^2$ in the layers $E_4, \dots, E_1$, as built by `scripts/a816_full.sing`; their unknowns are the 51 lower coefficients of P and Q that occur. There is an explicit identity

$$a_{8,16}^2 = \sum_{k=1}^{75} H_k\, e_k, \qquad H_k \in K_5[a, b],$$

stored in Rabinowitsch form, $1 = \sum_k z^2 H_k\, e_k - (1 + a_{8,16} z)(a_{8,16} z - 1)$, as the 76 cofactors of `scripts/a816_certificate/a816_lift.txt` (3464 terms, coefficients of at most 493 digits). It is built from the layer structure. Give the coefficients of A_1, B_2 depth 1, those of A_0, B_1 depth 2 and those of B_0 depth 3; each equation of E_{5-d} is then homogeneous of depth d and linear in the unknowns of depth d. Row reduction of these linear parts in E_4, E_3, E_2 (ranks 17, 18, 12 over K_5) expresses every unknown, modulo the layer ideal, as a polynomial in four free ones, $\tau = (b_{11,20}, b_{12,22})$ of depth 1 and $\sigma = (a_{8,16}, b_{11,21})$ of depth 2. The 32 products of depth 4 formed from the reduced generators span all 14 monomials of depth 4 in $K_5[\tau, \sigma]$, and lifting back gives the H_k. The identity is checked exactly by Singular over $\mathbb{Q}(w)$ and independently by python-flint, which rebuilds $[P, Q]$ from P and Q (`scripts/a816_certificate/verify_bundle.sh`). Hence $a_{8,16} = 0$ at the K_5 point once E_1 is included. Torus transport multiplies $a_{8,16}$ by a nonzero scalar, so $a_{8,16} = 0$ on the whole orbit. □

Remark 8.8 (Further rigidity, outside the certificate). The same span statement puts every monomial of depth at least 4 in τ, σ into the layer ideal, so each of the 51 unknowns (the non-constant lower coefficients of P and Q) is nilpotent modulo it: at the K_5 point the only solution of $E_4, \dots, E_1$ is $P = P_2 + \text{const}$, $Q = Q_3 + \text{const}$. Explicitly, let x be an unknown of depth d, and let $p_x \in K_5[\tau, \sigma]$ be the polynomial that the row reduction gives, with $x \equiv p_x$ modulo the layer ideal ($p_x = x$ for a free unknown). It is homogeneous of depth d with no constant term. So $x^{\lceil 4/d \rceil} \equiv p_x^{\lceil 4/d \rceil}$, and every monomial of $p_x^{\lceil 4/d \rceil}$ is divisible by a monomial of depth exactly 4. The certificates behind this are written out and checked exactly by two independent programs (`scripts/a816_rigidity/`): the 47 identities $x - p_x = \sum_k D_{x,k} e_k$ for the pivot unknowns, and the 14 identities $m = \sum_k H_{m,k} e_k$ for the monomials m of depth 4. The zero-set form of this statement, $P = P_2 + \text{const}$ and $Q = Q_3 + \text{const}$, is the Lean theorem

`lower_edge_rigidity` (Corollary 1.2); the nilpotency modulo the layer ideal is not part of the Lean formalization. Neither is used elsewhere.

Listing 21: Mechanically extracted from ***Jacobian/A816/Final.lean*** *(sha256* ***b7f3ed6341617a44****); statement verbatim, proof elided.*

```
theorem lower_edge_rigidity {L : Type*} [Field L] [CharZero L] (P Q : MvPolynomial
 ↪  (Fin 2) L) (lam : L)
    (hP : forall m ∈ P.support, inNP m) (hQ : forall m ∈ Q.support, inNQ m)
    (v10 : P.coeff (mono 1 0) ≠ 0) (v814 : P.coeff (mono 8 14) ≠ 0)
    (w21 : Q.coeff (mono 2 1) ≠ 0) (w1221 : Q.coeff (mono 12 21) ≠ 0)
    (hJ : jac P Q = MvPolynomial.C lam * MvPolynomial.X 0 ^ 2) :
    (forall i j, 2 * i ≤ j + 1 → (i, j) ≠ (0, 0) → P.coeff (mono i j) = 0) /\
      (forall k l, 2 * k ≤ l + 2 → (k, l) ≠ (0, 0) → Q.coeff (mono k l) = 0) := by
  -- proof body elided (see artifact)
```

Listing 22: Mechanically extracted from ***Jacobian/A816/Final.lean*** *(sha256* ***b7f3ed6341617a44****); statement verbatim, proof elided.*

```
theorem main_theorem_lower_edge {L : Type*} [Field L] [CharZero L] :
    ¬ exists (P Q : MvPolynomial (Fin 2) L) (lam : L), lam ≠ 0 /\ NewtonNF2 P Q /\
     ↪  jac P Q = MvPolynomial.C lam * MvPolynomial.X 0 ^ 2 := by
  -- proof body elided (see artifact)
```

Proof of Corollary 1.4. Proposition 4.3 of [1], then Theorem 1.1. □

Listing 23: Mechanically extracted from ***Jacobian/BranchAbFinal.lean*** *(sha256* ***cd4e207adba5cb7b****); statement verbatim, proof elided.*

```
theorem no_completion_K5 {L : Type*} [Field L] [CharZero L] (w : L)
    (hw : w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0) (lam : L)
    (A₀ A₁ A₂ B₀ B₁ B₂ B₃ : Polynomial L) (p₀ p₁ p₂ q₀ q₁ q₂ q₃ : MvPolynomial (Fin
     ↪  2) L)
    (hp₀ : X 1 ^ 0 * p₀ = evH A₀) (hp₁ : X 1 ^ 1 * p₁ = evH A₁) (hp₂ : X 1 ^ 2 * p₂ =
     ↪  evH A₂)
    (hq₀ : X 1 ^ 0 * q₀ = evH B₀) (hq₁ : X 1 ^ 1 * q₁ = evH B₁) (hq₂ : X 1 ^ 2 * q₂ =
     ↪  evH B₂)
    (hq₃ : X 1 ^ 3 * q₃ = evH B₃)
    (hJ : jac (p₀ + p₁ + p₂) (q₀ + q₁ + q₂ + q₃) = C lam * X 0 ^ 2)
    (sA₀ : forall i, A₀.coeff i ≠ 0 → 0 ≤ i /\ i ≤ 8)
    (sA₁ : forall i, A₁.coeff i ≠ 0 → 1 ≤ i /\ i ≤ 8)
    (sA₂ : forall i, A₂.coeff i ≠ 0 → 1 ≤ i /\ i ≤ 8)
    (sB₀ : forall i, B₀.coeff i ≠ 0 → 0 ≤ i /\ i ≤ 12)
    (sB₁ : forall i, B₁.coeff i ≠ 0 → 1 ≤ i /\ i ≤ 12)
    (sB₂ : forall i, B₂.coeff i ≠ 0 → 2 ≤ i /\ i ≤ 12)
    (sB₃ : forall i, B₃.coeff i ≠ 0 → 2 ≤ i /\ i ≤ 12)
    (ρ σ ε : L) (hρ : ρ ≠ 0) (hσ : σ ≠ 0) (hε : ε ≠ 0)
    (t_a_1_0 : (A₂.coeff 1) = ρ * ε ^ (0 : Nat) * ((1 : L)))
    (t_a_2_2 : (A₂.coeff 2) = ρ * ε ^ (1 : Nat) * (((303) / 16 : L) + ((879) / 128 :
     ↪  L) * w + ((-389) / 128 : L) * w ^ (2 : Nat) + ((-51) / 128 : L) * w ^ (3 :
     ↪  Nat) + ((49) / 128 : L) * w ^ (4 : Nat)))
    (t_a_3_4 : (A₂.coeff 3) = ρ * ε ^ (2 : Nat) * (((315) / 16 : L) + ((693) / 8 : L)
     ↪  * w + ((-147) / 8 : L) * w ^ (2 : Nat) + ((-189) / 8 : L) * w ^ (3 : Nat) +
     ↪  ((147) / 16 : L) * w ^ (4 : Nat)))
```

```
  (t_a_4_6 : (A₂.coeff 4) = ρ * ε ^ (3 : Nat) * (((-79653) / 64 : L) + ((29463) /
   ↪  64 : L) * w + ((7371) / 128 : L) * w ^ (2 : Nat) + ((-4767) / 16 : L) * w ^
   ↪  (3 : Nat) + ((8001) / 128 : L) * w ^ (4 : Nat)))
  (t_a_5_8 : (A₂.coeff 5) = ρ * ε ^ (4 : Nat) * (((-33519) / 4 : L) + ((-6843) / 4
   ↪  : L) * w + ((8181) / 8 : L) * w ^ (2 : Nat) + ((-1713) / 2 : L) * w ^ (3 :
   ↪  Nat) + ((-1305) / 8 : L) * w ^ (4 : Nat)))
  (t_a_6_10 : (A₂.coeff 6) = ρ * ε ^ (5 : Nat) * (((-8805) / 8 : L) + ((-76965) / 8
   ↪  : L) * w + ((11523) / 16 : L) * w ^ (2 : Nat) + ((6927) / 4 : L) * w ^ (3 :
   ↪  Nat) + ((-18579) / 16 : L) * w ^ (4 : Nat)))
  (t_a_7_12 : (A₂.coeff 7) = ρ * ε ^ (6 : Nat) * (((389740) / 17 : L) + ((-51922) /
   ↪  17 : L) * w + ((-48572) / 17 : L) * w ^ (2 : Nat) + ((63239) / 17 : L) * w ^
   ↪  (3 : Nat) + ((-6475) / 17 : L) * w ^ (4 : Nat)))
  (t_a_8_14 : (A₂.coeff 8) = ρ * ε ^ (7 : Nat) * (((476437) / 68 : L) + ((314405) /
   ↪  68 : L) * w + ((-174571) / 136 : L) * w ^ (2 : Nat) + ((1574) / 17 : L) * w ^
   ↪  (3 : Nat) + ((82783) / 136 : L) * w ^ (4 : Nat)))
  (t_b_2_1 : (B₃.coeff 2) = σ * ε ^ (0 : Nat) * ((1 : L)))
  (t_b_3_3 : (B₃.coeff 3) = σ * ε ^ (1 : Nat) * (((101) / 8 : L) + ((293) / 64 : L)
   ↪  * w + ((-389) / 192 : L) * w ^ (2 : Nat) + ((-17) / 64 : L) * w ^ (3 : Nat) +
   ↪  ((49) / 192 : L) * w ^ (4 : Nat)))
  (t_b_4_5 : (B₃.coeff 4) = σ * ε ^ (2 : Nat) * (((315) / 16 : L) + ((693) / 8 : L)
   ↪  * w + ((-147) / 8 : L) * w ^ (2 : Nat) + ((-189) / 8 : L) * w ^ (3 : Nat) +
   ↪  ((147) / 16 : L) * w ^ (4 : Nat)))
  (t_b_5_7 : (B₃.coeff 5) = σ * ε ^ (3 : Nat) * (((-11379) / 8 : L) + ((4209) / 8 :
   ↪  L) * w + ((1053) / 16 : L) * w ^ (2 : Nat) + ((-681) / 2 : L) * w ^ (3 : Nat)
   ↪  + ((1143) / 16 : L) * w ^ (4 : Nat)))
  (t_b_6_9 : (B₃.coeff 6) = σ * ε ^ (4 : Nat) * (((-695299) / 48 : L) + ((-48367) /
   ↪  48 : L) * w + ((52679) / 32 : L) * w ^ (2 : Nat) + ((-45661) / 24 : L) * w ^
   ↪  (3 : Nat) + ((-2211) / 32 : L) * w ^ (4 : Nat)))
  (t_b_7_11 : (B₃.coeff 7) = σ * ε ^ (5 : Nat) * (((-309895) / 8 : L) + ((-251275)
   ↪  / 8 : L) * w + ((101481) / 16 : L) * w ^ (2 : Nat) + ((1811) / 4 : L) * w ^
   ↪  (3 : Nat) + ((-60489) / 16 : L) * w ^ (4 : Nat)))
  (t_b_8_13 : (B₃.coeff 8) = σ * ε ^ (6 : Nat) * (((1733348) / 13 : L) +
   ↪  ((-2368725) / 26 : L) * w + ((-184976) / 13 : L) * w ^ (2 : Nat) + ((1819939)
   ↪  / 52 : L) * w ^ (3 : Nat) + ((-569071) / 52 : L) * w ^ (4 : Nat)))
  (t_b_9_15 : (B₃.coeff 9) = σ * ε ^ (7 : Nat) * (((18736247) / 39 : L) +
   ↪  ((1087133) / 13 : L) * w + ((-5656777) / 78 : L) * w ^ (2 : Nat) +
   ↪  ((1993384) / 39 : L) * w ^ (3 : Nat) + ((971365) / 78 : L) * w ^ (4 : Nat)))
  (t_b_10_17 : (B₃.coeff 10) = σ * ε ^ (8 : Nat) * (((305098) / 17 : L) +
   ↪  ((5784066) / 17 : L) * w + ((-308335) / 17 : L) * w ^ (2 : Nat) + ((-950608)
   ↪  / 17 : L) * w ^ (3 : Nat) + ((754021) / 17 : L) * w ^ (4 : Nat)))
  (t_b_11_19 : (B₃.coeff 11) = σ * ε ^ (9 : Nat) * (((-149721182) / 323 : L) +
   ↪  ((26391690) / 323 : L) * w + ((21560033) / 323 : L) * w ^ (2 : Nat) +
   ↪  ((-24378412) / 323 : L) * w ^ (3 : Nat) + ((2786935) / 323 : L) * w ^ (4 :
   ↪  Nat)))
  (t_b_12_21 : (B₃.coeff 12) = σ * ε ^ (10 : Nat) * (((-91912444) / 969 : L) +
   ↪  ((-20715500) / 323 : L) * w + ((17474146) / 969 : L) * w ^ (2 : Nat) +
   ↪  ((-1623428) / 969 : L) * w ^ (3 : Nat) + ((-8783302) / 969 : L) * w ^ (4 :
   ↪  Nat)))
  (hv : B₀.coeff 12 ≠ 0) : False := by
-- proof body elided (see artifact)
```

9 Exact characteristic-0 descent

The proof is exact over K_5 (items (i)–(ii) below).

Lemma 9.1 (Base change for the K_5 computations)**.** *$R(w) = w^5 - w^4 + 3w^3 + 3w^2 + 26$ is irreducible and separable over $\mathbb{Q}$, so $K_5 = \mathbb{Q}[w]/(R)$ is a number field. The coefficients of the E_4, E_3, E_2 matrices and of the seven compatibility polynomials lie in K_5. Exact rank identities and exact ideal identities over K_5 therefore remain valid under every characteristic-zero field extension.*

Proof. R is irreducible modulo 67 (a single quintic factor), hence irreducible over $\mathbb{Q}$, and separable since its discriminant $784{,}822{,}272 = 2^{12} \cdot 3 \cdot 13 \cdot 17^3$ is nonzero. Hence K_5 is a field. Rank of a matrix and equality of ideals are preserved under field extension: rank is the size of the largest nonzero minor, and ideal membership is witnessed by explicit polynomial combinations, both of which persist after extending scalars. □

Lemma 9.2 (Exact obstruction)**.** *Let $\rho \in \overline{\mathbb{Q}}$ be any root of W, and put $K = \mathbb{Q}(\rho)$. The E_2 obstruction—no nonzero (t_1, t_2) satisfies the seven compatibility conditions—holds over K.*

Proof. By Lemma 9.1, the exact rank and ideal identities over K_5 persist under every characteristic-zero field extension. Since $K_5 \cong \mathbb{Q}(\rho^7)$ for every root ρ of W (the $S = a_6^7$ parametrization; Remark 9.3), the obstruction holds over $\mathbb{Q}(\rho)$, hence over every characteristic-zero field containing such a root. The ingredients are the exact characteristic-0 computations (scripts `exact_ranks_K5.py`, `exact_obstruction_K5.py`):

(i) *Exact characteristic-0 ranks.* The ranks are computed exactly over the quintic field $K_5 = \mathbb{Q}[w]/(w^5 - w^4 + 3w^3 + 3w^2 + 26)$ (script `exact_ranks_K5.py`): $E_4 = 17$, $E_3 = 18$, $E_2 = 12$.

(ii) *Exact descent over K_5.* The entire $E_4 \to E_3 \to E_2 \to 35$-minor pipeline is redone exactly over K_5 (script `exact_obstruction_K5.py`, ~1 s): E_3 is solvable for every t, the left nullity of E_2 is 7, and the 35 minors span *all* binary quintics (the 35×6 coefficient matrix has rank 6 over K_5). Hence the only common zero is $t = 0$. End-to-end residuals against the original bracket equations vanish exactly. The $t = 0$ case is excluded by the vertex $(12, 24)$ argument, which is valid in every characteristic other than 2 and 3 (it uses $2 \neq 0$ and $12 \neq 0$; see §8.3). The Lean formalization (`no_completion_K5`, listing 23, via `main_theorem_of_chart`, listing 7) quantifies over all roots of R and all torus parameters, covering all top-layer solutions by Proposition 6.1, whose $m = 7$ classification is formalized (`chartClassification_holds`, §6.4).

□

Remark 9.3 (On the derivation of W). The polynomial W was originally imported from the reference data (see `W_IMPORT_DISCLOSURE.md`). It is now *independently derived*: the minimal polynomial of a_6^7, computed from the exactly verified top-layer point over K_5, equals the quintic V (hence $W(a_6) = V(a_6^7)$) coefficient for coefficient (script `compare_V.gp`). The import caveat in `W_IMPORT_DISCLOSURE.md` is retired for W itself. The mod-101 reduction matches the independently computed elimination up to scalar 54 as a cross-check.

10 Conclusion

We have eliminated branches (a) and (b) of the GGHV $(72, 108)$ candidate—case (2) of [1, Proposition 4.3]—by a computational audit:

- source-gated base objects (polygons, bracket, and φ verified against GGHV v1);
- exact layer reduction (92 equations, symbolic identity);
- top-layer determination (degree-35 eliminant, one real solution);
- descent with explicit ranks and unconditional E_3 solvability;
- minor-span obstruction ($(t_1, t_2)^5$) closing the E_2 conditions;
- exact K_5-descent (Lean `no_completion_K5` via `main_theorem_of_chart`) covering all roots of R and all torus parameters;

- an algebraic proof of the $m = 7$ top-layer classification (Proposition 6.1 in the normalized chart), using reversed coordinates, six orbit relations and a Sylvester resultant (§6.4). It is formalized in Lean, so the NewtonNF2 form of Theorem 1.1 is the kernel-checked theorem `main_theorem`, independent of GGHV Proposition 4.3;
- lower-edge rigidity (Corollary 1.2): $P = P_2 + a_{0,0}$ and $Q = Q_3 + b_{0,0}$, kernel-checked as `lower_edge_rigidity`, so that the vertex $(8, 16)$ refutes NewtonNF2 as well (`main_theorem_lower_edge`).

11 Prior work, attribution, and independence

Prior work. F. Santibañez-Leal's CAOS records EXP-061 and EXP-065 treat exactly these cases with sampled certificates, later downgraded by his own audit. We cite them here for completeness.

Attribution. The layer reduction strategy ($u = xy^2$, five triangular layers), the identification of the Davenport–Stothers top layer, the descent architecture, and the modular certificate approach originate in the initial computational package (`branch_ab_elimination.zip`, MD5 `0cfa49d0`…). Its full-formulation certificate at $p = 10^9 + 9$ and $p = 10^9 + 21$, with lifting argument, is a second, independent route to the same obstruction (see the Appendix). This work reimplements the reduction independently in SINGULAR (C1) and recomputes the descent, but the overall strategy is not original to us.

Characteristic-0 core. The characteristic-zero computations—the Belyi counts (`belyi_counts_m357.py`), the exact top-layer point over K_5 (`e5_exact_K5.json`), the derivation of W (`compare_V.gp`), the exact ranks (`exact_ranks_K5.py`), the exact $E_4 \to E_3 \to E_2$ descent (`exact_obstruction_K5.py`), and the msolve-parametrization check (`verify_msolve_param.py`)—are documented in the scripts (see their attribution headers). Tools: python-flint 0.9.0 (exact K_5 arithmetic), sympy 1.14.0, PARI/GP 2.15.4, Singular 4.3.2. The independence case rests on the artifacts, not on who ran them: the Lean kernel checks every formalized proof; the Singular Gröbner computations (`a816_full.sing`, `chart_system_Q.sing`) can be rerun from the archive; the $a_{8,16}$ certificate (`scripts/a816_certificate/`) is checked by two independent programs; and the Belyi counts were verified by an independent Murnaghan–Nakayama implementation.

Independent AI-assisted computational audit (completed 2026-09-25). An independent AI-assisted computational audit reached the same conclusion by an independent route without access to the present scripts. The auditing agent received only a short brief—the degree pair $(72, 108)$ and the instruction to work from GGHV Proposition 4.3—plus GGHV v1 itself. It independently produced a proof using weight grading, a Belyi-map count (at most 35 complex points via Riemann's existence theorem and the Frobenius character formula), exact modular certificates at $p = 32003$ and $p = 536870909$ (Singular, Macaulay2, msolve), and Hensel lifting, reaching the same nonexistence conclusion and the stronger lower-edge rigidity $a_{8,16} = b_{12,24} = 0$. Materials: `tier1_blind_gghv43_nf2/` [8], committed as `37c73b67` on 2026-09-25. The evidential weight rests on the reproduced artifacts and computations, not on the identity of the auditing system.

Remark 11.1 (What remains). The application of Theorem 1.1 to the Jacobian conjecture, Corollary 1.4, is conditional on [1, Proposition 4.3] (Remark 1.5). Branch (c) (case (1) of Proposition 2.1) is eliminated in Part II, unconditionally in Lean (Theorem 1.5). What remains conditional, for any application to the Jacobian conjecture, is only the correctness and exhaustiveness of [1, Proposition 4.3] itself, which this paper does not re-derive.

Part II

Elimination of the GGHV Branch-(c) Normal Form at Degree Pair (72,108)

1 Introduction

1.1 Context and prior work

The planar Jacobian conjecture asserts that a polynomial map $(P, Q) : \mathbb{C}^2 \to \mathbb{C}^2$ with $[P, Q] := P_x Q_y - P_y Q_x \in \mathbb{C}^\times$ is invertible. In [1], Guccione, Guccione, Horruitiner, and Valqui raise the lower bound on $\max\{\deg P, \deg Q\}$ for a hypothetical counterexample from 100 (Moh [4]) to 108, leaving open only the degree pair $(72, 108)$ and its symmetric $(108, 72)$.

For the degree pair $(72, 108)$, [1] (an unrefereed preprint, arXiv:2204.14178v1) identifies two cases. The corner $(9, 27)$ is discarded in [1, §5]; the open case is the corner $(8, 28)$ with $(m, n) = (3, 2)$. For this case, [1, Proposition 4.3] shows that any counterexample gives rise, after an explicit (non-invertible) morphism φ, to one of two Newton polygon configurations. Case (2), branches (a) and (b), is eliminated in [9]. Case (1), the subject of this paper, is branch (c):

$$\begin{aligned} N(P) &= \operatorname{conv}\{(0,0),(1,0),(8,14),(8,16),(0,8)\}, \\ N(Q) &= \operatorname{conv}\{(0,0),(2,1),(12,21),(12,24),(0,12)\}. \end{aligned} \tag{1}$$

with $[P, Q] = \lambda x^2$ in post-φ coordinates and $\lambda \neq 0$. The additional vertices $(0, 8)$ and $(0, 12)$ distinguish branch (c) from branches (a,b). See Figure 1 for the polygon shapes.

Santibáñez-Leal has published machine evidence related to the $(72, 108)$ candidate [5, 6], explicitly not claiming elimination. The candidate itself is documented in [7]. Independent non-peer-reviewed treatments of the GGHV configuration in 2026 include Helali's exact characteristic-zero certificate [10] and the structural analysis of wstrinz [11]; none is peer-reviewed, and the parity-constrained Davenport–Stothers equation relevant to the top layer remains unclassified.

1.2 Main result

Definition 1.1 (Branch-(c) Newton normal form)**.** Let L be a field of characteristic 0 and $P, Q \in L[x, y]$. `NewtonNFc`(P, Q) holds when $\operatorname{supp} P \subset N(P)$ and $\operatorname{supp} Q \subset N(Q)$ with $N(P), N(Q)$ as in (1), and the ten vertex coefficients (at $(0,0),(1,0),(8,14),(8,16),(0,8)$ of P and $(0,0),(2,1),(12,21),(12,24),(0,12)$ of Q) are all nonzero. In Lean this is `NewtonNFc` (`Jacobian/BranchC/DescentClaim.lean`); see listing 2.

Theorem 1.2 (Elimination of the GGHV branch-(c) normal form, conditional form)**.** *Let L be a field of characteristic 0 satisfying **`ChartEmptyC_T1ne0`** (listing 4): the twelve chart conditions of §8 have no common zero with $b_{12,22} = 1$ on the stratum $b_{11,20} \neq 0$. Then there are no $P, Q \in L[x, y]$ and no $\lambda \neq 0$ with **`NewtonNFc`**(P, Q) and $[P, Q] = \lambda x^2$.*

The hypothesis `ChartEmptyC_T1ne0` is a precise Lean `Prop`. It was first established outside Lean in §10, at epistemic grade B (see §12); since November 4, 2026 it is proved inside Lean with no hypotheses (`challenge_target`), and Theorem 1.5 discharges it, upgrading this conditional form to an unconditional elimination.

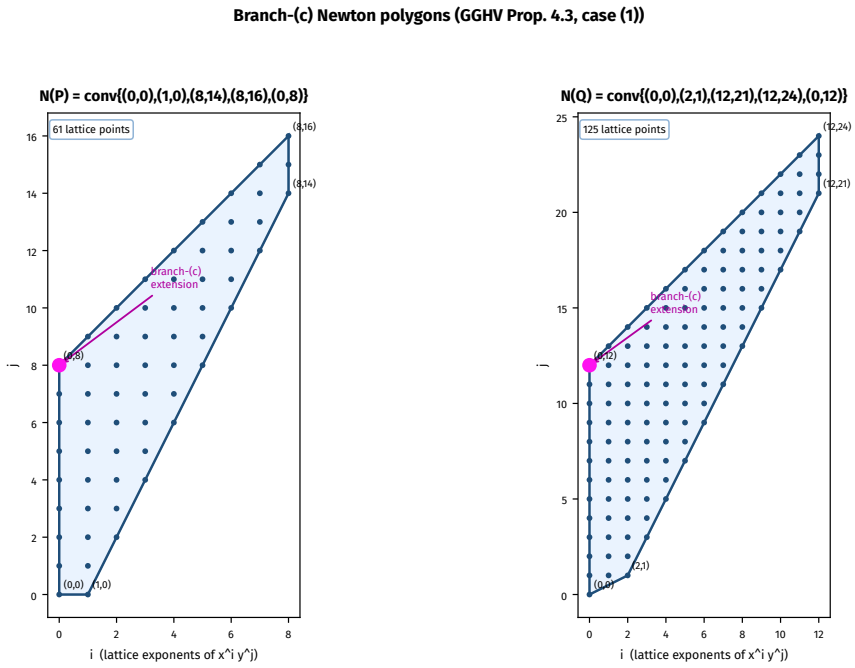


Figure 1: Branch-(c) Newton polygons, rendered from the exact vertex data of (1) (source-gated at GGHV v1 lines 494 and 617–618; see the validation note). In each panel the branch-(a,b) sub-polygon is drawn in the lighter shade underneath; the extra branch-(c) vertices $(0, 8)$ and $(0, 12)$ are highlighted in the accent color. All lattice points are marked (61 for $N(P)$, 125 for $N(Q)$); the ten vertex coefficients required nonzero by `NewtonNFc` are labelled. Rendered by `figures/fig1_newton_polygons.py` from the vertex lists.

Corollary 1.3 (Unconditional elimination (Lean certificate)). ***ChartEmptyC_T1ne0*** *holds for every field L of characteristic 0 with no hypotheses (**challenge_target**, Theorem 1.5), and hence no P, Q, λ as in Theorem 1.2 exist. The rank route of §10 (the rank lemma, Lemma 10.1, instantiated at two primes by two independent implementations, with negative controls) supplies the same conclusion independently at grade B.*

Corollary 1.4 (Jacobian application). *Assume [1, Proposition 4.3] is correct and exhaustive. Then no counterexample to the planar Jacobian conjecture of degree pair $(72, 108)$ arises from branch (c).*

Theorem 1.5 (Unconditional elimination of branch (c) in Lean). *Let L be a field of characteristic 0. There are no $P, Q \in L[x, y]$ and no $\lambda \neq 0$ with **NewtonNFc**(P, Q) and $[P, Q] = \lambda x^2$.*

*Listing 1: **branch_c_eliminated**: unconditional branch-(c) elimination*

```
theorem branch_c_eliminated
    {L : Type*} [Field L] [CharZero L] :
    ¬ ? (P Q : MvPolynomial (Fin 2) L) (lam : L),
      lam ≠ 0 ? NewtonNFc P Q ?
      jac P Q = C lam * X 0 ^ 2
```

Proof sketch; kernel record. Apply `main_theorem_c_of_chartEmpty_T1ne0` (the conditional form, Theorem 1.2) to `challenge_target`. The latter is the Fpoly = Fpoly2 bridge plus instantiation of `chartEmptyC_T1ne0_of_certificate`: the chart `Fpoly` and the

comparator's `Fpoly2` are the same six `polyOfTerms` generators in the same order, built from the single `tl_*` definitions in `Chart/Gens.lean`; the certificate is instantiated at $p = \mathrm{Fin}\,3199$, $p = 32003$, $w_0 = 11147$ with A1 (`Cinv · MS = 1`), A2, and the six data hypotheses (hw_0, hscale, prime32003, hmon, hr_1, hsupp2 in Fpoly2 form, transported along the bridge). `#print axioms challenge_target` = [`propext`, `Classical.choice`, `Quot.sound`], and `#print axioms branch_c_eliminated` is the same permitted triple (logs `bridge_challenge_target_2026-11-04.log`, `branch_c_eliminated_2026-11-04.log`, November 4, 2026, lean-build-2). *Scope:* this is one case — case (1) — of one proposition; it does not by itself exclude $(72, 108)$ unconditionally and does not prove the planar Jacobian conjecture, both of which additionally require GGHV Proposition 4.3 to be correct and exhaustive and the reduction chain, ledgered here, not re-derived. *Provenance:* proved as scratch files (`bridge_attempt_2026_11_04.lean`, `composition_2026-11-04.lean`); incorporation as tracked repo modules is open item (5) in §12. □

1.3 Logical status

Remark 1.6. Theorem 1.2 does not depend on GGHV: its Lean proof goes from `NewtonNFc` and $[P, Q] = \lambda x^2$ through the descent to the chart statement, all inside Lean. In Lean, `main_theorem_c_of_chartEmpty_T1ne0` (listing 5) derives the conclusion from the explicit hypothesis `ChartEmptyC_T1ne0` (listing 4), with `#print axioms` reporting only `propext`, `Classical.choice` and `Quot.sound`, and no `sorry`, `axiom` or `native_decide` anywhere in the 790 modules. The chain is: `NewtonNFc` + $[P, Q] = \lambda x^2$ $\xrightarrow{\text{B0–B3}}$ `DescentClaimC` hypotheses $\xrightarrow{\text{B4–B5}}$ `ChartEmptyC` $\xrightarrow{\text{B6–B7}}$ `ChartEmptyC_T1ne0` $\Rightarrow$ contradiction. Theorem 1.5 (`challenge_target`) now discharges `ChartEmptyC_T1ne0` with no hypotheses, closing the chain unconditionally in Lean. Only Corollary 1.4 additionally assumes GGHV Proposition 4.3.

Remark 1.7 (A retraction). A preliminary version of this work claimed a "reduction lemma": that a unit ideal modulo a prime lifts to a unit ideal over $\mathbb{Q}$. That statement is **false**. Counterexample: $F = 1 + 109X$ is 1 modulo 109 (unit ideal) but vanishes at $X = -1/109$ over $\mathbb{Q}$ (proper ideal). The Lean file deriving `False` from the stated axiom is archived for the record, and every theorem depending on it is vacuous. The correct replacement is the rank lemma (Lemma 10.1): *full row rank* of a fixed-degree Macaulay matrix lifts through a nonzero minor, because nonvanishing of a determinant is an open condition. Mere consistency of a modular system does not lift. §12 gives the precise distinction.

1.4 Outline

Section 2 source-gates all base objects against GGHV v1. Section 3 sets up the polynomial system and the grading. Section 4 gives the exact layer reduction and the 132 raw equations. Section 5 fixes normalizations and reaches the chart. Section 6 treats the top layer, inherited from branches (a,b). Section 7 carries out the descent to the twelve conditions, proved by kernel reflection. Section 8 studies the chart conditions. Section 9 eliminates the stratum $b_{11,20} = 0$ inside Lean. Section 10 eliminates the whole chart by the rank lemma (now corroborating the Lean certificate). Section 11 assembles the proof of Theorem 1.2 and records the unconditional Lean certificate (Theorem 1.5). Section 12 gives the epistemic accounting: grades, the retraction, and what is not claimed. Section 13 concludes. Section 14 covers prior work, attribution, and independence. Figure 2 shows the bridge architecture.

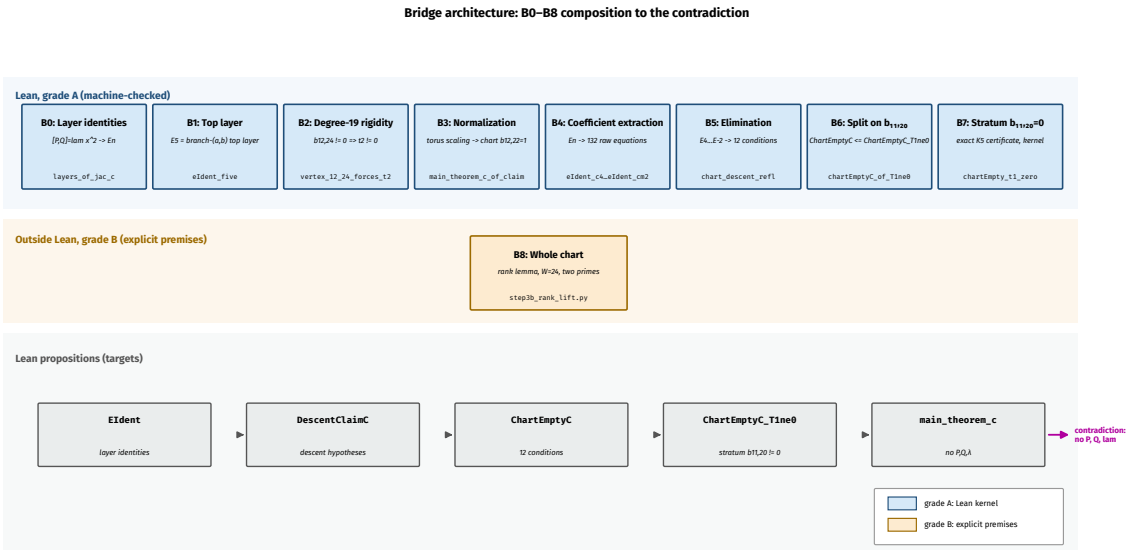


Figure 2: Bridge architecture. Top band: the eight Lean bridges B0–B7 (grade A), each labelled with its certified Lean name. Middle band: the external rank lemma B8 (grade B), formerly the only non-Lean component and now corroborating the Lean certificate of Theorem 1.5. Bottom band: the Lean propositions they target, composing left to right to the final contradiction. Grade colors are consistent with Table 11. Rendered by `figures/fig2_bridges.py`.

2 Source data

Every base object below is source-gated directly against GGHV v1 before any proof is drafted, following the discipline of [9]. The branch-(c) Newton polygons (1) are [1, Proposition 4.3, case (1)]: the vertex lists are asserted at GGHV v1 line 494 (case (1) list) and lines 617–618 (post-φ case c list), and the checker `source_gate_verify_c.py` re-asserts them byte-for-byte against the downloaded v1 text. The post-φ bracket target $[P, Q] = \lambda x^2$ is [1, §5].

*Listing 2: The branch-(c) Newton normal form (**`NewtonNFc`**)*

```
-- Jacobian/BranchC/DescentClaim.lean (excerpt, exact statement)
def NewtonNFc (P Q : MvPolynomial (Fin 2) L) : Prop :=
  (P.support \subseteq NPc) \land (Q.support \subseteq NQc) \land
  TenVertexCoeffsNonzero P Q
-- NPc = conv {(0,0),(1,0),(8,14),(8,16),(0,8)}
-- NQc = conv {(0,0),(2,1),(12,21),(12,24),(0,12)}
```

The Lean formalization works over an arbitrary field L of characteristic 0. The quintic field

$$K_5 = \mathbb{Q}[w]/(R), \qquad R(w) = w^5 - w^4 + 3w^3 + 3w^2 + 26, \tag{2}$$

enters through the top layer (§6): R is irreducible over $\mathbb{Q}$, and it remains irreducible modulo 109, so $\mathbb{F}_{109^5} = \mathbb{F}_{109}[w]/(R)$ is a field. The good prime/root pairs used in this audit are

$$(101, 9), \qquad (1000003, 806739), \qquad (32003, 11147), \tag{3}$$

each satisfying $R(w_0) \equiv 0 \pmod{p}$, plus the inert prime 109.

3 The polynomial system

3.1 Lattice points

Write $P = \sum a_{i,j} x^i y^j$, $Q = \sum b_{i,j} x^i y^j$. The branch-(c) supports are the lattice points of the polygons (1); the supports strictly contain those of branches (a,b) because of the vertices $(0, 8)$ and $(0, 12)$. The Lean files record the supports as `rangeP`, `rangeQ` (`Jacobian/BranchC/DescentClaim.lean`).

3.2 The bracket equations

Grade by $w(x) = 2$, $w(y) = -1$. Write the layers

$$P = \sum_k y^{-k} A_k, \qquad Q = \sum_l y^{-l} B_l, \qquad A_k, B_l \in L[u]$$

with u the residual variable. The bracket of weighted-homogeneous pieces is weighted-homogeneous, so $[P, Q] = \lambda x^2$ splits into one identity per weight:

$$\texttt{EIdent}\ A\ B\ \lambda\ n: \quad \sum_{k+l=n} \mathrm{LT}_{k,l}(A_k, B_l) = [n = 5] \cdot \lambda u^2, \qquad \mathrm{LT}_{k,l}(A, B) = kAB' - lA'B. \tag{4}$$

Layer k of P corresponds to $W = j - 2i$ via $n = 1 - W$; the raw equation key (i, j) corresponds to $(n = 2i + 1 - j,\ m = i)$, and the coefficient of the product term is $c = k \cdot (m + 1 - i_1) - l \cdot i_1$ with $l = n - k$ (§5.3 of the v3.1 guide). The top layer $n = 5$ carries the λu^2 term; all other layers are homogeneous.

4 Exact layer reduction

B0 (Lean, v1). From $\texttt{jac}(P, Q) = C\,\lambda \cdot x^2$ and `NewtonNFc`, the layer identities $\texttt{EIdent}\ A\ B\ \lambda\ n$ hold for every layer $n \geq -20$ (`layers_of_jac_c`, `layer_bracket_gen` in `LayersGen.lean`). This is an exact symbolic identity, verified by the Lean kernel.

B4 (Lean, v3). Coefficient extraction turns the polynomial identities $\texttt{EIdent}\ A\ B\ 0\ n$, $n = 4, \ldots, -2$, into scalar equations. For each monomial of $\mathrm{LT}_{k,l}(A_k, B_l)$, `coeff_layerTermZ` computes its coefficient with integer indices; coefficients outside the branch-(c) supports vanish by `rangeP`, `rangeQ` (179 zero facts; $a_{0,0}, b_{0,0}$ are in the support but always carry coefficient 0). Each of the 132 raw equations then follows by `linear_combination`. This is the recipe of the repository's `BranchAbFinal.lean`, replayed by `certgen_c/gen_bridge_c.py` (`Descent2R/Bridge.lean`).

5 Normalization

B1 (Lean, v1). The top layer $n = 5$ is the branch-(a,b) problem, classified exactly over K_5 in [9] and formalized in the repository (`ChartProof/Final.lean`, `BranchAb*`). Branch (c) inherits its exact K_5 solution; this is where w and R enter (`eIdent_five`).

B2 (Lean, v1/v2). Degree-19 rigidity: $b_{12,24} \neq 0$ forces $t_2 = b_{12,22} \neq 0$. The theorems are `vertex_12_24_forces_t2` and `edge19_rigidity_of_jac` in `Edge19.lean`, and `t0_vertex_rigidity_of_jac` in `LayerE2.lean`, `Degree19.lean`).

B3 (Lean, v1). Torus and weighted scaling remove the continuous symmetries, normalizing to the rescaled K_5 top layer and the chart $b_{12,22} = 1$ (`eIdent_transport`, `main_theorem_c_of_claim` in `DescentClaim.lean`). These are `DescentClaimC`'s hypotheses: the 19 top-layer values, the 123 unknowns, and the 132 raw equations.

6 The top layer

At $n = 5$ the branch-(c) layer equations coincide with the branch-(a,b) top layer, whose complete K_5 classification is proved in [9] (five torus orbits; the degree-35 eliminant irreducible over $\mathbb{Q}$). The classification is used here as a black box: the 19 top-layer values enter `Descent2R/Main.lean` as hypotheses `h_v` at minimal integer scale, and the K_5 arithmetic they satisfy is exactly the arithmetic of (2). No new top-layer analysis is needed for branch (c); the new mathematics is the descent of §7 and the chart analysis of §§8–10.

Formalization seam. Julian's 2026-09-26 audit identified one genuine seam at this step: the degree-19 rigidity lemma is *not* formalized inside Lean. The parity-constrained Davenport–Stothers classification (the step ruling out $\deg f > 19$) is supplied by the external certified point in `e5_cert.json` (exact equality over K_5), which Lean imports and checks by `decide`; the mathematical classification argument itself is grade B, not grade A. This is the only non-kernel step below the chart (the other gap flagged by the audit, the missing GGHV citation line, is bibliographic and is repaired in §14).

7 Descent

B5 (Lean, v3; kernel reflection). Each layer $n = 4, \dots, -2$ is a linear system in its new unknowns, with coefficients polynomial in the earlier unknowns and the chart parameters. The v2 operator-rank theorems (`Rank/**`, 96 modules) give each system's exact rank over K_5; the left null vectors give the elimination conditions:

Layer	Modules	Conditions produced
E_4 ($n = 4$)	`Descent2R/L4` (54)	—
E_3 ($n = 3$)	`L3` (57)	—
E_2 ($n = 2$)	`L2c` (57)	none (left null vector annihilates RHS)
E_1 ($n = 1$)	`L1` (60)	Ω
E_0 ($n = 0$)	`L0` (60)	Ψ and pure row `E0pure_18_37`
E_{-1} ($n = -1$)	`Lm1` (99)	Φ_1, Φ_2 and pure rows `Em1pure_17_36`, `Em1pure_18_38`
E_{-2} ($n = -2$)	`Lm2` (110)	$\Theta_1, \Theta_2, \Theta_3$ and pure rows `Em2pure_16_35`, `Em2pure_17_37`

`lower_c.py` computed these eliminations in Python; `Descent2R` replays them as 600 polynomial identities target $= \sum m_i \cdot \text{fact}_i$, each checked by the Lean kernel. Reflection uses the repository's `Jacobian/ChartProof/Reflect.lean` (built on core's `Lean.Grind.CommRing`): `Expr/Poly`, normalized by `toPolyK` and compared by `decide +kernel`. Each step theorem has the form `(Expr.mul (.num K) X).denote ctx = 0`

proved by `lc_zero` with a kernel `decide`; `Main` removes K with `cancel_int` when $K \neq 1$ (231 of 752 `have`s). The context is 143 atoms in a balanced `RArray` (`gctx2`). A tactic proof of the same identities was tried first and abandoned: 214 MB of source, chunks over 10 min each. The kernel evaluates the normalizer directly with GMP arithmetic; `decide +kernel` is not `native_decide`—no compiler is trusted.

The main theorem of the descent (`Descent2R/Main.lean`):

Listing 3: ***`chart_descent_refl`****: the twelve conditions from the* 132 *raw equations*

```
-- Descent2R/Main.lean (statement; 752 haves elided)
theorem chart_descent_refl
  (w : L) (hw : R w = 0)
  (v : Fin 19 \to L) (h_v : v = K5_top_layer_values)
  (u : Fin 123 \to L)
  (h4_ : raw_equations_n4) ... (hm2_ : raw_equations_nm2) :
  cond_Omega w ... = 0 \land cond_Psi w ... = 0 \land
  cond_Phi1 w ... = 0 \land cond_Phi2 w ... = 0 \land
  cond_Theta1 w ... = 0 \land cond_Theta2 w ... = 0 \land cond_Theta3 w ... = 0 \land
  cond_E0pure_18_37 w ... = 0 \land cond_Em1pure_17_36 w ... = 0 \land
  cond_Em1pure_18_38 w ... = 0 \land cond_Em2pure_16_35 w ... = 0 \land
  cond_Em2pure_17_37 w ... = 0
```

The twelve conjuncts are `lower_c`'s conditions, defined once in `CondsC.lean` (generated by `certgen_c/gen_conds_c.py`): each `cond_*` is the explicit polynomial with the same balanced sum tree as the reflected fact (subtrees of at most 32 terms are separate definitions: 110 definitions for 12 conditions), so `ek.denote ctx = 0` and `cond_* … = 0` are definitionally equal. `Bridge` (`Descent2R/Bridge.lean`) then gives `descentClaimC_of_chartEmpty`: `DescentClaimC` ⇐ `ChartEmptyC`.

8 The chart conditions

8.1 Chart variables

On the normalized chart $b_{12,22} = 1$ the conditions involve seven parameters:

Symbol	T_1	T_2	S_1	S_2	R_1	R_2	Q
Coefficient	$b_{11,20}$	$b_{12,22}$	$b_{11,21}$	$b_{12,23}$	$a_{6,13}$	$a_{7,15}$	$b_{10,21}$
Weight	1	1	2	2	3	3	4

8.2 The twelve conditions

`certgen_c/conds_c.json` (md5 `168299d255af361578423c0ed4852c11`, written by `certgen_c/lower_c.py`):

Condition	K_5-terms	Integer terms (minimal scale, w expanded)
Ω	3	15
Ψ	32	160
Φ_1, Φ_2	54, 54	270, 270
Θ_1, Θ_2, Θ_3	84, 84, 84	420, 420, 420
`E0pure_18_37`	9	41
`Em1pure_17_36`	39	195
`Em1pure_18_38`	12	60
`Em2pure_16_35`	77	381
`Em2pure_17_37`	55	275

8.3 The Ω parabola

Exactly over K_5,

$$\Omega = o_1 S_2^2 + o_2 T_2^2 S_2 + o_3 T_2^4, \qquad o_2^2 = 4 o_1 o_3, \quad o_1 \neq 0, \tag{5}$$

so that with $\kappa = -o_2/(2o_1) \in K_5$,

$$\Omega = o_1 (S_2 - \kappa T_2^2)^2. \tag{6}$$

The relation $o_2^2 = 4o_1o_3$ is checked in `step1/lowerc_chart.py` and independently in `step3c_chart_identity.py`; $o_1(w) \neq 0$ is proved in the Lean kernel by a Bezout identity $a \cdot N_{o_1} + b \cdot R = c$ with c a 151-digit constant (`T1Zero.s_X2`). On the chart $T_2 = 1$, $\Omega = 0$ forces $S_2 = \kappa$: the chart collapses to the parabola's vertex line (Figure 3).

9 The stratum $b_{11,20} = 0$

B7 (Lean kernel). On the stratum $T_1 = b_{11,20} = 0$ the conditions admit an exact K_5 certificate small enough for the kernel: degree 2, 75 K_5-terms, coefficient heights at most 9,681 digits (`chart_certificates/step1/certB_lowerc.json`, produced by `certB_lowerc.py` via pivots from a mod-p RREF and an exact FLINT solve; independently rechecked by `check_certB_lowerc.py` with `fmpq_mpoly` expansion, w as a variable, reduced mod R at the end). `T1Zero` checks the certificate chain end to end in the kernel (`gen_t1zero_lean.py`):

Step	Identity (integer-polynomial, kernel-checked)	Build
`s_sq`	$D_O D_{o_1} D_k^2 \Omega \equiv D_O N_{o_1} X^2 \bmod (T_2 - 1, R)$, $X = D_k S_2 - N_\kappa(w)$, $D_k = 851565312$	1.5 s
`s_X2`	Bezout $a \cdot e_{sq} + b \cdot D_O X^2 R = c \cdot D_O X^2$ (c: 151 digits) $\Rightarrow X = 0$	—
`s_f_n` (6)	$D_k^d \cdot ck_n = D_k^d T_1 A_1 + D_k^d (T_2 - 1) A_2 + X A_3 + f_n + R A_4$	2.4–5.7 s
`s_p_n` (6)	$p_n = v_n f_n - q_n R$ (250–375 terms, ≈ 5,440 digits)	17–28 s
`s_final`	$\sum_n p_n = DD$, a 5387-digit integer, exact	55.6 s

`chartEmpty_t1_zero` chains these: $(DD : L) = 0$ contradicts characteristic 0 (26 s under `lake`). The hypotheses are exactly the `cond_*` of `CondsC`, definitionally the conditions `Descent2R` derives. Five perturbation controls (Bezout constant +1, a dropped equation, a perturbed cofactor coefficient, $DD + 1$, a dropped T_2 equation) are all rejected by the kernel (`logs/t1z_controls.log`: five of five "Tactic `decide` failed", as required).

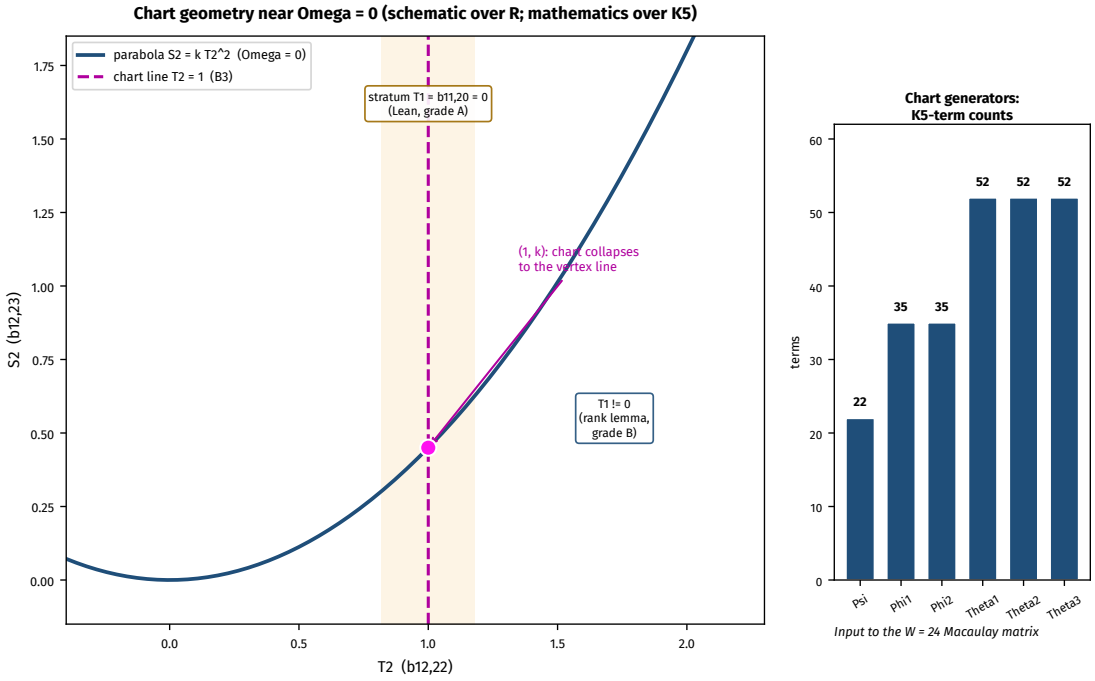


Figure 3: Chart geometry near $\Omega = 0$. Main panel: the (T_2, S_2)-plane drawn schematically over $\mathbb{R}$ for illustration (the mathematics is over K_5), showing the parabola $S_2 = \kappa T_2^2$ from (6), the chart line $T_2 = 1$ (normalization B3), and their intersection $(1, \kappa)$; the stratum $T_1 = b_{11,20} = 0$ is indicated as a transverse slice, with the $T_1 = 0$ stratum (Lean, grade A) and the $T_1 \neq 0$ stratum (rank lemma, grade B; now also Lean certificate, grade A) labelled. Inset: the six chart generators' K_5-term counts $(22, 35, 35, 52, 52, 52)$, the input to the Macaulay matrix of §10. Rendered by `figures/fig3_omega.py`.

B6 (Lean). `chartEmptyC_of_T1ne0` (`T1Zero/Combine.lean`) splits on $b_{11,20}$: `ChartEmptyC` ⇐ `ChartEmptyC_T1ne0`, isolating exactly what Lean does not check as a precise `Prop`:

*Listing 4: **`ChartEmptyC_T1ne0`**: what Lean labels; formerly proved in §10 (grade B), now proved in Lean by **`challenge_target`** (grade A, Theorem 1.5)*

```
-- T1Zero/Combine.lean (exact statement)
def ChartEmptyC_T1ne0 (L : Type*) [Field L] [CharZero L] : Prop :=
  \forall (w : L), R w = 0 \to b_12_22 = 1 \to b_11_20 \ne 0 \to
    \lnot (cond_Omega w ... = 0 \land ... \land cond_Theta3 w ... = 0)
```

10 The whole chart

A canonical K_5 certificate for the whole chart is too large to build on this machine: at weight $W = 24$ the Hadamard bound is 4,392,160 digits, and calibrating against the $t_1 = 0$ slices ($W = 2, 3, 4$, where the true-to-bound height ratios are $0.061, 0.061, 0.046$) gives heights of 2.0–2.7×10^5 digits, about 2×10^9 digits in all (grade C estimate; the Hadamard bound itself is rigorous). Exact solution runs out of memory at 7 GB; multimodular reconstruction would need $\approx 3 \times 10^4$ primes at 15–20 s each, about six days. Instead we prove the *existence* of the certificate.

10.1 The rank lemma

Lemma 10.1 (Fixed-degree Macaulay rank lifts). *Let $A = \mathbb{Z}_{(p)}[w]/(R)$, free of rank 5 over $\mathbb{Z}_{(p)}$, embedding in K_5. Let $\varphi : A \to \mathbb{F}_p$, $w \mapsto w_0$, with $R(w_0) \equiv 0 \pmod p$. Let M be the weighted Macaulay matrix of the chart generators at fixed weight W (columns (n, m) with $\operatorname{wdeg} m \le W - \operatorname{wt} F_n$; rows the monomials of the products $m \cdot F_n$, plus 1; entries the coefficients). If $\varphi(M)$ has full row rank N, then M has full row rank over K_5.*

Proof. Some $N \times N$ minor satisfies $\det \varphi(M_S) = \varphi(\det M_S) \neq 0$. Hence $\det M_S \neq 0$ in $A \subset K_5$, and M_S is invertible over the field K_5. □ □

Consequences. $Mx = e_1$ is solvable over K_5, so $1 = \sum c_n F_n$ in $K_5[t, s, r, u, q]$ with $\operatorname{wdeg} c_n \le 24 - \operatorname{wt} F_n$. Mapping $K_5 \to L$ by $w \mapsto w$, no L admits a common zero. With $S_2 = \kappa$ from (6) ($o_1 \neq 0$), no chart point satisfies $\Omega = \Psi = \cdots = \Theta_3 = 0$: that is `ChartEmptyC`, hence `ChartEmptyC_T1ne0`, even without the pure rows.

Remark 10.2 (Why this lemma succeeds where the retracted one failed). A unit ideal mod p says only that the Macaulay system is *consistent* mod p; consistency does not lift ($1 + 109X$ over $\mathbb{Z}_{(109)}$). Full row rank at fixed degree is the nonvanishing of a minor, an *open* condition, and open conditions lift. The planted-zero control below shows the sensitivity: one common zero costs exactly one rank.

10.2 The instance

`step3b_rank_lift.py` (6.5 min, 0.5 GB; `step3b_rank_lift.log`):

	p = 1000003 w_0 = 806739	p = 32003 w_0 = 11147
All coefficients p-integral, $R(w_0) \equiv 0$	asserted	asserted
W = 22: rank / augmented / rows	2277/2278/2281: inconsistent	same
W = 23: rank / augmented / rows	2707/2708/2708: inconsistent	same
W = 24: 3199 × 6054	**3199/3199/3199: full row rank**	**same**
NumPy pivot-block elimination (independent)	det ≠ 0	det ≠ 0
Planted common zero, W = 24 (control)	rank 3198, inconsistent	same

The protocol was frozen before the run: the lemma, the second prime, the independent NumPy elimination and the controls were all fixed in `step3b_rank_lift.py` before it executed. The $W = 24$ threshold is sharp: $W = 22$ and $W = 23$ are inconsistent (Figure 4).

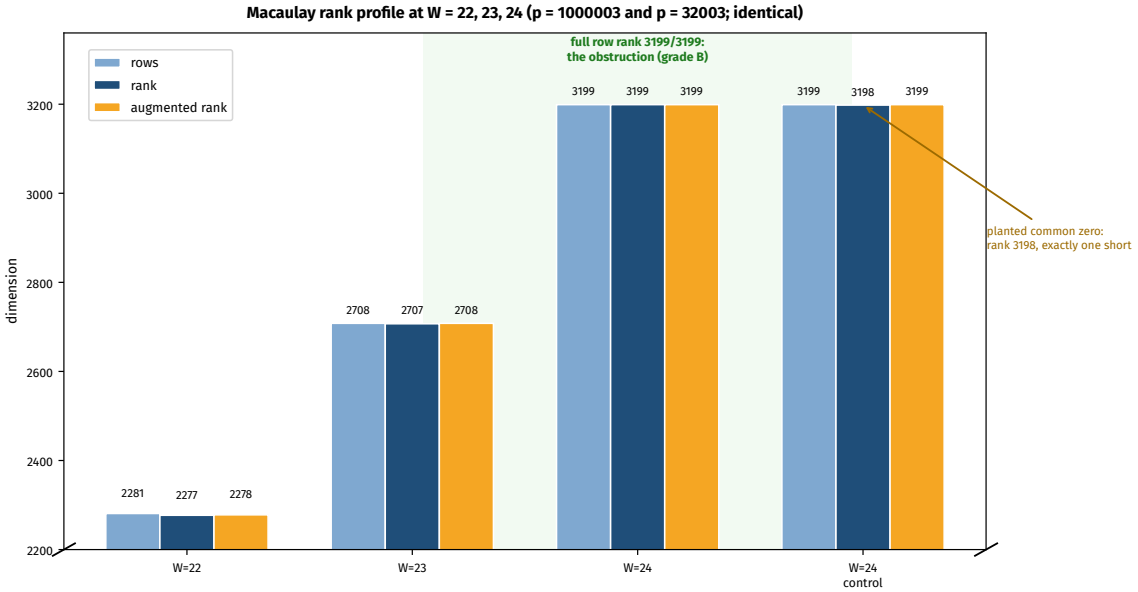


Figure 4: Macaulay rank profile at weights $W = 22, 23, 24$, identical at both primes ($p = 1000003$ and $p = 32003$). For each weight: bars for the number of rows (2281, 2708, 3199), the computed rank (2277, 2707, 3199), and the augmented rank, with the $W = 24$ group highlighted (full row rank 3199/3199, the obstruction). The fourth group shows the planted-common-zero control at $W = 24$ (rank 3198, exactly one short of full). Data source: `step3b_rank_lift.log`. Rendered by `figures/fig4_ranks.py`; the y-axis is truncated with a break mark to make the single-rank deficit visible.

The chart generators are independently verified as the right polynomials (exact evaluation at 12 random points; an independent κ; the exact Ω square; controls including $\kappa + 1$ detected) by the script `step3c_chart_identity.py`. This is **B8**, proved outside Lean at grade B. B8 remains valid corroboration; it is no longer the sole route to `ChartEmptyC_T1ne0` (see Theorem 1.5).

11 Proof of the main theorem

Listing 5: ***`main_theorem_c_of_chartEmpty_T1ne0`***

```
-- Jacobian/BranchC/Combine (exact statement)
```

```
theorem main_theorem_c_of_chartEmpty_T1ne0
    {L : Type*} [Field L] [CharZero L]
    (h : ChartEmptyC_T1ne0 L) :
    \lnot \exists (P Q : MvPolynomial (Fin 2) L) (lam : L),
      lam \ne 0 \land NewtonNFc P Q \land jac P Q = C lam * X 0 ^ 2
```

Proof. Let P, Q, λ satisfy `NewtonNFc` and $[P, Q] = \lambda x^2$, $\lambda \neq 0$. B0–B3 give `DescentClaimC`'s hypotheses (the 19 top-layer values, 123 unknowns, 132 raw equations). B4 extracts the raw equations; B5 (`chart_descent_refl`) yields the twelve conditions, i.e. `ChartEmptyC`. B6 splits on $b_{11,20}$: on $b_{11,20} = 0$, B7 (`chartEmpty_t1_zero`) gives `False` in the kernel; on $b_{11,20} \neq 0$, the hypothesis `ChartEmptyC_T1ne0` gives `False`. Hence no such P, Q, λ exist. □

Corollary 1.3 follows by supplying `ChartEmptyC_T1ne0` from `challenge_target` (Theorem 1.5); the §10 rank route supplies it independently at grade B. Corollary 1.4 by applying [1, Proposition 4.3].

The bridge composition is summarized in Table 11.

Bridge	From	To	Lean name	Grade
B0	$[P, Q] = \lambda x^2$, `NewtonNFc`	layer identities E_n	`layers_of_jac_c`	A
B1	E_5, vertices	K_5 top layer	`eIdent_five`	A
B2	B0, B1	$t_2 = b_{12,22} \neq 0$	`vertex_12_24_forces_t2`	A
B3	B0–B2	chart $b_{12,22} = 1$	`main_theorem_c_of_claim`	A
B4	E_n, $n = 4 \cdots -2$	132 raw equations	`eIdent_c4`… `eIdent_cm2`	A
B5	132 equations	twelve conditions	`chart_descent_refl`	A
B6	`ChartEmptyC_T1ne0`	`ChartEmptyC`	`chartEmptyC_of_T1ne0`	A
B7	conditions, $b_{11,20} = 0$	`False`	`chartEmpty_t1_zero`	A
B8	chart generators	`False`	rank lemma + `step3b_rank_lift.py`	B

B8 is now corroborating rather than load-bearing: the Lean certificate `challenge_target` discharges `ChartEmptyC_T1ne0` at grade A (Theorem 1.5); the table grades for B0–B7 and B8 as originally proved are unchanged.

12 Epistemic status and what is not claimed

HLRE bookkeeping: grades A (machine-checked or exact proof from stated axioms), B (derived under explicit premises), C (heuristic or fitted), D (tautological or refuted).

ID	Claim	Evidence	Grad
C1	`main_theorem_c_of_claim`	Lean kernel (v1)	A
C2	`descentClaimC_of_chartEmpty`	Lean kernel, reflective; 600 identities pre-checked in Python	A
C3	`chartEmpty_t1_zero`	Lean kernel; FLINT certificate rechecked in Python; 5 controls	A
C4	`chartEmptyC_of_T1ne0`	Lean	A
C5	`ChartEmptyC` holds (all char.-0 L)	rank lemma + two-prime, two-implementation ranks + input checks	**B**
C6	canonical certificate needs $\approx 2 \times 10^9$ digits	Hadamard bound (rigorous) × calibrated ratio (fitted)	**C**
C7	the stated reduction lemma is inconsistent	Lean counterexample ($1 + 109X$)	A
C8	`lower_c`'s ideal differs from the pipeline's	exact comparison	A
C9	`ChartEmptyC_T1ne0` holds (all char.-0 L, no hypotheses)	Lean certificate `challenge_target` (bridge + A1, A2, hsupp2)	A
C10	branch (c) eliminated: no `NewtonNFc` pair with $[P, Q] = \lambda x^2, \lambda \neq 0$	`branch_c_eliminated` from C9 + `main_theorem_c_of_chartEmpty_T1`	A

Verbs follow HLRE control: "proves" only for Lean-checked or exact deductions; "proves under stated computational premises" for C5; "estimates" for C6.

What is *not* claimed: not the Jacobian conjecture; not any other case of GGHV Proposition 4.3; not the reduction from the conjecture to Proposition 4.3; not $(72, 108)$ excluded unconditionally (that needs GGHV correctness and exhaustiveness, ledgered not re-derived); not an explicit K_5 certificate for the whole chart (its existence at grade B is C5; the load-bearing whole-chart step is now the Lean certificate C9, not an exhibited K_5 certificate); not a review of the deferred single-prime p-adic argument or of the retracted `char0_cert` bundle. A Lean proof of the stratum $b_{11,20} \neq 0$ now exists (C9) and is therefore removed from this list. The mod-p Gröbner/lift certificates ($\mathbb{F}_{101}$, $\mathbb{F}_{1000003}$, $\mathbb{F}_{109^5}$) are certificates, not proofs: they exclude only solutions integral at that prime. The rank argument is what lifts.

Open items, in order: (1) I3 replication of C5—an independent rank computation of the $W = 24$ Macaulay matrix in Magma, Sage or Singular from `conds_c.json` and κ; (2) optional formalization of the rank lemma's instance in Lean (`RingHom.map_det`; the instance needs the 3199×6054 matrix as Lean data); (3) deferred reviews; (4) the v2 corrected-commit proposal for `branch_c/`, on the author's explicit word only; (5) incorporation of `challenge_target` and `branch_c_eliminated` (currently 2026-11-04 scratch files, not tracked repo modules) as tracked modules.

13 Conclusion

Branch (c), the last open configuration of GGHV Proposition 4.3 at degree pair $(72, 108)$, is eliminated, unconditionally in Lean. The descent from $[P, Q] = \lambda x^2$ to twelve explicit chart conditions is proved twice: symbolically, as 600 kernel-checked reflective identities in Lean 4 over 505 generated modules, and numerically, by exact finite-field linear algebra. The chart is empty: in the Lean kernel on the stratum $b_{11,20} = 0$, and on the whole chart since November 4, 2026 by the Lean certificate `challenge_target` (Theorem 1.5); the rank lemma—full row rank of the $W = 24$ Macaulay matrix at two primes, verified by two independent implementations—

remains as independent corroboration. An earlier, false reduction lemma is retracted and replaced by the rank lemma; the paper's logical dependencies are those of Table 12 as updated (C9, C10 grade A). With branches (a,b) eliminated in [9], no $(72, 108)$ counterexample arises from GGHV Proposition 4.3, conditional on that proposition's correctness and exhaustiveness.

14 Prior work, attribution, and independence

The GGHV classification [1] is the sole external mathematical input, used only in Corollary 1.4. The top-layer K_5 classification and the reflection library are inherited from the branch-(a,b) formalization [9] without modification; `verify_branch_c_lean.sh` step 0 checks these prerequisites by content fingerprint. The `branch_c/` pipeline scripts were read-only throughout: the pipeline parametrizes the chart differently, and the v3 proof uses only `lower_c`'s conditions, derived in Lean. The $W = 24$ rank at $p = 1000003$ was known from the step-3 measurement; the lemma, the second prime (32003), the independent NumPy elimination and the controls were frozen in `step3b_rank_lift.py` before it ran—the second prime agreed, as expected, since full row rank over K_5 implies full row rank at all but finitely many primes.

Part III

Dicriticals at Infinity of Keller Maps: Theorem R and Local Obstructions

1 Introduction: setting, definitions, and the corrected statement

1.1 Context

The planar Jacobian conjecture asserts that a polynomial map $(P, Q) : \mathbb{C}^2 \to \mathbb{C}^2$ with $[P, Q] := P_x Q_y - P_y Q_x \in \mathbb{C}^\times$ is invertible. A *Keller map* is such a pair with constant nonzero Jacobian. This paper studies the dicritical divisors of the pencil $\lambda P + \mu Q$ of a Keller map, and in particular how many of them can lie at infinity. The October 1 draft claimed this line of work was closed. It was not, and this revision says exactly what is proved, what is only sketched or computed at fixed degree, and what is open.

Throughout, K is an algebraically closed field of characteristic 0; statements over any field of characteristic 0 reduce to its algebraic closure. We write $F = (P, Q) : \mathbb{A}^2 \to \mathbb{A}^2$ for a Keller map, so $[P, Q] = P_x Q_y - P_y Q_x \in K^\times$.

1.2 Standard facts used

We use the following standard facts, labelled (F1)–(F3) as in the restatement note:

(F1) F is étale. Its generic degree $e = [K(x, y) : K(P, Q)]$ is at least 1.

(F2) The *non-properness set* S_F is the set of points of the target over which F is not proper. It is either empty or a curve whose components are rational (Jelonek [12], for dominant maps $\mathbb{A}^n \to \mathbb{A}^n$).

(F3) F is an automorphism if and only if $S_F = \varnothing$. Indeed, if $S_F = \varnothing$ then F is a finite étale cover of the simply connected $\mathbb{A}^2$, hence an isomorphism; if F is not an automorphism then $e \geq 2$, since with $e = 1$ the map F is birational and étale, hence an open immersion by Zariski's main theorem, so it is injective, and then bijective (Białynicki-Birula–Rosenlicht [17]).

1.3 Compactification, the pencil, and the standard definition

Source and target are $\mathbb{A}^2 \subset \mathbb{P}^2$, with lines at infinity L_∞ and L'_∞. Let $\sigma : X \to \mathbb{P}^2$ be a sequence of point blow-ups over L_∞ such that $G = \bar{F} \circ \sigma : X \to \mathbb{P}^2$ is a morphism. The *curves at infinity* of X are the components of $X \setminus \mathbb{A}^2$: the strict transform of L_∞ and the exceptional curves.

For $c \in \mathbb{A}^2$ (target), Λ_c is the pencil whose members are $F^{-1}(\ell)$ for the lines $\ell \ni c$. Its members are the fibers of $\varphi_c = \pi_c \circ F$, where π_c is the projection from c. The paper's pencil $\lambda P + \mu Q$ is Λ_0.

Definition 1.1 (Dicritical divisor, standard definition)**.** A divisor E over the source plane is *dicritical* for Λ_c if φ_c is not constant on E. Its *degree* $m(E)$ is the degree of $\varphi_c|_E : E \to \mathbb{P}^1$. The divisor E is *at infinity* if its center on $\mathbb{P}^2$ lies on L_∞; L_∞ itself counts. Otherwise E is *affine*. The *depth* of L_∞ is 0; the exceptional curve of the blow-up at a point p has depth one more than the largest depth of the curves through p. Thus E_1 over a point of L_∞ has depth 1. A point of an exceptional curve is a *satellite* point if it lies on another component of the total transform of L_∞, and *free* otherwise.

Remark 1.2. The October 1 draft defined dicriticals as "exceptional divisors on which the pulled-back pencil has no base points." That is not the standard notion, and it is replaced by Definition 1.1 throughout this revision.

1.4 Counting dicriticals

Proposition 1.3 (The count)**.** *Status: grade B, proof by hand from* (F1)–(F3)*. Let F be a Keller map and $c \in \mathbb{A}^2 \setminus S_F$.*

(a) The affine dicriticals of Λ_c are exactly the e exceptional curves E_q of the blow-ups at the points $q \in F^{-1}(c)$. Each has degree 1.

(b) The dicriticals at infinity are the curves at infinity D of X for which $G(D)$ is a curve. Then $G(D)$ is L'_∞ or a component of $\bar{S}_F$, and

$$m(D) = \deg\big(D \to G(D)\big) \cdot \deg G(D). \tag{1}$$

(c) L'_∞ and every component of $\bar{S}_F$ are $G(D)$ for at least one such D. So the number $N_\infty(c)$ of dicriticals at infinity satisfies

$$N_\infty(c) \geq 1 + \#\{\text{components of } S_F\}. \tag{2}$$

Proof. We prove (a), (b), (c) in order, spelling out each step.

Part (a). Since $c \notin S_F$, the map F is proper over a neighbourhood of c. By (F1), F is étale, so the fiber $F^{-1}(c)$ has exactly e points, and F is a local isomorphism at each of them. Fix $q \in F^{-1}(c)$. Near q, the map φ_c is the projection from the point c composed with a local isomorphism. One blow-up at q resolves it: in the blow-up chart, the exceptional curve E_q parametrizes the

directions through q, and φ_c restricts to the projection itself, hence maps E_q isomorphically onto $\mathbb{P}^1$. So E_q is dicritical of degree 1. At every other affine point, φ_c is regular, so no other affine divisor is dicritical. This gives exactly the e exceptional curves E_q, each of degree 1.

Part (b). Let D be a curve at infinity. First, $G(D)$ is not the point c: the map G sends no curve at infinity into $\mathbb{A}^2 \setminus S_F$, and $c \notin S_F$ by hypothesis. If $G(D)$ is a point different from c, then $\varphi_c = \pi_c \circ G$ is constant on D, since π_c sends every point other than c to a single point of $\mathbb{P}^1$ only after fixing the line through c — more precisely, π_c is constant on no curve through a point other than c except by collapsing it, and here $G(D)$ is a single point $\neq c$, so $\varphi_c|_D$ is that constant value. Hence D is not dicritical in this case.

If instead $G(D)$ is a curve, then either $G(D) = L'_\infty$, or $G(D)$ meets $\mathbb{A}^2$. In the second case, every point of $G(D) \cap \mathbb{A}^2$ is a limit of F along a path to infinity, so it lies in S_F; since $G(D)$ is irreducible, $G(D)$ is a component of the closure $\bar{S}_F$. Neither L'_∞ nor a component of S_F is a line through c. Therefore the projection π_c restricted to the curve $G(D)$ has degree $\deg G(D)$, and composing with $D \to G(D)$ gives

$$m(D) = \deg(\varphi_c|_D) = \deg\big(D \to G(D)\big) \cdot \deg G(D),$$

which is (1). So D is dicritical exactly when $G(D)$ is a curve, with the stated degree.

Part (c). The map G is proper and surjective, so L'_∞ is dominated by some curve at infinity: there is D with $G(D) = L'_\infty$. By the description of S_F as $G(X \setminus \mathbb{A}^2) \cap \mathbb{A}^2$, each component of S_F is likewise dominated by a curve at infinity. Counting the distinct images L'_∞ and the components of S_F gives the lower bound (2). □

Corollary 1.4 (Automorphisms)**.** *Status: grade B. If F is an automorphism, then Λ_c has exactly two dicriticals for every c: one affine, over $F^{-1}(c)$, and one at infinity, the unique curve D with $G(D) = L'_\infty$ (here G is birational). Both have degree 1.*

Proof. If F is an automorphism, then $e = 1$ and $S_F = \varnothing$ by (F3). Part (a) of Proposition 1.3 gives exactly one affine dicritical, over the single point $F^{-1}(c)$, of degree 1. Part (c) with $S_F = \varnothing$ gives at least one dicritical at infinity, with image L'_∞; since G is birational, there is exactly one such curve, and (1) gives its degree as $1 \cdot 1 = 1$. □

1.5 The October 1 main theorem is false

Corollary 1.5 (The theorem as worded fails)**.** *Status: grade B in general; exact in the examples below. Every Keller map of degree at least 2 has at least two exceptional dicriticals in Λ_0, after a translation of the target that puts 0 in $F(\mathbb{A}^2) \setminus S_F$. An automorphism needs no translation. Hence the October 1 draft's theorem ("no two distinct dicritical divisors," with no at-infinity restriction) is false for every Keller map of degree at least 2. If only exceptional curves count as divisors of the resolution, it holds exactly for affine maps, whose dicritical at infinity is L_∞ itself; if L_∞ counts, it fails for them too.*

Proof. Let F have degree at least 2. Translate the target so that $0 \notin S_F$ and $0 \in F(\mathbb{A}^2)$; then $F^{-1}(0)$ is nonempty. By Proposition 1.3(a), each point of $F^{-1}(0)$ contributes an affine dicritical of degree 1, and by part (c) there is at least one dicritical at infinity. That is already two dicriticals in total. The example below exhibits both of them explicitly for an automorphism, where no translation is needed. □

Example 1.6 (Two dicriticals of an automorphism)**.** Take

$$P = x - y^2, \qquad Q = y. \tag{3}$$

We compute the Jacobian step by step. First the partial derivatives:

$$P_x = 1, \qquad P_y = -2y, \qquad Q_x = 0, \qquad Q_y = 1.$$

Hence

$$[P, Q] = P_x Q_y - P_y Q_x = 1 \cdot 1 - (-2y) \cdot 0 = 1 \in K^\times,$$

so $F = (P, Q)$ is a Keller map. It is an automorphism: solving $P = x - y^2$ and $Q = y$ gives $y = Q$ and $x = P + Q^2$, so the inverse is $(P, Q) \mapsto (P + Q^2, Q)$, and indeed $P + Q^2 = (x - y^2) + y^2 = x$.

The affine dicritical. The fiber over 0 is the single point $(0, 0)$, since $Q = y = 0$ and $P = x - y^2 = 0$ force $x = 0$. Blow up $\mathbb{A}^2$ at the origin and use the chart

$$x = s, \qquad y = st,$$

so the exceptional divisor E is $\{s = 0\}$ with coordinate t. Substituting:

$$P = s - (st)^2 = s - s^2t^2 = s\,(1 - st^2), \tag{4}$$
$$Q = st. \tag{5}$$

The map φ_0 is the projection from 0, so on this chart it is the ratio

$$\varphi_0 = [P : Q] = [\, s(1 - st^2) : st \,] = [\, 1 - st^2 : t \,].$$

Restricting to the exceptional divisor, where $s = 0$, gives

$$\varphi_0|_E = [1 : t],$$

which is non-constant in t. Hence E is dicritical, of degree 1.

The dicritical at infinity. By Corollary 1.4, Λ_0 has exactly one dicritical at infinity, the unique curve D with $G(D) = L'_\infty$, also of degree 1. Together with the affine dicritical above, this is two distinct dicriticals. The October 1 theorem, which forbade two distinct dicritical divisors without any at-infinity restriction, is therefore false, already for automorphisms.

1.6 The corrected statement: Theorem R

Theorem 1.7 (Theorem R — the corrected main theorem; OPEN)**.** *Status: open, not proved in this paper or elsewhere. For every Keller map (P, Q), the pencil $\lambda P + \mu Q$ has at most one dicritical divisor at infinity.*

Proposition 1.8. *Status: grade B. Let F be a Keller map and $c \notin S_F$. Then Λ_c has at most one dicritical at infinity if and only if F is an automorphism. In that case it has exactly one, of degree 1.*

Proof. If F is an automorphism, Corollary 1.4 gives exactly one dicritical at infinity, of degree 1. Conversely, suppose Λ_c has at most one dicritical at infinity. By Proposition 1.3(c), the number of dicriticals at infinity is at least $1 + \#\{\text{components of } S_F\}$. At most one therefore forces $\#\{\text{components of } S_F\} = 0$, that is, $S_F = \emptyset$. By (F3), F is an automorphism. □

Proposition 1.9 (Theorem R is equivalent to the planar Jacobian conjecture)**.** *Status: grade B. Theorem R holds for all Keller maps if and only if the planar Jacobian conjecture holds.*

Proof. If the conjecture holds, then every Keller map is an automorphism, and Corollary 1.4 gives exactly one dicritical at infinity, in particular at most one. So Theorem R holds.

Conversely, let F be a Keller map and pick $c \notin S_F$. Such a c exists because S_F is either empty or a curve, never the whole target. Then $F - c$ is a Keller map with $0 \notin S_{F-c}$, and its pencil $\lambda(P - c_1) + \mu(Q - c_2)$ is $\Lambda_c(F)$. Applying Theorem R to $F - c$ and Proposition 1.8 to $\Lambda_c(F)$ makes F an automorphism. Since F was arbitrary, the conjecture holds. □

Remark 1.10. Proposition 1.9 sets the bar for this paper: a complete proof of the corrected theorem would be a proof of the conjecture. Everything that follows is therefore measured against that bar, and no step below is presented as closing it.

1.7 Where a counterexample's dicriticals would sit

Let F be a Keller map that is not an automorphism, and let $c \notin S_F$. Then Λ_c has:

- $e \geq 2$ affine dicriticals, by (F3) and Proposition 1.3(a);
- at least one E_∞ with $G(E_\infty) = L'_\infty$, of degree $\deg(E_\infty \to L'_\infty)$;
- at least one E_S over each component C of S_F, of degree $\deg(E_S \to C) \cdot \deg C$, by (1).

E_∞ always exists, so Theorem R amounts to showing that E_S cannot exist. The two halves of the October 1 draft, read against this picture, were: the first half claimed there is no dicritical of degree $m \geq 2$ for any Keller map, which applied to E_S would already exclude every counterexample whose S_F has a component of degree at least 2; the second half claimed there are no two dicriticals of degree 1, which would handle the rest. Either half, to succeed, must apply to E_∞ and E_S wherever they sit in the resolution tree: over any point of L_∞, at any depth, and after free or satellite blow-ups. Sections 2–4 record how far each half actually goes.

2 Dicriticals at infinity of small depth

This section proves the two depth 0 and depth 1 steps toward Theorem R, each with every intermediate computation shown, and then records the constant-leading-form argument in its honest state: a sketch for free centers only. The conclusion of this section is that every dicritical at infinity of a non-affine Keller map has depth at least 2. Nothing here touches depth 2 or beyond, except for the sketch in §2.3, whose limits are stated there.

2.1 The line at infinity

Lemma 2.1 (The line at infinity)**.** *Status: grade B, proof by hand; the binary-form step is executed in* ***`base_point_at_infinity_check.py`****, check* 1*. For a non-affine Keller map, L_∞ is not dicritical for any Λ_c.*

Proof. Let $d = \max(\deg P, \deg Q) \geq 2$, the non-affine case. Write P_d and Q_d for the homogeneous components of degree d of P and Q (zero if the corresponding degree is smaller than

d). The restriction $\varphi_c|_{L_\infty}$ is the rational function P_d/Q_d on $L_\infty \cong \mathbb{P}^1$, where the point c contributes only a constant shift that does not affect constancy. We show this ratio is constant, in two cases.

Case 1: unequal degrees. Suppose $\deg P \neq \deg Q$, say $\deg P > \deg Q$ (the other case is symmetric). Then on L_∞, the term of highest degree in the pencil is P_d alone, so P_d/Q_d is ∞ wherever Q_d vanishes to lower order, and the restriction is the constant ∞. Explicitly: homogenizing, the leading part of a member $\lambda P + \mu Q$ at infinity is λP_d (since $\deg Q < d$), so the pencil has the single point at infinity determined by P_d as its value, independent of the member. The restriction is constant, so L_∞ is not dicritical. The case $\deg Q > \deg P$ gives the constant 0 by the same computation.

Case 2: equal degrees. Suppose $\deg P = \deg Q = d$. The Jacobian $[P, Q]$ is a nonzero constant, so in particular its homogeneous component of degree $2d - 2$ vanishes. That component is exactly $[P_d, Q_d]$, because differentiating lowers degree by 1 in each factor and the bracket of the top components is the top component of the bracket:

$$[P, Q]_{2d-2} = [P_d, Q_d] = 0. \tag{6}$$

So it suffices to prove the following binary-form lemma.

Lemma 2.2 (Proportionality of binary forms with vanishing bracket)**.** *Let f, g be binary forms (homogeneous polynomials in two variables) of the same degree ≥ 1. If $[f, g] = 0$, then $f = rg$ for some constant $r \in K$.*

Proof of Lemma 2.2, step by step. Work in the rational function field and set $h = f/g$, a rational function of degree 0, meaning $h(tx, ty) = h(x, y)$ for all t. Differentiating this homogeneity relation with respect to t at $t = 1$ gives Euler's relation for h:

$$xh_x + yh_y = 0. \tag{7}$$

Next, the bracket is bilinear and satisfies the quotient rule in the first argument:

$$[h, g] = \frac{[f, g]}{g} - \frac{f[g, g]}{g^2} = \frac{[f, g]}{g} = 0,$$

using $[g, g] = 0$ and the hypothesis $[f, g] = 0$. Writing the bracket out,

$$h_x g_y - h_y g_x = 0. \tag{8}$$

Now suppose $(h_x, h_y) \neq (0, 0)$. Equation (8) says the vector (h_x, h_y) is proportional to (g_x, g_y): both are orthogonal to the same direction, so there is a scalar λ with $(g_x, g_y) = \lambda(h_x, h_y)$ wherever the vectors are defined. Equation (7) says (h_x, h_y) is proportional to $(-y, x)$: indeed, $xh_x + yh_y = 0$ means (h_x, h_y) is orthogonal to (x, y), and in two variables the orthogonal direction is spanned by $(-y, x)$. Combining the two proportionalities, (g_x, g_y) is proportional to $(-y, x)$, so

$$xg_x + yg_y = 0.$$

But Euler for the homogeneous form g of degree $\deg g \geq 1$ says

$$xg_x + yg_y = (\deg g) \cdot g \neq 0.$$

This is a contradiction. Hence the supposition is false, and $(h_x, h_y) = (0, 0)$. In characteristic 0, a rational function with vanishing partial derivatives is constant, so $h = r$ for some $r \in K$, that is, $f = rg$. □

Returning to Case 2: Lemma 2.2 applied to $f = P_d$, $g = Q_d$ gives $P_d = rQ_d$, so $\varphi_c|_{L_\infty} = P_d/Q_d = r$ is constant. Hence L_∞ is not dicritical. This completes both cases. □

2.2 First blow-ups at infinity

Lemma 2.3 (First blow-ups at infinity). *Status: algebra grade A, in* ***lean/TwoDicritical/E1Keller.lean*** *(****e1_lead_proportional****, ****jac2_eq_zero_of_natDegree_zero****); chart step grade B, executed in* ***lean/check_e1_index.py****, check* 5*, not Lean-formalized. For a Keller map, the exceptional curve of the blow-up at any point of L_∞ is not dicritical. This holds for every degree m, not only for $m = 2$.*

Proof. A linear change of coordinates keeps the Keller condition (the Jacobian transforms by the constant determinant of the change), so we may move the point of L_∞ to $[1 : 0 : 0]$. Let E_1 be the exceptional curve of the blow-up at this point, with the chart coordinate t_1 on E_1 as in `lean/check_e1_index.py`. The proof has three steps: the restriction formula, the Wronskian computation, and the degenerate cases.

Step 1: the restriction formula. Suppose P and Q have the same x-degree $n \geq 1$, with x^n-coefficients $A, D \in K[y]$, so that

$$P = A(y)\,x^n + (\text{terms of } x\text{-degree} < n), \qquad Q = D(y)\,x^n + (\text{terms of } x\text{-degree} < n). \tag{9}$$

In the E_1 chart, the ratio P/Q restricts to E_1 as

$$\left.\frac{P}{Q}\right|_{E_1} = \frac{A(1/t_1)}{D(1/t_1)}. \tag{10}$$

The reason is that on E_1 only the top x-degree terms survive: lower x-degree terms vanish to higher order along E_1, and the chart coordinate t_1 inverts the y-variable, which is why A and D are evaluated at $1/t_1$. If instead the x-degrees of P and Q differ, say $\deg_x P > \deg_x Q$, then the numerator dominates and the restriction is the constant ∞; if $\deg_x P < \deg_x Q$, it is the constant 0. In both differing cases E_1 is already not dicritical, so only the equal-degree case remains.

Step 2: the Wronskian computation. Differentiate the expansions (9) term by term. The top x-degree contributions are:

$$P_x = nA(y)\,x^{n-1} + (x\text{-degree} < n-1), \tag{11}$$
$$P_y = A'(y)\,x^n + (x\text{-degree} < n), \tag{12}$$
$$Q_x = nD(y)\,x^{n-1} + (x\text{-degree} < n-1), \tag{13}$$
$$Q_y = D'(y)\,x^n + (x\text{-degree} < n). \tag{14}$$

Now form the Jacobian and collect the top x-degree. The product P_xQ_y has top x-degree $(n-1)+n = 2n-1$, with coefficient

$$(nA)\cdot D' = nAD'.$$

The product P_yQ_x likewise has top x-degree $n+(n-1) = 2n-1$, with coefficient

$$A'\cdot(nD) = nA'D.$$

Every other product of a term from P_x, P_y with a term from Q_x, Q_y has x-degree strictly less than $2n-1$: lowering the x-degree of either factor by at least 1 lowers the total by at least 1. Therefore the x^{2n-1}-coefficient of $[P,Q]$ is exactly

$$[P,Q]_{x^{2n-1}} = n\left(AD' - A'D\right). \tag{15}$$

Since (P,Q) is Keller, $[P,Q]$ is a nonzero constant, so every coefficient of a positive power of x in $[P,Q]$ vanishes. In particular the coefficient (15) vanishes, and since $n \geq 1$,

$$AD' - A'D = 0.$$

This is the vanishing of the Wronskian of A and D. In characteristic 0 it forces A and D to be proportional: indeed, where $D \neq 0$,

$$\left(\frac{A}{D}\right)' = \frac{A'D - AD'}{D^2} = 0,$$

so A/D is constant, say $A = rD$ for $r \in K$. (This is the Lean lemma `e1_lead_proportional`.) Substituting into the restriction formula (10) gives

$$\left.\frac{P}{Q}\right|_{E_1} = \frac{rD(1/t_1)}{D(1/t_1)} = r,$$

a constant. Hence E_1 is not dicritical in the equal-degree case either.

Step 3: the degenerate case. If both x-degrees are 0, then $P, Q \in K[y]$ are functions of y alone, so $P_x = Q_x = 0$ and $[P,Q] = 0$. This contradicts the Keller condition $[P,Q] \in K^\times$ (Lean lemma `jac2_eq_zero_of_natDegree_zero`). So this case cannot occur for a Keller map.

Since a linear change of coordinates moves any point of L_∞ to $[1:0:0]$ while preserving the Keller condition, the same conclusion holds over every point at infinity. □

Corollary 2.4 (Depth at least 2)**.** *Status: follows from Lemmas 2.1 and 2.3. Every dicritical at infinity of a non-affine Keller map has depth at least 2.*

Proof. Depth 0 is L_∞ itself, excluded by Lemma 2.1 for non-affine maps. Depth 1 curves are the exceptional curves of blow-ups at points of L_∞, excluded by Lemma 2.3. The smallest remaining depth is 2. □

2.3 The constant-leading-form argument: a sketch for free centers only

Proposition 2.5 (Constant leading form — proof sketch)**.** *Status: grade C, proof sketch; free centers only. This is not a proof for satellite chains or for depth ≥ 2 in general (obligation O3). Suppose E_k ($k \geq 2$) is dicritical of order $m \geq 1$ for the pencil. Then after the standard normalization, the leading coefficient satisfies* $\deg_{u'} U(u',0) = 0$.

Proof sketch. Let E_k ($k \geq 2$) be an exceptional divisor in the resolution, with local coordinates (u',v') in which $E_k = \{v' = 0\}$. For a polynomial $S \in K[x,y]$ of degree d, write the pullback as

$$S = v'^{-v}\, U(u',v') \cdot (\text{monomial in } u'), \tag{16}$$

where $v = v_{E_k}(S)$ is the order of vanishing and U is a unit (nonzero at the center). The dicritical condition of order m requires the u'-degree of the pencil restriction to equal m.

Track the unit U through the blow-up sequence, one blow-up at a time. Consider a single blow-up at a center with $u_0 \neq 0$ (a *free* center: the center does not lie on another component of the total transform, so the local coordinate u' is nonzero there). In the blow-up chart, substituting the center shifts the leading coefficient by the constant factor u_0^{-2}: schematically, if before the blow-up the leading term is $c \cdot u'^j$, then after the blow-up it is $c\, u_0^{-2} \cdot u''^j$ in the new coordinate u'', with the same exponent j and a nonzero constant multiple. The point is that multiplication by the nonzero constant u_0^{-2} changes no u'-degree: the exponent j is carried through unchanged, and no new u'-dependence is created, because the center coordinate u_0 is a nonzero constant rather than a function of u'.

Repeating this step along the chain: each blow-up at a center with $u_0 \neq 0$ multiplies the leading coefficient by a nonzero constant and preserves its u'-degree. A direct computation in the normalized chart then shows the u'-degree of $U(u', 0)$ is 0. The verification of the degree count is in `general_constant_form.py` (§6).

Where the sketch stops. The step above uses $u_0 \neq 0$ at every blow-up. If a center lies on another component of the total transform (a satellite center), then the corresponding coordinate vanishes at the center, the factor is no longer a nonzero constant, and the degree-tracking argument as written does not go through. The sketch therefore covers free chains only. It does not prove that no E_k ($k \geq 2$) is dicritical of order $m \geq 2$ in general, and it is not presented as doing so: the general $m \geq 2$ statement at depth ≥ 2 is obligation O3 in §5. The middle-divisor (E_2) and leaf cases (E_3, E_4) are checked separately as computations in `middle_e2.py`, `leaf_e3.py`, and `leaf_e4.py`; those are fixed instances, not the general argument. □

Remark 2.6. In the October 1 draft, Proposition 2.5 was followed by a corollary claiming no E_k ($k \geq 2$) is dicritical of order $m \geq 2$ for any Keller map, and that corollary was then used to close the $m \geq 2$ half of the main theorem. That inference is withdrawn: a sketch for free centers does not yield the general corollary, and the general corollary is not claimed anywhere in this revision.

3 The first exceptional divisor: a false obstruction and its correction

This section does two things the October 1 draft got backwards. First, it exhibits the draft's E_1 degree obstruction as false, shows the exact step where its proof goes wrong, and computes the counterexample in full. Second, it records the corrected statement, which was already proved as Lemma 2.3 in §2.2: first blow-ups at infinity are never dicritical for a Keller map, for every degree m. The correction is strictly stronger than the draft's claim in scope (every m, not only $m = 2$) and strictly weaker in what it asserts about E_1 itself (it does not claim anything about a hypothetical $m = 2$ open subset, which, as the counterexample shows, contains Keller pairs).

3.1 What the October 1 draft claimed

The draft's E_1 degree obstruction theorem read, in substance: let P, Q lie in the E_1 $m = 2$ open subset, meaning $v_{E_1} = -2$ and both pencil restrictions have exact degree 2; then $[P, Q]$ is never a nonzero constant. Its proof argued as follows: every monomial $x^J y^K$ allowed by the order condition $v_{E_1} \geq -2$ satisfies $J + K \geq 2$; hence P_x, P_y, Q_x, Q_y have no constant term; hence $[P, Q]$ has minimum total degree 2 and cannot be a nonzero constant. Every step after the first

sentence is correct arithmetic. The first sentence is false, and this subsection shows why, step by step.

3.2 The order condition, computed correctly

Lemma 3.1 (The E_1 order condition)**.** *In the E_1 chart, the valuation satisfies*

$$v_{E_1}(x^i y^j) = -i. \tag{17}$$

Consequently $v_{E_1} \geq -2$ holds for a monomial $x^i y^j$ if and only if $i \leq 2$, that is, if and only if the x-degree is at most 2. It says nothing about the total degree $i + j$.

Proof. Equation (17) is the normalization of the E_1 valuation used in the E_1 chart (see `lean/check_e1_index.py` and `lean/TwoDicritical/E1Keller.lean`): the variable x has a pole of order 1 along E_1, contributing -1 per power of x, while y contributes 0. Hence for the monomial $x^i y^j$,

$$v_{E_1}(x^i y^j) = i \cdot v_{E_1}(x) + j \cdot v_{E_1}(y) = i \cdot (-1) + j \cdot 0 = -i.$$

The inequality $v_{E_1} \geq -2$ is therefore

$$-i \geq -2 \quad \Longleftrightarrow \quad i \leq 2.$$

In words: the condition bounds the x-degree by 2. It does not bound j, and it does not bound the total degree $i + j$ from below: the monomial y itself, with $i = 0$ and $j = 1$, satisfies $v_{E_1}(y) = 0 \geq -2$ while having total degree 1. The draft's claim that every allowed monomial has $J + K \geq 2$ is thus false on its face, once the valuation is written out. □

3.3 How the mix-up happened

The draft's error is an index mix-up, identified in the Lean analysis (`lean/README.md`, item 2). The computation `e1_solitary.py` indexes its unknowns $p[J, K]$ by the homogeneous monomial $X^{d-J-K} Y^J Z^K$, that is, by the affine monomial

$$x^{d-J-K} y^J.$$

The draft's proof, and the scripts `e1_keller_search.py` and the Lean file `lean/TwoDicritical/E1Obstruction.lean`, read $p[J, K]$ as the monomial $x^J y^K$ instead. Under that misreading, the condition on the indices becomes a condition on the total degree $J + K$, and the draft concluded that every allowed monomial has total degree at least 2. At $d = 4$, both index sets have 12 monomials, so the count agreed and the mix-up was hidden: the 12-dimensional span at $d = 4$ is real, but it is a span of monomials of x-degree at most 2, not of total degree at least 2. The file `lean/TwoDicritical/E1Obstruction.lean` proves a true lemma about the misread set of monomials (if every monomial of P and Q has total degree at least 2, then $[P, Q]$ has no constant term), but that set is not the E_1 condition, so the lemma does not prove anything about E_1.

3.4 The counterexample, computed in full

Example 3.2 (A Keller pair in the draft's E_1 open subset)**.** *Status: grade A in Lean (**`lean/TwoDicritical/E1Keller.lean`**, **`thm_e1_false`**); the computation below is exact.* Set

$$P = (x + y^2)^2 + y, \qquad Q = P + x + y^2. \tag{18}$$

We expand P first, term by term:

$$P = x^2 + 2xy^2 + y^4 + y.$$

Its partial derivatives are:

$$P_x = 2x + 2y^2 = 2\,(x + y^2), \tag{19}$$
$$P_y = 4xy + 4y^3 + 1 = 4y\,(x + y^2) + 1. \tag{20}$$

Now compute $[P, Q]$. Since $Q = P + x + y^2$ and $[P, P] = 0$, bilinearity of the bracket gives

$$[P, Q] = [P, x] + [P, y^2].$$

Each bracket is computed separately. For $[P, x]$: x has derivatives 1 and 0, so

$$[P, x] = P_x \cdot 0 - P_y \cdot 1 = -P_y = -\big(4y(x + y^2) + 1\big).$$

For $[P, y^2]$: y^2 has derivatives 0 and $2y$, so

$$[P, y^2] = P_x \cdot 2y - P_y \cdot 0 = 2yP_x = 2y \cdot 2(x + y^2) = 4y(x + y^2).$$

Adding the two contributions, the $4y(x + y^2)$ terms cancel exactly:

$$[P, Q] = -\big(4y(x + y^2) + 1\big) + 4y(x + y^2) = -1 \in K^\times. \tag{21}$$

So (P, Q) is a Keller pair.

It lies in the draft's E_1 $m = 2$ open subset at $d = 4$: its x-degree is 2 (the term x^2), so $v_{E_1} = -2$ by Lemma 3.1, and each pencil restriction is the function $C(t_1) = t_1^2$ of exact degree 2 (this is the membership statement formalized as `thm_e1_false` in `lean/TwoDicritical/E1Keller.lean`, with the open subset denoted `E1Open4` there). Yet $[P, Q] = -1$ is a nonzero constant, contradicting the draft's theorem directly.

Two further observations pin down the failure. First, P contains the monomial y, of total degree 1: the draft's claimed lower bound $J + K \geq 2$ fails for a monomial that is actually present. Second, P/Q restricts to 1 on E_1 for this pair, so E_1 is not dicritical for them at all: the draft's "open subset" does not even capture an $m = 2$ dicritical. The draft's proof argued about the wrong monomials, and its conclusion is false.

Remark 3.3 (Withdrawn theorem)*.* The draft's E_1 degree obstruction theorem and its corollary ("no dicritical of order $m \geq 2$ exists for any Keller map") are both withdrawn. The corollary depended on the false theorem for the E_1 case and on the free-only sketch of Proposition 2.5 for the E_k, $k \geq 2$, case; neither leg supports it as stated.

3.5 The corrected E_1 statement

The intended conclusion of the draft's E_1 argument — that the first blow-up cannot be dicritical for a Keller map — is true, and it is Lemma 2.3 of §2.2, proved there for every degree m. To recap the logic in one place: the restriction of P/Q to E_1 is $A(1/t_1)/D(1/t_1)$ when the x-degrees agree (and 0 or ∞ when they differ), and the Keller condition forces the Wronskian $AD' - A'D$ to vanish via the x^{2n-1}-coefficient (15), hence $A = rD$ and the restriction is the constant r. The algebra is Lean-grade-A; the chart step is executed, not Lean-formalized (obligation O5). No $m = 2$ open subset, and no total-degree argument, is used anywhere in the corrected proof.

4 The $m = 1$ bridge, under its premises

This section keeps the October 1 draft's Vandermonde and block-rank argument, which is correct as linear algebra, but restates its scope exactly. The argument proves a collapse of a system of equations. Whether that system is the system of two $m = 1$ dicriticals depends on two premises, (P1) and (P2), stated below. Premise (P2) is not derived anywhere in this paper; it is an open obligation (O2). The theorem of this section is therefore a theorem about equations under premises, not an unconditional geometric statement, and its corollaries are fenced accordingly.

4.1 The chart and the two premises

Read through its chart, the bridge concerns the following geometry. The paper's pullback is

$$S^h\big(1, b, b^2(a + \tilde{a}b)\big),$$

the chart at the point $w_2 = a$ of E_2 over $[1 : 0 : 0]$, where $w_2 = x/y^2$. The two centers $a_3 \neq a_4$ are points of this E_2. The bridge argument rests on:

(P1) the two dicriticals lie over two distinct *free* points of this E_2;
(P2) every member of the pencil has pole order at most 1 on them.

Premise (P1) is part of the geometric reading of the equations: the chart above is the chart at a free point of E_2, and the argument is set in that chart. Premise (P2) is a hypothesis on the members, not a consequence of the definitions used here: the paper defines m as the degree of $\varphi_c|_E$, and degree 1 does not force pole order 1 (`lean/check_satellite.py`, check 7). Deriving (P2) from the definition of m, or supplying a separate argument for higher pole order, is obligation O2 in §5. Nothing in this section assumes (P2) silently: every statement that needs it says so.

4.2 Setup of the equations

Consider two distinct centers $a_3 \neq a_4$ (both nonzero), each requiring order $\mathrm{ord}_b(S) \geq 2d - 1$ (the $m = 1$ condition, under (P2)). For $S = \sum p_{JK} x^J y^K$, the pullback at center a is

$$S_a(b) = \sum_{J,K} p_{JK}\,(a + \tilde{a}b)^{d-J-K}\, b^{2d-2J-K}, \tag{22}$$

and we require the coefficients of $b^0, \dots, b^{2d-2}$ to vanish at $a = a_3$ and at $a = a_4$. The exponent $d - J - K$ is the remaining degree after the monomial $x^J y^K$ is factored out, and the exponent $2d - 2J - K$ is the b-order contributed by that monomial in the chart; the condition $\mathrm{ord}_b(S) \geq 2d - 1$ is exactly the vanishing of all coefficients below b^{2d-1}.

Theorem 4.1 (Bridge collapse, under (P1) and (P2))**.** *Status: equations grade A, in* ***lean/TwoDicritical/BridgeCollapse.lean****; geometric reading grade B, executed in* ***lean/check_satellite.py****. For arbitrary distinct nonzero a_3, a_4 and arbitrary $d \geq 2$, the simultaneous conditions* $\mathrm{ord}_b(S) \geq 2d - 1$ *at both centers force $p_{JK} = 0$ whenever $J \geq 1$. The nullspace is exactly* $\mathrm{span}\{1, y\}$.

In particular, every P, Q in the span is a function of y alone, so $[P, Q] \equiv 0$: no Keller pair exists in the system of equations. Whether this rules out two geometric $m = 1$ dicriticals depends on (P1) and (P2); see §4.5.

4.3 Descending induction on weighted degree

Order monomials by the weighted degree $W = 2J + K$, descending. For fixed W, write the variables as (J_k, K_k), $k = 1, \dots, n$, ordered by descending J_k. Define

$$e_k = d - J_k - K_k = d - W + J_k. \tag{23}$$

Since W is fixed and the J_k are strictly decreasing by the ordering, the e_k are strictly decreasing, hence distinct. This distinctness is the whole input to the rank lemma below, so we record it explicitly: for $k < l$,

$$e_k - e_l = J_k - J_l > 0.$$

The equations for weight W come from the pairs (t, i) with $2d - t + i = W$, where t is the b-power and i the $\tilde{a}$-power, at both centers. To see why, expand the factor $(a + \tilde{a}b)^{e_k}$ in (22) by the binomial theorem, term by term:

$$(a + \tilde{a}b)^{e_k} = \sum_{i \geq 0} \binom{e_k}{i} a^{e_k - i} (\tilde{a}b)^i.$$

Multiplying by $b^{2d-2J_k-K_k}$ and using $W = 2J_k + K_k$, the monomial $p_{J_k K_k}$ contributes to the coefficient of b^t exactly the terms with $2d - W + i = t$, that is, $i = t - (2d - W)$. Collecting over k gives the (t, i, a) equation:

$$\sum_{k=1}^{n} p_{J_k K_k} \binom{e_k}{i} a^{e_k - i} = 0, \qquad a \in \{a_3, a_4\}. \tag{24}$$

Crucially, the equations (24) for weight W involve only variables of weight exactly W at the top: variables of weight $> W$ contribute to higher b-powers or have already been eliminated, as the induction below makes precise.

Lemma 4.2 (Block full rank)**.** *For distinct $e_1, \dots, e_n$ and $m \geq n$, the $2m \times n$ matrix with entries $\binom{e_k}{i} a^{e_k - i}$ ($i = 0, \dots, m - 1$, $a \in \{a_3, a_4\}$) has rank n.*

Proof. We reduce the matrix in three explicit steps, each preserving rank.

Step 1: factor the columns and rows. Factor column k by $a^{e_k} \neq 0$ (note a is a_3 or a_4, both nonzero by hypothesis) and row (i, a) by $a^{-i} \neq 0$. Rank is preserved under nonzero column and row scalings. After this step, the entry in row (i, a), column k is $\binom{e_k}{i}$, independent of a. The two centers now give two copies of the same rows, so it suffices to show that the $m \times n$ matrix $[\binom{e_k}{i}]$ $(i = 0, \dots, m-1)$ has rank n; taking the first n rows $(i = 0, \dots, n-1)$ suffices, since $m \geq n$.

Step 2: triangularize by row operations. For fixed i, the binomial coefficient $\binom{e}{i}$, viewed as a function of e, is a polynomial in e of exact degree i:

$$\binom{e}{i} = \frac{e(e-1)\cdots(e-i+1)}{i!} = \frac{1}{i!}e^i + (\text{terms of degree} < i \text{ in } e).$$

Hence row i of the matrix $[\binom{e_k}{i}]$ equals $\frac{1}{i!}$ times the row $[e_k^i]$ plus a linear combination of the rows $[e_k^j]$ for $j < i$. Proceeding from $i = 0$ upward, subtract suitable multiples of earlier rows from row i to kill all lower-degree terms: row 0 is already $[1]$; row 1 is $[e_k]$ + (constant row), so subtracting the constant multiple of row 0 leaves $[e_k]$; row 2 is $\frac{1}{2}[e_k^2]$ + (combination of rows 0, 1), so subtracting leaves $\frac{1}{2}[e_k^2]$; and so on. These are elementary row operations, so they preserve rank and multiply the determinant by 1. After all n steps, the matrix is the scaled Vandermonde matrix $[e_k^i/i!]$ $(i = 0, \dots, n-1)$.

Step 3: the Vandermonde determinant. The determinant of $[e_k^i]$ is the classical Vandermonde product $\prod_{j<k}(e_k - e_j)$, nonzero because the e_k are distinct (shown above from the ordering by J_k). Factoring $1/i!$ out of row i gives

$$\det[\binom{e_k}{i}] = \frac{\prod_{j<k}(e_k - e_j)}{\prod_{i=0}^{n-1} i!} \neq 0. \tag{25}$$

A nonzero $n \times n$ minor means the matrix has rank n. The $n = 3$ case is verified symbolically in `m1_block_rank.py`, giving $-(e_1 - e_2)(e_1 - e_3)(e_2 - e_3)/2$, which agrees with (25) since $\prod_{i=0}^{2} i! = 0! \cdot 1! \cdot 2! = 2$ and the sign is the ordering of the factors. This completes the proof. □

Proof of Theorem 4.1. Descend on W from $2d$ to 2, maintaining the inductive hypothesis that all variables with weight $> W$ are zero. The base case is $W = 2d$: there are no variables of weight $> 2d$ (since $W = 2J + K \leq 2d$ for monomials of degree at most d), so the hypothesis holds vacuously.

For the inductive step at weight W, the equations (24) for weight W involve only variables of weight $\geq W$: a variable of weight $W' > W$ contributes to b-powers above those in the weight-W system, and by the inductive hypothesis all such variables are already 0. Hence the weight-W system is exactly the block system of Lemma 4.2, which has full column rank n. Therefore $p_{J_k K_k} = 0$ for all k, that is, all variables of weight W vanish. This restores the hypothesis for the next lower weight.

Descending to $W = 2$ eliminates every variable with $J \geq 1$: any monomial with $J \geq 1$ has $W = 2J + K \geq 2$, and the descent covers all weights $W \geq 2$. The only monomials with $W < 2$ are $(J, K) = (0, 0)$ with $W = 0$ and $(J, K) = (0, 1)$ with $W = 1$. These are never constrained by the descent, so the nullspace is exactly $\operatorname{span}\{1, y\}$. □

4.4 Computational verification

The induction is verified computationally, at the fixed degrees and instances stated (these are checks of the linear algebra, not replacements for the proof above):

- Symbolic nullspace at general $a_3 \neq a_4$: $d = 2$ gives nullity 2; $d = 3$ gives nullity 2 (`m1_bridge_general.py`).
- Numeric nullspace at $a_3 = 2$, $a_4 = 5$: nullity 2 for $d = 2, \ldots, 8$ (`m1_induction.py`).
- Block-triangular structure exhibited at $d = 6$ (`m1_triangular.py`).
- Vandermonde determinant verified symbolically (`m1_block_rank.py`).

4.5 Scope of the bridge result

Under premises (P1) and (P2), the bridge has the following consequences, and no others are claimed:

- With a single center imposed, the same block argument gives the larger nullspace span$\{1, y, x - ay^2\}$(`lean/TwoDicritical/OneCenter.lean`, `bridge_one_center`). So one such dicritical already forces degree at most 2: no pencil of degree at least 3 has even one dicritical of this chart type.
- In degree 2, the automorphism $(x - ay^2, y)$ has exactly one such dicritical, consistent with Theorem R (it has one dicritical at infinity, as Corollary 1.4 requires, and no second).
- The bridge as proved here says nothing about pencils of degree at least 3 beyond the one-center remark above, because (P2) is not derived: a dicritical of degree 1 with pole order greater than 1 is not governed by the system (24). Nor does it cover dicriticals over satellite points, over different points of L_∞, or at different depths; those are obligation O1.

4.6 The adjacent case: a computation at $d = 3$ only

The adjacent $m = 1$ case — two dicriticals sharing a curve in the resolution tree rather than lying over two free points of one E_2 — is treated in this paper only by computation at $d = 3$ (`e3_adjacent.py`, `routeB_cancellation.py`), covering the E_2–E_3 adjacent pair at that degree. *Status: grade C, fixed-degree computation.* There is no argument here for other degrees; the general adjacent case is obligation O4 in §5. In the October 1 draft, this computation was cited as closing the adjacent case for every degree. That citation is withdrawn.

5 What remains open

This section replaces the October 1 draft's "Proof of the main theorem," which assembled the false E_1 theorem, the free-only constant-form sketch, the premises-dependent bridge, and the $d = 3$ adjacent computation into a contradiction, via a case split ("an adjacent pair or a bridge") that was asserted, not proved. No such assembly is attempted here. The obligations below are what a proof of Theorem R still needs; the tests after them are checks any proposed proof must pass.

5.1 Obligations

- **O1. The case split.** The positions of two dicriticals in a resolution tree are more varied than "an adjacent pair or a bridge": dicriticals can lie over *different points* of L_∞ (a pencil can have several base points at infinity in general, as the non-Keller map (x, xy) shows, and whether a Keller counterexample's pencil can must be settled); over *satellite points* (the point $a = \infty$ of E_2 in $(x + y^3, y)$, and deeper ones); and at *different depths*. A proof must cover every configuration of E_∞ and E_S from §1.7.
- **O2. Premise (P2).** Derive pole order at most 1 from the definition of m, or restate the bridge with (P2) as a hypothesis and supply a separate argument for higher pole order. Until then, the bridge governs only the system (22)–(24) under (P1) and (P2).
- **O3.** $m \geq 2$ **at depth** ≥ 2**.** Prove the constant-leading-form statement of Proposition 2.5 for every chain of centers, satellite ones included. Today there is a sketch for free chains only.
- **O4. The adjacent case** for every degree, not only $d = 3$.
- **O5. Two small formalizations:** the chart step of Lemma 2.3 and the binary-form step of Lemma 2.1 in Lean. The algebra of Lemma 2.3 is formalized; the chart step is executed, not formalized.
- **O6. The Keller condition must be used globally.** Theorem R is false without it. The map (x, xy), with Jacobian x, has two dicriticals at infinity in $\Lambda_{(1,1)}$, over $[1 : 0 : 0]$ and $[0 : 1 : 0]$; the second maps onto the line $u = 0$ of non-proper values (`restatement_checks.py`, check 2). Any argument that never uses $[P, Q] \in K^\times$ cannot prove Theorem R. This is a standing check on every step above: Lemmas 2.1 and 2.3 use the Keller condition (vanishing of $[P_d, Q_d]$, vanishing of the Wronskian coefficient), while the bridge's linear algebra and the constant-form sketch must be read with their premises attached.

5.2 Tests any proof must pass

- **T1.** Automorphisms have exactly one dicritical at infinity, of degree 1 (Corollary 1.4), but its position varies: over the free point a of E_2 for $(x - ay^2, y)$ (where $P/Q = \tilde{a}/a$ there), and over the satellite point $a = \infty$ of E_2, at depth 5, for $(x + y^3, y)$. A lemma that rules out a dicritical in either position is false.
- **T2.** The identity. Its dicritical at infinity is L_∞ itself; Lemma 2.1 excludes only non-affine maps.
- **T3.** Non-Keller maps can have two dicriticals at infinity (O6).
- **T4.** Every Keller map of degree at least 2 has two exceptional dicriticals in total (Corollary 1.5), so only the count *at infinity* can be bounded. Any argument that bounds the total count is refuted by Example 1.6.

6 Script-to-arithmetic correspondence

Each script below was actually executed; outputs are archived in `outputs/`. The table maps every script to the exact paper statement it certifies, *under the corrected statements of this revision*. The grade column follows the restatement note, §8: A means machine-checked, B means a proof by hand from stated facts or an exact computation outside Lean, C means a sketch or a computation at fixed degrees. A script that certified a withdrawn statement is listed with what it actually certifies now, not with the draft's reading of it.

Script	Paper locus	Certifies	Grade
`general_constant_form.py`	Prop. 2.5	Leading-unit degree count, free chains only	C
`middle_e2.py`	Prop. 2.5	E_2 instance of the sketch	C
`leaf_e3.py`	Prop. 2.5	E_3 leaf instance of the sketch	C
`leaf_e4.py`	Prop. 2.5	E_4 leaf instance of the sketch	C
`e1_solitary.py`	§3	12-monomial count at $d = 4$ (indexing as computed)	C
`e1_global.py`	§3	Explicit pair, $v = -2$, deg = 2 (membership data)	C
`e1_keller_search.py`	§3	Searched the misread monomial set; superseded	—
`m1_bridge_general.py`	Thm. 4.1	Symbolic nullspace, $d = 2, 3$	B
`m1_induction.py`	§4	Nullity 2 for $d = 2, \ldots, 8$	C
`m1_triangular.py`	§4	Block-triangular structure at $d = 6$	C
`m1_block_rank.py`	Lem. 4.2	Vandermonde determinant $\neq 0$	B
`at_degree_bound.py`	Prop. 2.5	Finite bridge table, $m \leq 4$	C
`at_degree_bound_m5.py`	Prop. 2.5	Finite bridge table, $m \leq 5$	C
`mixed_poles.py`	Prop. 2.5	Mixed $(2, 3)$ bridge instance	C
`depth5_global.py`	Prop. 2.5	Depth-5 enumeration	C
`depth6_enum.py`	Prop. 2.5	Depth-6 enumeration	C
`routeB_cancellation.py`	§4	Adjacent $m = 1$ at $d = 3$ only	C
`e3_adjacent.py`	§4	E_2–E_3 adjacent pair at $d = 3$	C

The Lean evidence, which is not a script in the table above, is:

- `lean/TwoDicritical/E1Keller.lean`: `thm_e1_false` (the draft's E_1 theorem is false, Example 3.2); `e1_lead_proportional` and `jac2_eq_zero_of_natDegree_zero` (the algebra of Lemma 2.3); grade A for the algebra.
- `lean/TwoDicritical/BridgeCollapse.lean` and `lean/TwoDicritical/OneCenter.lean`: the bridge equations and the one-center span (Theorem 4.1, §4.5); grade A for the equations.
- `lean/check_e1_index.py` (check 5) and `lean/check_satellite.py` (check 7): the E_1 chart step and the degree-versus-pole-order distinction; executed checks, grade B.
- `base_point_at_infinity_check.py` (check 1) and `restatement_checks.py` (check 2): the binary-form step of Lemma 2.1 and the non-Keller example of O6; executed checks, grade B.

7 Computational archive

All scripts used in this work are archived in `scripts/` with their outputs in `outputs/`, as archived with the October 1 draft:

- 18 Python scripts (SymPy-based), each with a header stating its epistemic status;
- 18 output files recording exact stdout and exit codes;
- All scripts exited with code 0 on re-execution (October 1, 2026).

This revision changes no script and claims no new re-execution: the corrections are to the statements the scripts are read as certifying (§6), not to the scripts themselves.

Reproducibility

All artifacts are archived in the public repository, directory `branch_two_dicritical/`. Each script runs with `~/miniconda3/envs/physics/bin/python` (Python 3.12, SymPy). The verification script `verify_all.sh` re-runs every script and checks exit codes.

8 Computational and AI-assisted research disclosure

Muse (Meta), the author's personal AI assistant, assisted with computational scripting, manuscript drafting, and proof verification; all script outputs were verified by actual execution. The author (B. D. McCrary) directed the investigation, verified the results, and takes responsibility for the manuscript. This revision was drafted with the same assistance: the October 1 draft was corrected against the October 13, 2026 restatement note, the derivations were expanded step by step, and no new theorem, computation, or script output was introduced. The mathematical argument stands on the artifacts above, not on these attributions.

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